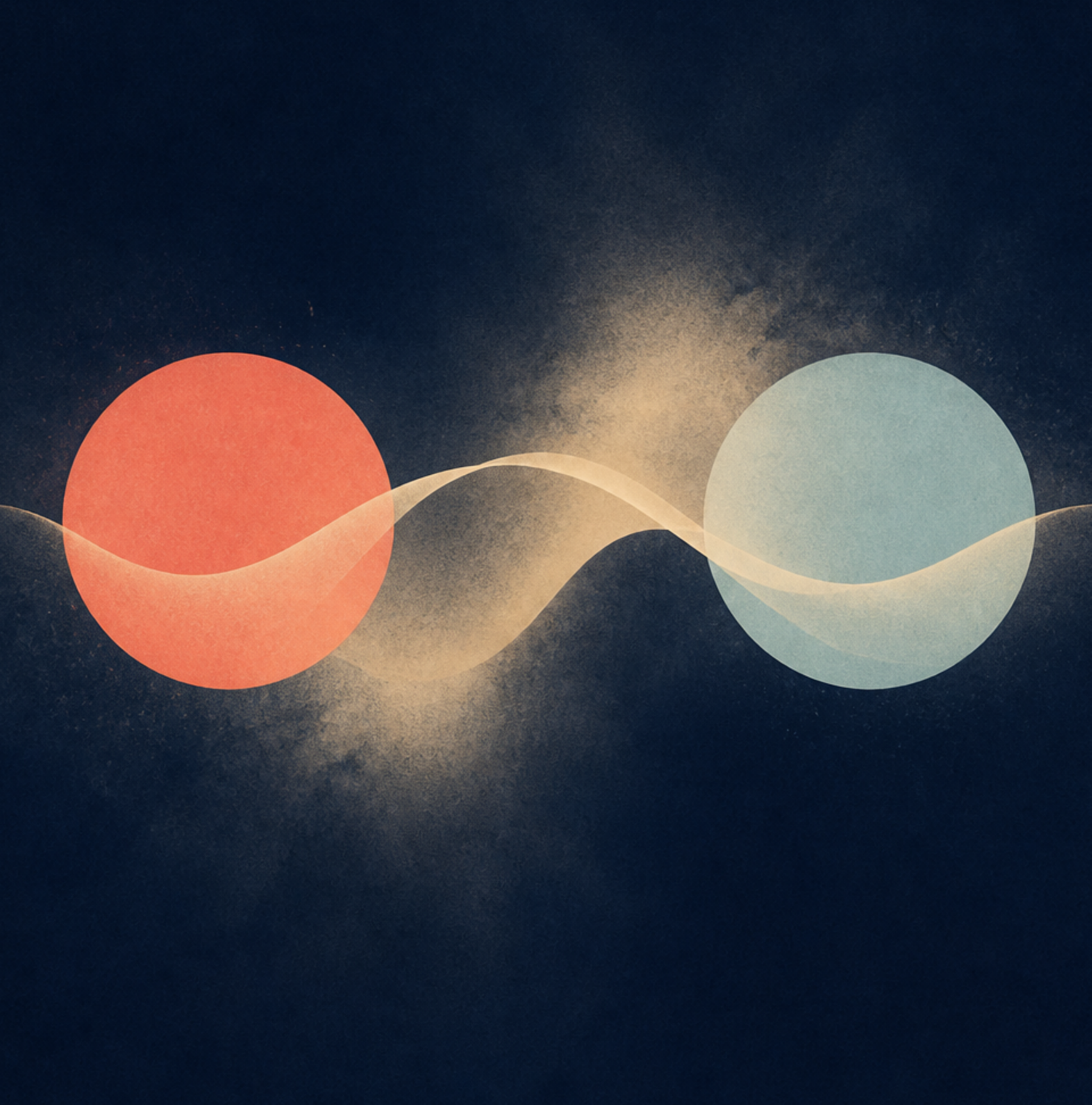
Quantum Entanglement

Supplementary track, 2006 Fall Part 1



Original lectures by Leonard Susskind

Companion edition by LazyingArt LLC with Video2Book.

ABOUT THESE NOTES

These independently edited companion notes follow lectures by Leonard Susskind. They were reconstructed from machine transcripts, subtitles, and selected blackboard frames, then checked against the available source material. They are not an original manuscript by Professor Susskind and have not been reviewed or endorsed by him.

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CHAPTER 1

CLASSICAL INFORMATION, REVERSIBLE DYNAMICS, AND MATRICES

Overview. Quantum mechanics asks us to reason outside the range in which ordinary intuition was trained. We therefore begin one step earlier, with classical information: bits, configurations, and reversible update laws. The same discussion leads naturally to vectors and matrices, which will provide the language for quantum bits.

1.1 Intuition Built for the Wrong World

Our physical intuition is old. Long before anyone wrote an equation, an animal chasing prey had to estimate relative speed and direction. If a lion is gaining on an antelope, the lion is responding to a changing separation and therefore, in an implicit way, to relative velocity. Nervous systems that handled such judgments well were favored; those that handled them badly did not leave many descendants. Much of what we call common sense about motion was trained in this ordinary world.

That inheritance is useful, but narrow. Nothing in ordinary life moves close to the speed of light, so evolution supplied no instinct for relativistic spacetime. A physicist first learns the abstract geometry and only later acquires a new intuition for it. Quantum mechanics is harder still. Our familiar picture of a particle comes from rocks and arrows: at each instant the object has a position, the succession of positions forms a trajectory, and changes of position define a velocity. An electron does not oblige us by fitting that picture. Position uncertainty and the absence of a classical trajectory have no analogue in the experience that shaped our senses.

There is no paradox in this failure of intuition. There would be a paradox only if instincts adapted to stones, animals, and modest speeds somehow knew in advance about electrons. The right response is to build a different intuition from a small set of precise rules. Here the emphasis will not be the usual route through wave mechanics and the Schrödinger equation. We will concentrate on the logical structure of quantum mechanics and on quantum information. Before asking what a quantum bit is, we must understand exactly what an ordinary bit does.

1.2 Bits and the Amount of Information

Physics is inseparable from information. Sometimes that information is expressed by a real number, but the simplest possible piece of information is an answer to a question with only two alternatives: yes or no, heads or tails, up or down.

Definition 1.1. A *classical bit* is a physical question with exactly two possible answers.

We may label those answers 0 and 1. The labels carry no intrinsic meaning; any two distinct symbols would do. Since Dirac notation will soon become indispensable, we introduce it even while it is redundant:

$$|0\rangle\,, \qquad |1\rangle\,. \tag{1.1}$$

The vertical bar and angle bracket form a *ket*, the right half of a bra–ket. Nothing quantum has entered merely because we have changed notation.

For several distinguishable bits, write the answers in a fixed order. Six distinguishable coins, for example, may have the configuration

$$|0\,1\,1\,0\,1\,0\rangle\,. \tag{1.2}$$

Commas are optional once the ordering is understood. One bit has two possible states, two bits have four, and every added bit doubles the count. Thus an n-bit system has

$$N_s = 2^n \tag{1.3}$$

states. Conversely,

$$n = \log_2 N_s. \tag{1.4}$$

Equation (1.4) suggests a definition that is useful even when the number of states is not a power of two:

$$I = \log_2 N_s. \tag{1.5}$$

A six-sided die therefore carries $\log_2 6$ bits of information. The answer need not be an integer. Systems assembled literally from binary switches will be convenient examples, but the logarithm is the more general measure.

1.3 Encoding an Apparently Continuous World

Bits are not confined to coins. Any measured quantity known to finite accuracy can be written as a finite binary string. Suppose, for example, that the temperature of a room is required only to a chosen number of significant figures. Express the measured value in base two and retain the digits needed for that accuracy. Greater precision requires a longer string, but no change of principle.

A spatially varying field can be treated similarly. Divide the room into a fine ordered lattice of cells, record the field value in each cell to finite precision, and concatenate the resulting binary strings. Making the cells smaller improves spatial resolution; keeping more binary digits improves the field-value resolution. Temperature fields, electric fields, and magnetic fields can all be encoded this way to any prescribed finite accuracy.

Particle configurations give an even more literal binary picture. Subdivide space into cells small enough that each contains at most one particle. Assign

$$n_i = \begin{cases} 1, & \text{cell } i \text{ is occupied,} \\ 0, & \text{cell } i \text{ is empty.} \end{cases} \tag{1.6}$$

The complete configuration is the ordered string

$$|n_1 n_2 n_3 \cdots\rangle\,. \tag{1.7}$$

It is a very long answer to a sequence of yes-or-no questions: is there a particle in the first cell, in the second, and so on? This is the physical version of the game *Twenty Questions*, except that the number of questions may be enormous.

The construction explains why digital computers can simulate physics. A computer stores finite strings, while finite resolution turns measured quantities and fields into precisely such strings. The approximation can be improved systematically by refining the lattice, the time step, or the number of retained digits.

[a] **Question from the audience.** Is the accuracy of the temperature representation arbitrary?

Response. It is chosen according to the approximation we want. A finite string gives finite accuracy; retaining more binary places gives a finer approximation. There is no claim here that a finite string specifies an arbitrary classical real number exactly.

[a]Lecture timestamp: 00:29:43.

[a] **Question from the audience.** What happens for a general real number, especially an irrational one?

Response. An exact real number may require infinitely many bits. Some rational binary expansions repeat, and a generic irrational expansion never terminates. A real computer always keeps finitely many digits. The point is that any specified finite accuracy can be encoded, not that every classical real number has a finite exact description.

[a]Lecture timestamp: 00:30:11.

1.4 A State and a Law of Motion

So far we have described a system at one instant. Motion requires a second ingredient.

Definition 1.2. A *configuration* is the complete state of a system at a specified instant. A discrete-time law of motion is a rule that maps each configuration to the configuration one time step later.

The distinction is elementary but essential. A snapshot of an electron distribution is a configuration; it is not yet an orbit. If space is replaced by a lattice and time by a sequence $t_0, t_1, t_2, \ldots$, then motion is the succession

$$|s(t_0)\rangle \longmapsto |s(t_1)\rangle \longmapsto |s(t_2)\rangle \longmapsto \cdots . \tag{1.8}$$

A physical model therefore needs both a set of possible states and an update rule.

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[a] **Question from the audience.** How does this cell description account for an electron moving in an orbit?

Response. First specify only the configuration at an instant: which cells are occupied, with the nucleus fixed if that is part of the model. Then discretize time. The orbital motion is the rule that updates one occupancy string to the next. The rule may be complicated because of the ordering of the cells, but it is still an update of information.

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[a] Lecture timestamp: 00:31:12.

It helps to draw an abstract *state space*. Each point in this space denotes an entire configuration, not a location in ordinary space. A 10×10 occupancy lattice contains one hundred bits and hence 2^{100} possible configurations.[1] Each one of those configurations is a single point in the state space.

For one bit there are only two points. The identity law leaves both fixed:

$$0 \mapsto 0, \qquad 1 \mapsto 1. \tag{1.9}$$

The other reversible possibility swaps them:

$$0 \mapsto 1, \qquad 1 \mapsto 0. \tag{1.10}$$

[1] Editorial clarification: The spoken lecture briefly says 10^{100} while developing the example; the bit count fixes the intended value as 2^{100}.

This is a flip-flop. A freely evolving physical example is less immediate than a manually operated light switch, because intervention would enlarge the system. Day and night provide an approximate two-state cycle; a pendulum also alternates between extremes, although its full state includes all the intermediate positions and velocities.

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[a] **Question from the audience.** Would an ordinary light switch be an example of the flip-flop law?

Response. Not if a person is toggling it, because then the switch is not evolving by itself; the person belongs to the enlarged dynamical system. What we seek is a closed system whose own law alternates its state. Day and night suggest the idea, while a pendulum reminds us that a real example usually has more than two states.

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[a]Lecture timestamp: 00:40:38.

A two-bit system has four configurations. One update law may carry all four around a single cycle. Another may consist of two separate two-cycles. These are loops in the logical space of configurations, not necessarily loops in physical space. The arrows specify the law: wherever the system starts, following the arrows tells us its subsequent history.

1.5 Reversibility and the Preservation of Distinctions

Not every arrow diagram has the same physical status. Consider a rule in which two different states merge into the same successor:

$$a \mapsto c, \qquad b \mapsto c. \tag{1.11}$$

The next state is defined whether we start at a or b, but after reaching c we cannot determine which past occurred. The distinction between a and b has disappeared.

Fundamental classical dynamics does not permit this loss when every degree of freedom is retained. The required property

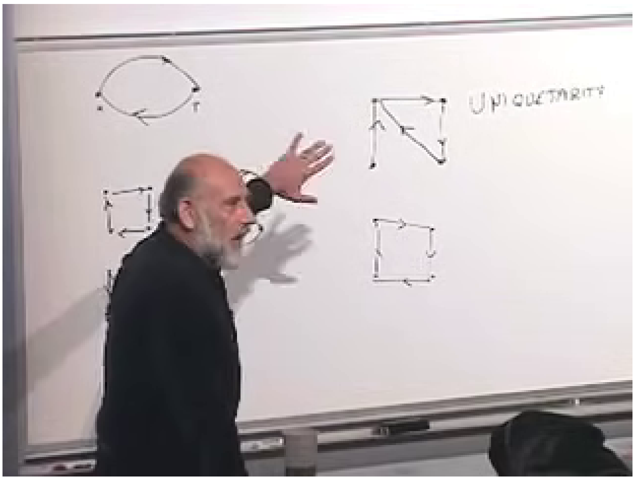

Figure 1.1: Several update diagrams remain on the board during the discussion of reversibility. Closed cycles coexist with a merging pattern, and the quantum term *unitarity* is written beside them.

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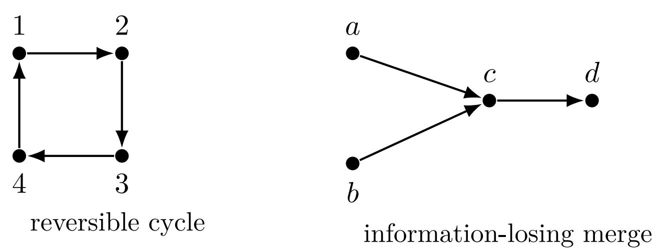

Figure 1.2: A reversible rule assigns one predecessor and one successor to every state. A merging rule has a unique future but an ambiguous past.

has two halves:

$$\boxed{\begin{array}{c}\text{one unique future and one unique past}\\ \text{for every state}\end{array}}. \tag{1.12}$$

For a finite state space, the update must therefore be a permutation. Every state has exactly one outgoing arrow and exactly one incoming arrow, so the diagram decomposes into disjoint cycles.

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[a] **Question from the audience.** What is different about the merging diagram?

Response. It does not consist entirely of loops. More fundamentally, finding the system at the merged state does not tell us which predecessor it had. We can construct the future, but we cannot reconstruct the past. One bit of historical distinction has been lost.

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[a] Lecture timestamp: 00:46:32.

This notion is *time reversibility*, not necessarily exact time-reversal symmetry. A clockwise cycle has a definite forward orientation. The original law does not send the system counterclockwise, but an inverse law exists that does. Exact time-reversal symmetry asks an additional question: after applying the appropriate reversal operation, are the same dynamical laws obtained? Reversibility alone asks only whether the history can be uniquely run backward.

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[a] **Question from the audience.** Is the uniqueness of the past just time-reversal symmetry?

Response. Not exactly. The arrows of a cycle may point only one way, so the law has an orientation. Reversibility means that a well-defined inverse law can be constructed by reversing every arrow. It does not by itself say that the inverse law is identical to the original law.

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[a] Lecture timestamp: 00:49:58.

Nor should we repair an irreversible diagram by placing two outgoing arrows at the merged state. That would make the backward motion ambiguous in a different way: the system would need a probabilistic instruction, perhaps choosing one branch 30% of the time and the other 70%. We are still constructing deterministic classical dynamics, for which one present state determines one future and one past.

Every real system is ultimately quantum mechanical. Why large objects appear classical, and why their quantum alternatives become difficult to observe, is a later question. The corresponding quantum principle will preserve the same logical core. Half jokingly, one might call uniqueness of past and future *unique-tarity*. Its actual name is *unitarity*: quantum evolution can be run uniquely forward or backward.

1.6 Apparent Information Loss and Coarse-Graining

Macroscopic experience seems to contradict reversibility. Put all the molecules of a gas in the left corner of a sealed room and release them. Soon the gas fills the room. Start instead from the right corner and the final macroscopic picture looks the same. Have the two distinct initial states merged?

Only after we have thrown away detail. A microscopic state specifies every molecule. If every position and velocity could be followed with sufficient precision, reversing those motions would distinguish the two preparations. In practice the calculation is prohibitively difficult and the required precision grows rapidly, but practical inability is not fundamental erasure. The macroscopic statement *air fills the room* identifies an enormous number of distinct microscopic configurations as one coarse-grained state.

This is where the second law of thermodynamics enters. We lose the ability to distinguish detailed states because we do not keep track of all microscopic variables. Information appears to disappear because our description is coarse, not because the fine-grained deterministic law merges distinct points.

[a] **Question from the audience.** Would tracking all the molecules in a classical room require infinitely many bits?

Response. For an exact classical description, yes. Positions and velocities are real numbers, and following a chaotic many-particle system for longer times demands ever greater precision. Quantum mechanics changes this conclusion: for a sealed region with bounded energy, the physically distinguishable description can be discrete rather than an arbitrary list of exact classical real numbers.

[a] Lecture timestamp: 00:56:11.

The issue is not merely academic. Whether a black hole can destroy information was disputed for decades; Stephen Hawking originally argued that it could. That controversy asks whether the apparent merging is only another coarse-grained description or a genuine violation of quantum unitarity. We will not settle it before quantum mechanics has supplied the precise language, but it reveals how much rests on the simple arrow diagrams.

1.7 Rows, Columns, and Inner Products

Before replacing classical bits by quantum bits, we need a small amount of linear algebra. For present purposes a vector may be treated simply as a finite ordered list of real numbers. A four-component vector can be written horizontally,

$$A^{\mathsf{T}} = (A_1, A_2, A_3, A_4), \tag{1.13}$$

or vertically,

$$A = \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{pmatrix}. \tag{1.14}$$

The word *four-dimensional* means only that there are four components. There is no need to visualize an arrow in ordinary space. The horizontal object is a row vector and the vertical object a column vector.

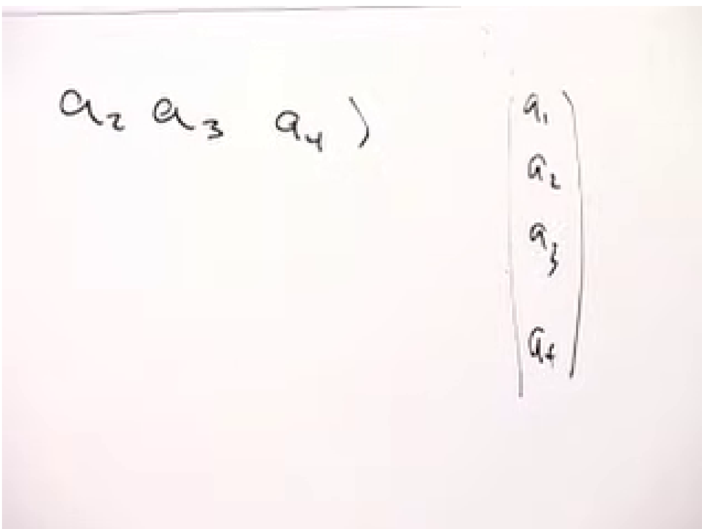

Figure 1.3: The same components are written as a row and as a column. At this stage they are real numbers, so no complex conjugation is involved.

Lecture frame: 01:03:05.

[a] **Question from the audience.** Isn't the row vector the dagger of the column vector?

Response. That will be the right language once the components are complex. Then transposition must be accompanied by complex conjugation. For the moment the entries are real, so we need only distinguish the row arrangement from the column arrangement.

[a]Lecture timestamp: 01:03:14.

Let

$$B = (B_1, B_2, B_3, B_4) \tag{1.15}$$

be a row with the same number of entries as the column A. Their product is defined by matching positions and summing:

$$BA = (B_1, B_2, B_3, B_4) \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{pmatrix} \tag{1.16}$$

$$= B_1A_1 + B_2A_2 + B_3A_3 + B_4A_4. \tag{1.17}$$

The result is a number, not another vector or a matrix. In an abstract vector space this operation is called an inner product.

[a] **Question from the audience.** Is this the dot product?

Response. Yes. For ordinary vectors in three-dimensional space it is the familiar dot product. The name *inner product* emphasizes that the same row-by-column operation applies in vector spaces of any finite dimension.

[a]Lecture timestamp: 01:06:44.

1.8 Matrices as Machines for Updating States

A matrix is a square array of numbers that acts on a vector to produce another vector. Think of it as a machine: feed in the old column, apply a fixed rule, and obtain a new column.

In four dimensions,

$$M = \begin{pmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ M_{41} & M_{42} & M_{43} & M_{44} \end{pmatrix}. \tag{1.18}$$

The first index labels the row and the second the column. One may regard M as a collection of columns or as a collection of rows; both viewpoints will be useful.

To calculate $A' = MA$, take the inner product of each row of M with the input column:

$$A'_i = (MA)_i = \sum_{j=1}^{4} M_{ij} A_j. \tag{1.19}$$

For example,

$$A'_1 = M_{11}A_1 + M_{12}A_2 + M_{13}A_3 + M_{14}A_4, \tag{1.20}$$
$$A'_2 = M_{21}A_1 + M_{22}A_2 + M_{23}A_3 + M_{24}A_4. \tag{1.21}$$

The third and fourth entries follow by the same pattern.

1.8.1 Encoding a Classical Configuration

Suppose a system has four possible states. Represent them by the one-hot columns

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad e_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \tag{1.22}$$

The location of the single 1 tells us which classical configuration is present.

Consider the cyclic update

$$1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4 \longrightarrow 1. \tag{1.23}$$

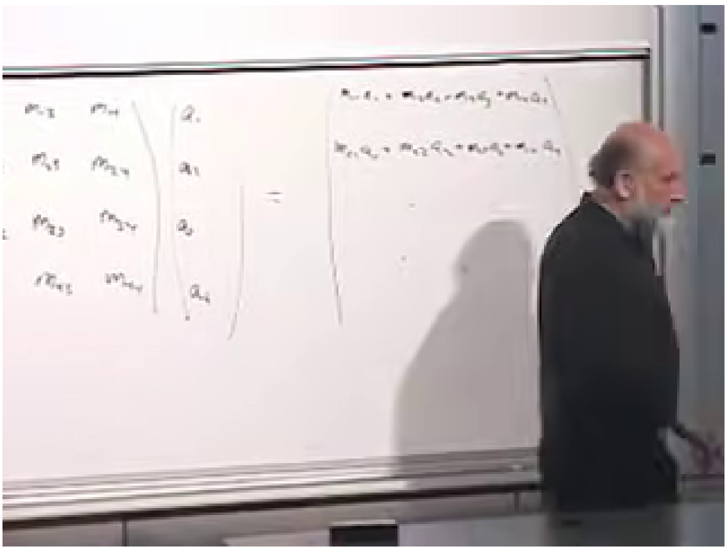

Figure 1.4: A matrix acts on a column vector row by row. The first two output components are expanded explicitly on the board.

Lecture frame: 01:12:45.

It is represented by

$$P = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{1.24}$$

Indeed,

$$\begin{aligned} Pe_1 &= e_2, \quad Pe_2 = e_3, \\ Pe_3 &= e_4, \quad Pe_4 = e_1. \end{aligned} \tag{1.25}$$

The board construction is corrected while it is being written, first with five states and then with four. Equation (1.24) records the settled cyclic rule rather than the abandoned intermediate array.

The same matrix acts on an arbitrary column:

$$P \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} = \begin{pmatrix} D \\ A \\ B \\ C \end{pmatrix}. \tag{1.26}$$

For a classical state only one component is 1, but allowing general entries exposes the underlying linear operation. Classical deterministic updates use special zero-one matrices with one 1 in every row and every column. Quantum mechanics will retain matrix evolution while greatly enlarging the allowed entries.

1.8.2 Composition Is Matrix Multiplication

If one time step is

$$V' = MV, \tag{1.27}$$

then a second application gives

$$V'' = MV' = M(MV) = M^2V. \tag{1.28}$$

Multiplying matrices is therefore not an arbitrary algebraic pastime. It composes physical updates. Five time steps are represented by M^5.

For two 2×2 matrices

$$M = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix}, \qquad N = \begin{pmatrix} N_{11} & N_{12} \\ N_{21} & N_{22} \end{pmatrix}, \tag{1.29}$$

the product $C = MN$ has entries

$$C_{ij} = \sum_k M_{ik} N_{kj}. \tag{1.30}$$

In particular,

$$C_{11} = M_{11}N_{11} + M_{12}N_{21}, \tag{1.31}$$
$$C_{12} = M_{11}N_{12} + M_{12}N_{22}, \tag{1.32}$$
$$C_{21} = M_{21}N_{11} + M_{22}N_{21}, \tag{1.33}$$
$$C_{22} = M_{21}N_{12} + M_{22}N_{22}. \tag{1.34}$$

Each entry is a row of the left matrix multiplied by a column of the right matrix. Equivalently, act with M on each column of N and place the resulting columns side by side.

A row vector can also act from the left. If $B = (B_1, \ldots, B_4)$, then BM is another row vector with

$$(BM)_j = \sum_i B_i M_{ij}. \tag{1.35}$$

Again the rule is row times column. The shapes keep track of the result:

$$\begin{array}{cc} \text{row} \times \text{column} & = \text{number}, \\ \text{matrix} \times \text{column} & = \text{column}, \\ \text{row} \times \text{matrix} & = \text{row}, \\ \text{matrix} \times \text{matrix} & = \text{matrix}. \end{array} \tag{1.36}$$

These operations are worth practicing with small numerical 2×2 and 3×3 examples. Together with a little calculus, row–column multiplication is the main new mathematical habit needed for the quantum mechanics ahead.

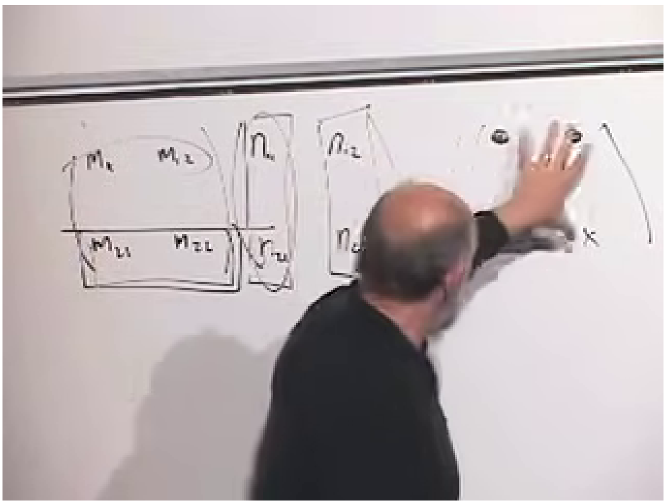

Figure 1.5: Matrix multiplication organized as repeated matrix–column multiplication. The first row and the columns of the second matrix are marked to show where each entry of the product comes from.

Lecture frame: 01:26:30.

[a] **Question from the audience.** What restriction does reversibility place on the update matrix?

Response. It must have an inverse. The inverse matrix implements the reverse update and takes every state back to its unique predecessor. Not every matrix is invertible, so the algebra reproduces the distinction made earlier with arrow diagrams. For the cyclic example, $P^{-1} = P^3$ and $P^4 = \mathbb{1}$.

[a] Lecture timestamp: 01:31:50.

1.9 Why Prepare the Algebra Before the Qubit?

The first hour gave a qualitative account of deterministic classical physics in terms of bits and reversible arrows. The remaining mathematics translates that account into a form that can survive the transition to quantum mechanics. A configuration becomes a column; a one-step law becomes a matrix; repeated time evolution becomes a product of matrices; reversibility becomes the existence of an inverse.

We have not yet used a quantum rule. That delay is deliberate. Next we replace the classical alternatives $|0\rangle$ and $|1\rangle$ by a qubit and ask what new states are possible. The row vectors, column vectors, inner products, and matrices introduced here will then reveal why quantum information differs so sharply from a classical string of bits.

[a] **Question from the audience.** Why introduce all this matrix algebra before saying what it is for?

Response. The qualitative problem came first: how should deterministic physics be represented by states and update laws? The algebra is the preparation for the next step. Once qubits are introduced, these manipulations will no longer look detached; they will be the natural language of quantum states and their evolution.

[a] Lecture timestamp: 01:34:00.

CHAPTER 2

QUBITS, COMPLEX STATE VECTORS, AND OBSERVABLES

Overview. A single electron spin is the first genuinely quantum bit. It may be prepared relative to any spatial direction, yet a fixed detector returns only two outcomes. We let the classical magnetic picture fail, then construct the complex vector space, probability rule, and matrix language needed to describe what the experiment actually does.

2.1 One Spin and Two Meanings of Vector

A classical bit has two possible states and nothing between them. We may call them 0 and 1, heads and tails, or up and down; the names are labels for two distinct points in a classical state space. A quantum bit also yields two alternatives when measured in a fixed way, but it cannot be described by only two state-space points. The simplest example is the spin of an electron.

There are two different notions of vector in this example, and confusing them makes everything that follows unnecessarily mysterious. First there is an ordinary *spatial vector*: an arrow in three-dimensional space with a magnitude and a direction. The electron has a magnetic moment that behaves, for the present purpose, like a tiny magnet. Its magnitude is fixed, while a spatial direction can be assigned to it in a preparation procedure.

The second object is an abstract vector in a complex vector space. This *state vector* will describe the quantum state. It is not simply the magnetic arrow written in another notation. The relation between spatial directions and state vectors is real and important, but it is not the identity of two objects. We shall earn that relation gradually.

2.2 Letting the Classical Picture Fail

Begin deliberately with a classical story. Put a tiny magnet in a strong external magnetic field. Unless it is already aligned, it precesses about the field. As it radiates energy, the precession damps and the magnet settles into its lowest-energy orientation. By choosing the direction of the applied field, waiting for this relaxation, and then removing the field, we seem able to prepare the magnetic moment in any desired direction.

The same apparatus suggests a method of detection. Turn on a field in a chosen direction and observe the radiation produced while the little magnet settles. A classical magnet aligned with the field should emit no energy. One initially opposed to the field should emit the maximum amount, and an intermediate angle should produce an intermediate amount. In that picture the radiated energy varies continuously with the angle.

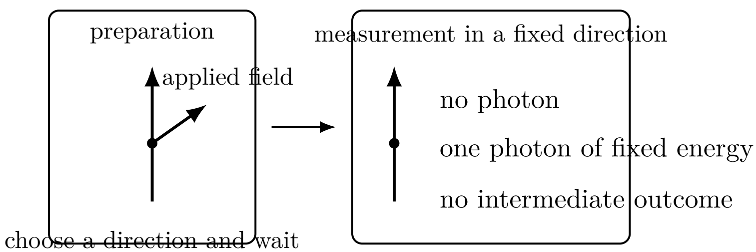

Figure 2.1: The operational puzzle. A continuously variable preparation is followed by a detector with only two possible outputs.

That classical prediction is false for an electron. When the detector field is turned on, only two results occur:

1. no photon is emitted;
2. one photon is emitted with the fixed energy separating the down and up levels.

There is never a photon carrying an arbitrary fraction of that energy. The continuous classical response has collapsed into a binary quantum answer.

The direction used in preparation has not become irrelevant. It controls the *probabilities* of the two answers. Prepare the electron up relative to the detector and the up result occurs with certainty. Prepare it down and the transition occurs with certainty. Prepare it horizontally and a vertical detector returns the two answers equally often:

$$P(\text{up}) = \frac{1}{2}, \qquad P(\text{down}) = \frac{1}{2}. \tag{2.1}$$

Thus a continuously variable family of preparations leads to a two-valued measurement whose statistics vary continuously.

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[a] **Question from the audience.** Could another small object, perhaps a second electron, supply the magnetic field instead of a macroscopic magnet?

Response. It could, but then the two electrons form a joint quantum system and can become entangled. That is precisely where the course is going. We postpone it until the state space of one electron is under control.

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[a] Lecture timestamp: 00:21:23.

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[a] **Question from the audience.** Must the probability distribution be found by repeating the entire preparation and measurement? What happens if we measure the same electron again?

Response. Yes, the original distribution requires repeated, identically prepared trials. After the first measurement, repeating the same measurement on that electron gives the same result; the measurement has prepared it relative to that detector. Waiting for a radiative transition is itself one possible preparation procedure.

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[a] Lecture timestamp: 00:22:17.

[a] **Question from the audience.** Are we imagining a free electron, and is the magnetic field left on continuously?

Response. We isolate the spin degree of freedom and ignore the electron's position, whether it is freely moving or otherwise constrained. A large electromagnet may be switched on quickly for preparation or detection. The simplified experiment is designed to ask only about spin.

[a]Lecture timestamp: 00:23:48.

[a] **Question from the audience.** If the electron is prepared exactly down and the field is reversed, is a photon emitted with certainty, and where does its energy come from?

Response. For exact opposition the transition occurs with probability one. For an oblique preparation the probability is smaller. The energy ultimately comes from the work done in establishing or reversing the external magnetic field; the apparatus is part of the energy accounting.

[a]Lecture timestamp: 00:25:22.

[a] **Question from the audience.** When does the photon appear? Is the transition instantaneous?

Response. The emission time is random, like the decay time of an unstable state. One can specify a characteristic lifetime or half-life, and a stronger field shortens that scale. Energy–time uncertainty enters the finer description, but the present operational facts are the two possible outputs and their measured frequencies.

[a]Lecture timestamp: 00:27:31.

[a] **Question from the audience.** Would a beam containing a million identically prepared electrons simply produce a measurable light intensity?

Response. Yes. The ensemble intensity reveals the average transition probability, while the emission time of any particular electron remains random. An ensemble displays the distribution that a single trial cannot.

[a]Lecture timestamp: 00:29:17.

This is the point of departure. Two isolated classical points can encode the detector answers, but they cannot encode all the preparations that alter the probabilities of those answers. We need a different state space.

2.3 Complex Numbers and the Geometry of State Space

The required state space is linear: its elements can be added and multiplied by scalars. Its scalars are complex numbers. Write

$$z = a + ib, \qquad i^2 = -1, \qquad z^* = a - ib, \tag{2.2}$$

where a and b are real. Complex conjugation produces the positive quantity

$$\begin{aligned} z^* z &= (a - ib)(a + ib) \\ &= a^2 + b^2. \end{aligned} \tag{2.3}$$

This simple cancellation is the algebraic seed of the probability rule.

A complex vector space contains objects $|a\rangle, |b\rangle, \ldots$ for which

$$c\,|a\rangle\,, \qquad |a\rangle + |b\rangle \tag{2.4}$$

are again vectors whenever c is complex. Column vectors give

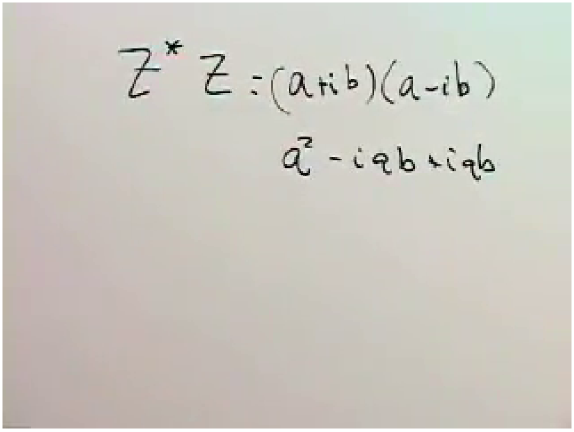

Figure 2.2: Complex conjugation and z^*z are developed on the board before amplitudes are interpreted as probabilities.

Lecture frame: 00:37:10.

a concrete realization:

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}, \qquad c\,|a\rangle = \begin{pmatrix} ca_1 \\ ca_2 \\ \vdots \\ ca_n \end{pmatrix}, \\ |a\rangle + |b\rangle = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{pmatrix}. \tag{2.5}$$

Even the complex numbers themselves form a one-dimensional complex vector space. For an electron spin we shall need two dimensions.

To each column ket we associate a conjugate row bra:

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \quad \longleftrightarrow \quad \langle a| = \begin{pmatrix} a_1^* & a_2^* \end{pmatrix}. \tag{2.6}$$

The inner product is then

$$\langle b \mid a \rangle = b_1^* a_1 + b_2^* a_2, \tag{2.7}$$

and the squared norm is

$$\langle a \mid a \rangle = |a_1|^2 + |a_2|^2. \tag{2.8}$$

It is real and nonnegative because every term has the form $z^* z$.

---

[a] **Question from the audience.** Why is there no explicit star on the bra symbol?

Response. Turning a ket into its bra already means transpose and complex conjugate. The reversal of the bracket is notation for that operation, so writing another star would repeat what is implicit.

---

[a]Lecture timestamp: 00:45:24.

2.4 The Qubit Basis and the Born Rule

Return now to the detector. Let $|+\rangle$ denote its up state and $|-\rangle$ its down state. A general spin state is a linear combination

$$|\psi\rangle = a_+ |+\rangle + a_- |-\rangle \quad \longleftrightarrow \quad \begin{pmatrix} a_+ \\ a_- \end{pmatrix}. \tag{2.9}$$

The plus and minus signs inside the kets are labels, not arithmetic operations. Once this ordered basis is chosen,

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{2.10}$$

Indeed,

$$a_+ \begin{pmatrix} 1 \\ 0 \end{pmatrix} + a_- \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a_+ \\ a_- \end{pmatrix}. \tag{2.11}$$

The coefficients are probability amplitudes. The probability of each detector outcome is the squared magnitude of the corresponding amplitude:

$$P(+) = |a_+|^2, \qquad P(-) = |a_-|^2. \tag{2.12}$$

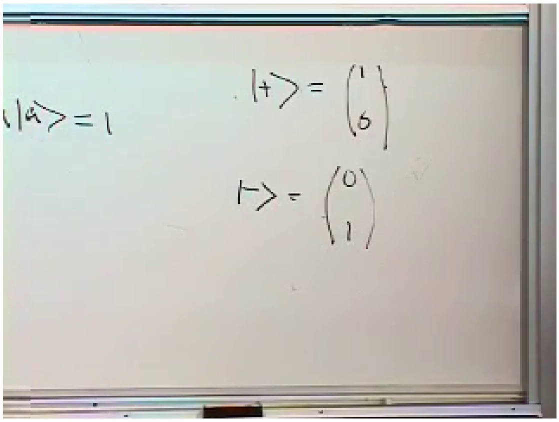

Figure 2.3: The detector states are fixed as the two standard basis columns.

Lecture frame: 00:58:10.

Because these are the only outcomes,

$$|a_+|^2 + |a_-|^2 = 1, \qquad \text{or equivalently} \qquad \langle\psi\,|\,\psi\rangle = 1. \tag{2.13}$$

A physical state vector is therefore normalized.

---

[a] **Question from the audience.** Do the plus and minus signs have an algebraic meaning in the components?

Response. Here they label the two basis states and their coefficients. We could call them 1 and 2, or up and down. In particular, $-\,|+\rangle$ is a valid scalar multiple of $|+\rangle$; it is not the state $|-\rangle$.

---

[a] Lecture timestamp: 00:56:54.

Multiplying a state by a complex number of unit magnitude leaves all its probabilities unchanged:

$$|\psi\rangle \longmapsto e^{i\phi}\,|\psi\rangle\,, \qquad |e^{i\phi}a_\pm|^2 = |a_\pm|^2. \tag{2.14}$$

Thus two vectors differing only by an overall phase represent the same physical state. A *relative* phase is different.

Replacing only a_- by $-a_-$, for example, cannot generally be removed by multiplying the whole vector by one phase; it can change the results of measurements in another basis.

---

[a] **Question from the audience.** Are $|+\rangle$, $-|+\rangle$, and $e^{i\phi}|+\rangle$ different states? What about changing the sign of one term in a superposition?

Response. They are different mathematical vectors but, when the factor multiplies the entire vector and has unit magnitude, they describe the same physical state. A relative sign or phase between the up and down terms is observable in suitable measurements and therefore changes the state.

---

[a]Lecture timestamp: 00:59:54.

The basis is orthonormal:

$$\begin{aligned} \langle + | + \rangle &= 1, & \langle - | - \rangle &= 1, \\ \langle + | - \rangle &= 0, & \langle - | + \rangle &= 0. \end{aligned} \tag{2.15}$$

For example,

$$\langle + | - \rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0. \tag{2.16}$$

An equal-weight state provides the first useful correction at the board. The column

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{2.17}$$

has squared norm 2, not 1, and therefore is not normalized. Dividing by $\sqrt{2}$ repairs it:

$$|\rightarrow\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{|+\rangle + |-\rangle}{\sqrt{2}}. \tag{2.18}$$

It gives $P(+) = P(-) = 1/2$ in the vertical detector and corresponds to one horizontal preparation. The correction from $(1,1)^T$ to $2^{-1/2}(1,1)^T$ is not cosmetic: normalization is

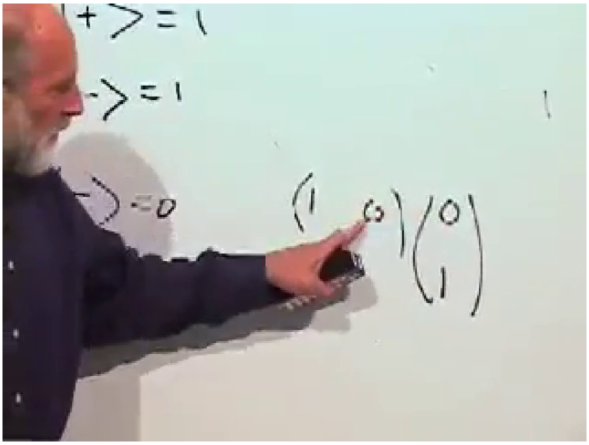

Figure 2.4: The four inner products of the up/down basis are checked directly on the board.

Lecture frame: 01:05:00.

what turns squared amplitudes into a complete probability distribution.

After the break it is worth returning to the two meanings of vector. A spatial arrow has three real components. A qubit state vector has two complex components, hence four real numbers before imposing normalization and identifying overall phase. The objects are related, but one is not merely a relabeling of the other. We do not yet need the full geometric map between them.

---

[a] **Question from the audience.** How do we know the dimension of the state space for a different quantum system?

Response. Experiment tells us how many mutually exclusive outcomes a complete measurement can distinguish. Electron spin has two such outcomes and therefore a two-dimensional complex state space. A spin-one system has three outcomes in the corresponding measurement and requires three components.

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[a] Lecture timestamp: 01:10:56.

2.5 Classical Observables and Ensemble Averages

Having described states, turn to measurable quantities. In a classical system with discrete states n, an observable is a real-valued function f_n on those states. If the state is uncertain and described by probabilities p_n, its ensemble average is

$$\langle f\rangle_{\text{ens}} = \sum_n p_n f_n. \tag{2.19}$$

The notation does not imply that a single measurement must ever equal this number.

For a coin, assign $f_+ = +1$ and $f_- = -1$. Then

$$\langle f\rangle_{\text{ens}} = p_+ - p_-. \tag{2.20}$$

A fair coin has average zero. If $p_+ = 3/4$ and $p_- = 1/4$, the average is $1/2$. Neither 0 nor $1/2$ is an outcome of a single toss; each summarizes many $+1$ and -1 results.

---

[a] **Question from the audience.** How can the expected value be zero or one half when neither value can appear in a single trial?

Response. The phrase expected value is potentially misleading. It means the average of a large ensemble, not the value one should expect to see on the next run. A fair sequence of $+1$ and -1 results averages to zero, and a three-to-one imbalance averages to one half.

---

[a]Lecture timestamp: 01:23:30.

An especially useful observable is the indicator for one state:

$$\chi_k(n) = \begin{cases} 1, & n = k, \\ 0, & n \neq k. \end{cases} \tag{2.21}$$

Its average is

$$\langle \chi_k\rangle_{\text{ens}} = p_k. \tag{2.22}$$

Thus knowledge of the averages of all classical observables includes, in particular, knowledge of the probability distribution.

2.6 Matrices and Quantum Observables

In quantum mechanics observables are represented not by functions on state-space points but by linear operators. In a finite-dimensional state space, those operators are matrices. A matrix M maps one vector to another:

$$M\,|a\rangle = |a'\rangle\,. \tag{2.23}$$

For two components,

$$\begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} m_{11}a_1 + m_{12}a_2 \\ m_{21}a_1 + m_{22}a_2 \end{pmatrix}. \tag{2.24}$$

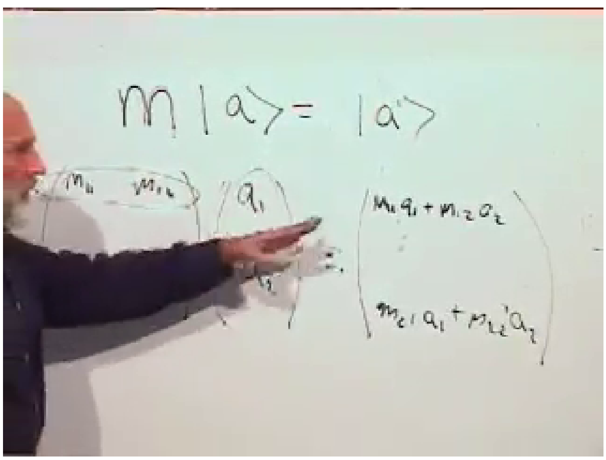

Figure 2.5: Matrix action is expanded row by row into the two components of the transformed vector.

Lecture frame: 01:32:10.

Real two-dimensional vectors make the geometry visible. The matrices

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \tag{2.25}$$

respectively stretch every direction uniformly, stretch only the

y-component, and stretch only the x-component. The matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{2.26}$$

interchanges the components and reflects the plane across $x = y$. Meanwhile,

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} v_y \\ -v_x \end{pmatrix} \tag{2.27}$$

rotates vectors through a clockwise quarter-turn in the convention used at the board. A shear supplies another linear example, although its matrix is left as practice. The essential restriction is linearity: not every imaginable transformation of the plane is represented by a matrix.

Quantum observables belong to a special class.

Definition 2.1. A matrix M is *Hermitian* when

$$M_{ij} = M_{ji}^*. \tag{2.28}$$

The diagonal condition $M_{ii} = M_{ii}^*$ forces every diagonal entry to be real. Off-diagonal entries occur in conjugate pairs. For example,

$$M = \begin{pmatrix} r & 4+i \\ 4-i & s \end{pmatrix}, \qquad r, s \in \mathbb{R}, \tag{2.29}$$

is Hermitian. If all entries happen to be real, Hermitian is the same as symmetric.

Hermitian matrices will represent measurable quantities, including spin along the x, y, and z directions. They should not be confused with the operators that describe a physical kick or time evolution.

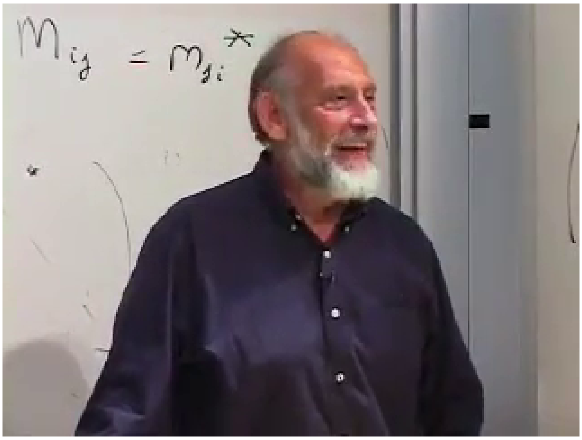

Figure 2.6: The defining conjugation across the diagonal, $M_{ij} = M^*_{ji}$, is isolated at the end of the lecture.

Lecture frame: 01:42:50.

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[a] **Question from the audience.** If a matrix kicks the state into another state, must it preserve normalization? Does a Hermitian matrix do that?

Response. A physical kick must preserve the length of the state vector, but a general Hermitian matrix does not. Length-preserving transformations are unitary. Hermitian operators represent observables; unitary operators represent the reversible transformations that act on states. Their detailed relation comes next.

---

[a]Lecture timestamp: 01:45:49.

We have reached the minimum language needed for the next step. A qubit is a normalized vector in a two-dimensional complex vector space; amplitudes determine probabilities; and observables are Hermitian matrices. With those pieces in place, spin along arbitrary directions and eventually entangled pairs can be discussed without returning to the classical picture that has already failed.

CHAPTER 3

HERMITIAN OBSERVABLES, SPIN, AND THE PROBABILITY RULE

Overview. Quantum mechanics is a calculus for probabilities. To use it without replacing understanding by slogans, we need a compact mathematical chain: complex phases, normalized states, Hermitian observables, eigenvalues and eigenvectors, orthogonality, and the Born rule. The three Pauli matrices then turn that chain into concrete predictions for electron spin.

3.1 Complex Numbers in Polar Form

Begin by removing one small source of algebraic friction. A complex number

$$z = x + iy \tag{3.1}$$

is a point in the complex plane, with x on the real axis and y on the imaginary axis. If the point has polar coordinates (r, θ), then

$$x = r\cos\theta, \qquad y = r\sin\theta, \tag{3.2}$$

and therefore

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}. \tag{3.3}$$

Euler's notation is useful because the angle-addition formulas become the exponential law

$$e^{i\theta}e^{i\phi} = e^{i(\theta+\phi)}. \tag{3.4}$$

Indeed, multiplying the two trigonometric forms gives

$$\begin{aligned}(\cos\theta + i\sin\theta)&(\cos\phi + i\sin\phi)\\ &= \cos(\theta+\phi) + i\sin(\theta+\phi).\end{aligned} \tag{3.5}$$

The familiar sum formulas are all contained in this one multiplication.

Complex conjugation reflects the point across the real axis:

$$z^* = x - iy, \qquad (e^{i\theta})^* = e^{-i\theta}. \tag{3.6}$$

Consequently,

$$e^{i\theta} e^{-i\theta} = 1. \tag{3.7}$$

Numbers of the form $e^{i\theta}$ lie on the unit circle and are called *pure phases*. Every nonzero complex number is a positive modulus times a pure phase:

$$z = |z|e^{i\theta}, \qquad |z|^2 = z^* z = x^2 + y^2. \tag{3.8}$$

---

[a] **Question from the audience.** Does changing θ to $-\theta$ correspond to reversing the direction of rotation in the complex plane?

Response. Yes. Positive and negative phase angles trace the unit circle in opposite senses. Complex conjugation changes $e^{i\theta}$ into $e^{-i\theta}$, which is the reflected point and the inverse phase.

---

[a]Lecture timestamp: 00:06:36.

That is the whole complex-number runway needed today: modulus, phase, conjugation, and the unit circle.

3.2 Quantum Mechanics as a Probability Calculus

We now turn to the postulates. A physical system is prepared in a *state*; a measurement asks for the value of an *observable*; quantum mechanics supplies probabilities for the possible answers.

States are vectors in a complex vector space. For a two-level spin system,

$$|\psi\rangle = \alpha\,|u\rangle + \beta\,|d\rangle\,, \qquad |u\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |d\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.9}$$

The coefficients determine the up/down probabilities,

$$P(u) = |\alpha|^2, \qquad P(d) = |\beta|^2, \tag{3.10}$$

so the state must be normalized:

$$\langle \psi \,|\, \psi \rangle = |\alpha|^2 + |\beta|^2 = 1. \tag{3.11}$$

The dimension refers to the number of independent alternatives in the state space, not to the dimension of ordinary space. A system with three mutually exclusive basis states would use three orthonormal columns and three amplitudes.

Probabilities do not determine the amplitudes completely. Multiplication by a pure phase leaves a magnitude unchanged:

$$|e^{i\theta}\alpha|^2 = |\alpha|^2. \tag{3.12}$$

The amplitudes therefore contain information beyond the probabilities for this one basis. Its role will become visible when different measurements and, later, time evolution are considered.

The basis states are orthonormal:

$$\langle u \,|\, u \rangle = \langle d \,|\, d \rangle = 1, \qquad \langle u \,|\, d \rangle = \langle d \,|\, u \rangle = 0. \tag{3.13}$$

Their orthogonality belongs to the complex space of states. It must not yet be confused with two arrows at right angles in ordinary three-dimensional space; that distinction will become decisive near the end of the lecture.

3.3 Why Observables Are Hermitian

Experimental results can be recorded as real numbers. We may package two real numbers into one complex number for convenience, but the pointer readings themselves can always be separated into real quantities. In finite dimensions, quantum observables are represented by a special class of linear operators: Hermitian matrices.

To define them, first transpose a matrix by interchanging rows and columns:

$$(M^T)_{ij} = M_{ji}. \tag{3.14}$$

A real matrix satisfying $M^T = M$ is symmetric. For complex matrices the corresponding operation is the Hermitian conjugate,

$$M^\dagger = (M^T)^*, \qquad (M^\dagger)_{ij} = M^*_{ji}. \tag{3.15}$$

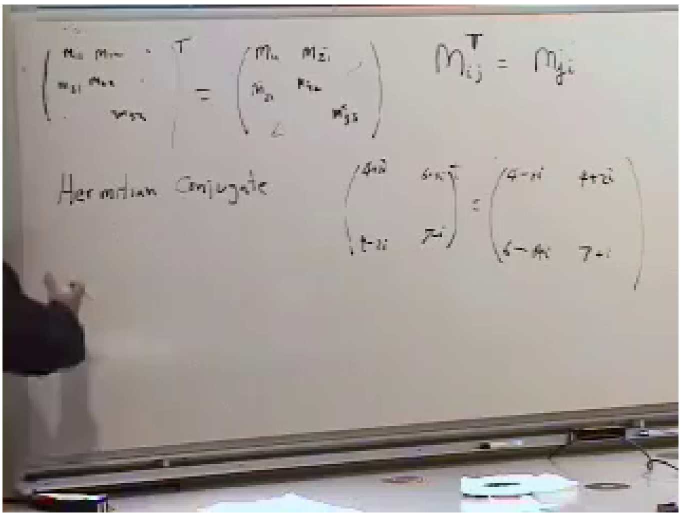

Figure 3.1: Transpose, symmetry, and the transition to Hermitian conjugation share one blackboard layout.

Lecture frame: 00:23:06.

Definition 3.1. A matrix M is *Hermitian* if

$$M^\dagger = M, \qquad \text{equivalently} \qquad M_{ij} = M^*_{ji}. \tag{3.16}$$

The diagonal entries are real, since $M_{ii} = M^*_{ii}$. Reflected off-diagonal entries are conjugate pairs. Thus the general two-by-two form is

$$M = \begin{pmatrix} a & z \\ z^* & b \end{pmatrix}, \qquad a, b \in \mathbb{R}. \tag{3.17}$$

Hermiticity is the matrix analogue of a reality condition; it does not require every entry separately to be real.

The first payoff follows from two elementary conjugation identities:

$$\langle B \mid A \rangle = \langle A \mid B \rangle^* , \tag{3.18}$$

and, for any M,

$$\langle B | M | A \rangle = \left(\langle A | M^\dagger | B \rangle \right)^* . \tag{3.19}$$

The expression $\langle B | M | A \rangle$ is a number, the BA matrix element of M. If M is Hermitian and $B = A$, then

$$\langle A | M | A \rangle = \left(\langle A | M | A \rangle \right)^* . \tag{3.20}$$

It is therefore real.

Proposition 3.2. *For every Hermitian M, the expectation value*

$$\langle M \rangle_A = \langle A | M | A \rangle \tag{3.21}$$

is real in every normalized state $|A\rangle$.

The probabilistic meaning of expectation value will be developed further, but its reality already explains why Hermitian operators are natural candidates for observables.

---

[a] **Question from the audience.** Does the angle-bracket notation for an average come from the bra–ket notation?

Response. Yes. Dirac's bracket notation supplies the same brackets. In $\langle A | M | A \rangle$, the operator is sandwiched between the bra and ket representing the state, and the result is the expectation value.

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[a]Lecture timestamp: 00:33:22.

[a] **Question from the audience.** Can a physical quantity itself have an imaginary measured value? If measurements are real, why are complex numbers needed in quantum mechanics?

Response. A set of measured results can always be recorded as real numbers, even if we choose to package a pair as one complex number. The sharper question is why the theory is driven to complex amplitudes. Hold that question: the answer is tied to reversible time evolution, which has not yet been introduced. Restricting the theory to real vectors would lose important possibilities.

[a] Lecture timestamp: 00:34:31.

The mathematics is not an ornament. Without it, entanglement can be made to sound mysterious while remaining undefined. We continue in the simplest possible setting, one two-level system, so that two such systems can later be combined honestly.

3.4 Eigenvalues Are Possible Measurement Results

Definition 3.3. A nonzero vector $|a\rangle$ is an eigenvector of M with eigenvalue λ_a if

$$M\,|a\rangle = \lambda_a\,|a\rangle\,. \tag{3.22}$$

The matrix may change the vector's length or phase, but it does not turn it into a different direction in state space.

For a Hermitian matrix the eigenvalue is real. Multiplying Equation (3.22) on the left by $\langle a|$ gives

$$\langle a|\,M\,|a\rangle = \lambda_a\,\langle a\,|\,a\rangle\,. \tag{3.23}$$

The left side is real by Hermiticity, and the norm on the right is real and positive. Hence

$$\lambda_a = \frac{\langle a|\,M\,|a\rangle}{\langle a\,|\,a\rangle} \in \mathbb{R}. \tag{3.24}$$

Now comes the physical statement: when M represents an observable, its eigenvalues are the values that a measurement of M can return. The eigenvector belonging to λ_a is a state in which that value occurs with certainty.

The simplest example is a diagonal matrix:

$$M = \begin{pmatrix} m_{11} & 0 \\ 0 & m_{22} \end{pmatrix}. \tag{3.25}$$

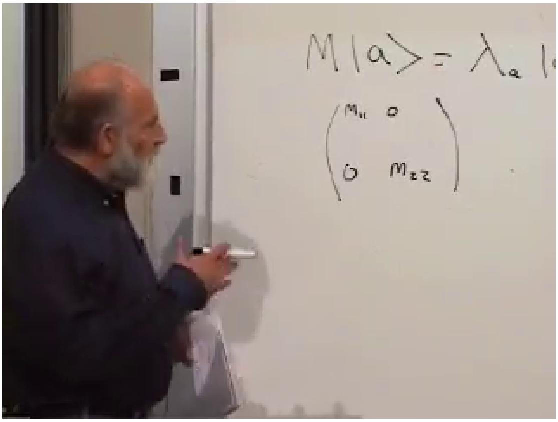

Figure 3.2: The abstract eigenvalue equation is placed above the first diagonal example.

Lecture frame: 00:44:04.

Direct multiplication gives

$$M \begin{pmatrix} 1 \\ 0 \end{pmatrix} = m_{11} \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$
$$M \begin{pmatrix} 0 \\ 1 \end{pmatrix} = m_{22} \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.26}$$

For a larger diagonal matrix, the same rule holds: its diagonal entries are eigenvalues and its standard one-hot columns are eigenvectors.

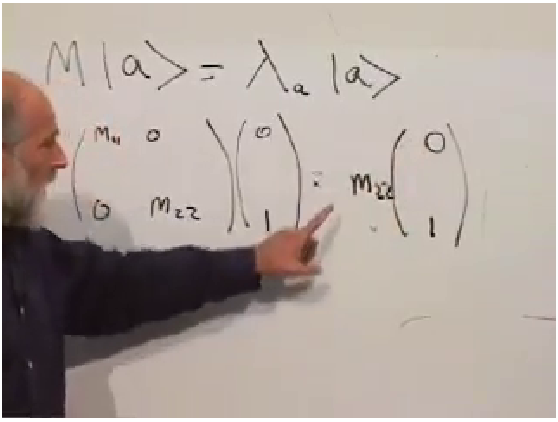

Figure 3.3: The lower basis vector returns multiplied by the lower diagonal entry.

Lecture frame: 00:45:58.

3.5 The Three Spin Matrices

For electron spin along the third axis, choose

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{3.27}$$

Its eigenstates are $|u\rangle = (1,0)^T$ with eigenvalue $+1$ and $|d\rangle = (0,1)^T$ with eigenvalue -1. Thus the two possible detector values are the two eigenvalues.

The first-axis spin matrix is

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{3.28}$$

Its candidate eigenvectors are first written without normalization, since multiplying an eigenvector by any nonzero scalar leaves it an eigenvector. Physical states, however, must be normalized:

$$|x+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \qquad \sigma_1\,|x+\rangle = +\,|x+\rangle\,,$$
$$|x-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}, \qquad \sigma_1\,|x-\rangle = -\,|x-\rangle\,. \tag{3.29}$$

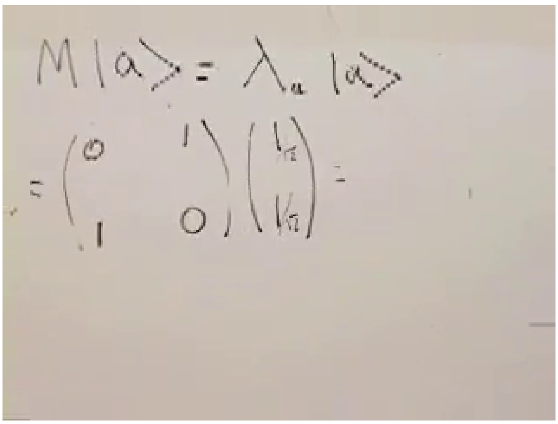

Figure 3.4: The off-diagonal matrix and equal-weight candidate appear before the product is completed.

Lecture frame: 00:50:29.

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[a] **Question from the audience.** Can the σ_1 eigenvectors be seen geometrically rather than found by algebra?

Response. In the real two-component picture, σ_1 interchanges the basis directions E_1 and E_2. The diagonal direction $(1,1)^T$ is fixed, while the other diagonal $(1,-1)^T$ reverses. They are therefore eigenvectors with eigenvalues $+1$ and -1. This is a picture in component space, not yet a statement about the physical spin pointer.

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[a] Lecture timestamp: 00:56:45.

The remaining matrix is

$$\sigma_2 = \begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}, \tag{3.30}$$

with normalized eigenvectors

$$|y+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}, \qquad |y-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}. \tag{3.31}$$

Their eigenvalues are again $+1$ and -1.

All three matrices square to the identity:

$$\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = I. \tag{3.32}$$

If $\sigma_i |a\rangle = \lambda |a\rangle$, then applying σ_i twice gives $\lambda^2 = 1$, so $\lambda = \pm 1$. The identical spectra are the first sign that these matrices represent spin components along equivalent spatial axes.

3.6 Distinct Outcomes Require Orthogonal States

Before proving the next theorem, state its meaning. If one observable gives two different definite answers in two states, then a measurement of that observable can distinguish the states perfectly. Quantum mechanics expresses this perfect distinguishability as orthogonality.

Theorem 3.4. *Let M be Hermitian, with*

$$M |A\rangle = \lambda_A |A\rangle, \qquad M |B\rangle = \lambda_B |B\rangle. \tag{3.33}$$

If $\lambda_A \neq \lambda_B$, then

$$\langle B \mid A\rangle = 0. \tag{3.34}$$

Proof. Multiplying the first eigenvalue equation by $\langle B|$ gives

$$\langle B| M |A\rangle = \lambda_A \langle B \mid A\rangle. \tag{3.35}$$

Hermiticity and the second eigenvalue equation give the same matrix element as

$$\langle B| M |A\rangle = \lambda_B \langle B \mid A\rangle, \tag{3.36}$$

because λ_B is real. Subtraction yields

$$(\lambda_A - \lambda_B) \langle B \,|\, A\rangle = 0. \tag{3.37}$$

The eigenvalues are different, so the inner product must vanish. □

The σ_1 eigenvectors provide an immediate check:

$$\langle x+ \,|\, x-\rangle = \frac{1}{2}\begin{pmatrix} 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} - \frac{1}{2} = 0. \tag{3.38}$$

Conversely, for two orthogonal states there exists a measurement that distinguishes them. Orthogonality is therefore not merely a geometric word; it has an operational meaning.

3.7 The Probability Postulate

Suppose the system is prepared in a normalized state $|B\rangle$. Let M be an observable, and let $|A\rangle$ be a normalized eigenvector of M with eigenvalue λ_A. Then the probability that measuring M returns λ_A is

$$P(\lambda_A \mid B) = |\langle A \,|\, B\rangle|^2 = \langle A \,|\, B\rangle \langle B \,|\, A\rangle\,. \tag{3.39}$$

The product is real and nonnegative. For normalized vectors it is at most one.

[a] **Question from the audience.** Is this the cosine squared of the angle between the two states?

Response. For real unit vectors in ordinary Euclidean space, the inner product is the cosine of an angle. Here the inner product is generally complex, so there is no ordinary spatial angle whose cosine it is. The useful analogue is the magnitude of the overlap, and probability is its absolute square.

[a] Lecture timestamp: 01:19:19.

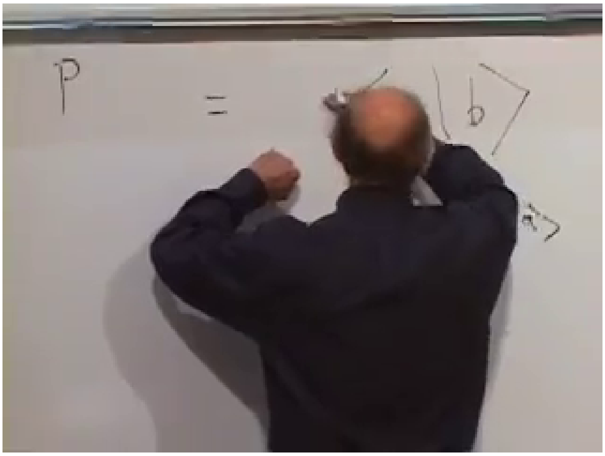

Figure 3.5: The probability rule is written as an overlap multiplied by its conjugate.

Lecture frame: 01:16:40.

[a] **Question from the audience.** What if the prepared state is another eigenvector of M with a different eigenvalue?

Response. Then the two eigenvectors are orthogonal. Their overlap is zero, so the probability of obtaining λ_A is zero. Preparing a state with one definite value excludes a different definite value of the same observable.

[a]Lecture timestamp: 01:20:18.

[a] **Question from the audience.** Can the same rule handle a preparation in which no outcome is certain?

Response. Yes. That is the generic case: $|B\rangle$ need not be an eigenstate of M. Its overlaps with the normalized eigenvectors give the probabilities of the corresponding outcomes. We are beginning with a two-outcome example before treating more elaborate distributions.

[a]Lecture timestamp: 01:21:23.

The lecture trains the rule on several preparations rather than hiding it inside one formula.

First prepare $\sigma_3 = +1$:

$$|B\rangle = |u\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{3.40}$$

Measure σ_1. For its $+1$ eigenvector,

$$\langle x+ \,|\, u\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}, \tag{3.41}$$

so

$$P(\sigma_1 = +1 \mid \sigma_3 = +1) = \frac{1}{2}. \tag{3.42}$$

Replacing $|x+\rangle$ by $|x-\rangle$ gives the same probability. Reversing preparation and measurement also gives one half because the absolute square of the overlap is symmetric.

For a genuinely complex example, prepare $|x+\rangle$ and measure σ_2. Then

$$\langle y+ \,|\, x+\rangle = \frac{1}{2} \begin{pmatrix} 1 & -i \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1-i}{2},$$
$$P(\sigma_2 = +1 \mid \sigma_1 = +1) = \left| \frac{1-i}{2} \right|^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}. \tag{3.43}$$

Preparing spin along one axis and measuring along a perpendicular axis therefore gives equal probabilities for the two outcomes.

3.8 Pointers, States, and Arbitrary Axes

The preceding sentence can be misunderstood unless the vocabulary is repaired. A magnetic moment has a spatial direction, which we shall call a *pointer*. A ket is a vector in the complex state space. Two spatial pointers may be perpendicular, while two state vectors may be orthogonal. These are different statements in different spaces.

[a] **Question from the audience.** If two physical spin directions are ninety degrees apart, should their state vectors be orthogonal?

Response. No. Spin-up states along perpendicular spatial axes have overlap-squared 1/2, not zero. The state orthogonal to spin-up along one axis is spin-down along that same axis. We reserve *perpendicular* for pointers in ordinary space and *orthogonal* for perfectly distinguishable states.

[a] Lecture timestamp: 01:29:25.

[a] **Question from the audience.** Could we use *independent* instead of *orthogonal*?

Response. No. Linear independence only says that no vector is a scalar multiple of the others; it does not require zero inner products. Orthogonality is a stronger condition and here carries the physical meaning of perfect distinguishability.

[a] Lecture timestamp: 01:31:25.

Let $\hat{n} = (n_1, n_2, n_3)$ be a unit pointer in ordinary space:

$$n_1^2 + n_2^2 + n_3^2 = 1. \tag{3.44}$$

The operator for spin along that direction is the matrix-valued analogue of a dot product:

$$\boldsymbol{\sigma} \cdot \hat{n} = n_1\sigma_1 + n_2\sigma_2 + n_3\sigma_3 = \begin{pmatrix} n_3 & n_1 - in_2 \\ n_1 + in_2 & -n_3 \end{pmatrix}. \tag{3.45}$$

It is Hermitian: the diagonal entries are real and the off-diagonal entries are conjugates.

To find its possible values, square it before inserting matrix entries. Matrix order matters, so the cross terms occur as anticommutators:

$$\begin{aligned}(\boldsymbol{\sigma} \cdot \hat{n})^2 = n_1^2\sigma_1^2 + n_2^2\sigma_2^2 + n_3^2\sigma_3^2 \\ + n_1n_2(\sigma_1\sigma_2 + \sigma_2\sigma_1) \\ + n_2n_3(\sigma_2\sigma_3 + \sigma_3\sigma_2) \\ + n_1n_3(\sigma_1\sigma_3 + \sigma_3\sigma_1).\end{aligned} \tag{3.46}$$

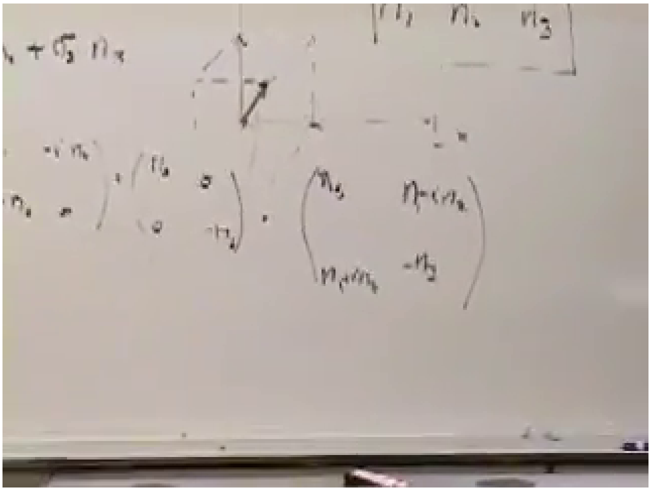

Figure 3.6: The spatial unit pointer, the operator sum, and its explicit Hermitian matrix are developed together.

Lecture frame: 01:38:00.

Different Pauli matrices anticommute:

$$\sigma_i\sigma_j + \sigma_j\sigma_i = 0 \qquad (i \neq j). \tag{3.47}$$

The lecture checks one case explicitly:

$$\sigma_1\sigma_3 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_3\sigma_1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -\sigma_1\sigma_3. \tag{3.48}$$

The cross terms in Equation (3.46) vanish, while $\sigma_i^2 = I$. Hence

$$(\boldsymbol{\sigma} \cdot \hat{n})^2 = (n_1^2 + n_2^2 + n_3^2)I = I. \tag{3.49}$$

Every eigenvalue therefore satisfies $\lambda^2 = 1$:

$$\lambda = \pm 1. \tag{3.50}$$

Whichever spatial axis the apparatus selects, the detector still has only the two outcomes $+1$ and -1. The relation between nearby directions appears not as an intermediate measured value, but in the probabilities of those two fixed answers.

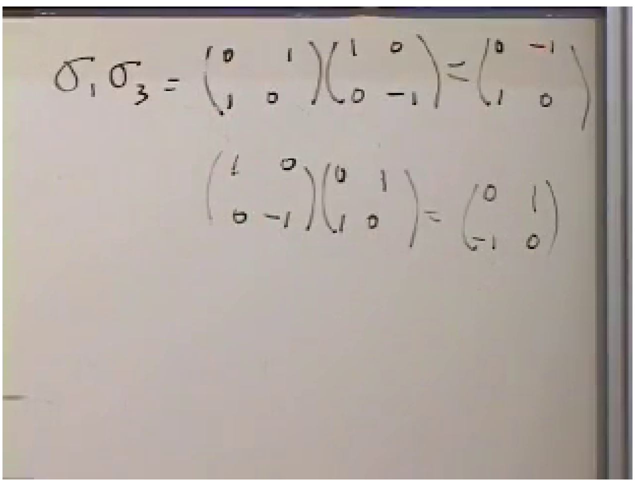

Figure 3.7: Multiplying σ_1 and σ_3 in opposite orders produces matrices that differ by a minus sign.

Lecture frame: 01:44:50.

CHAPTER 4

SPIN GEOMETRY AND THE FIRST ENTANGLED STATE

Overview. Entanglement belongs to systems with parts, but the two-spin problem cannot be read honestly until the one-spin machinery is complete. We therefore begin with spin along an arbitrary axis, derive the transition probability between two directions, interpret measurement as a filter, and identify commutation as the criterion for compatible observables. A counting argument then reveals that two spins possess more pure states than two separately prepared spins can supply. The excess states are entangled, and the simplest example already has definite pair correlations while each electron separately is completely unpolarized.

4.1 Finishing the One-Spin Problem

Bell's theorem can wait. Entanglement itself begins with two electrons, but the remaining structure of one electron will enter every calculation. Let us therefore finish the two-state system before combining two copies of it.

Choose an ordinary spatial unit vector

$$\hat{n} = (n_1, n_2, n_3), \qquad n_1^2 + n_2^2 + n_3^2 = 1. \tag{4.1}$$

The labels $1, 2, 3$ may equally be called x, y, z. What matters is that the axes form one consistent right-handed coordinate system.

[a] **Question from the audience.** Is the coordinate triad drawn on the board right-handed?

Response. The first sketch is left-handed. Reversing the second axis, or relabeling two axes consistently, repairs it. The calculation below uses only a fixed right-handed convention; the accidental orientation of the sketch carries no physics.

[a] Lecture timestamp: 00:02:33.

The spin has three observable components, represented by the Pauli matrices

$$\begin{aligned} \sigma_1 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \\ \sigma_3 &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \end{aligned} \tag{4.2}$$

The component along $\hat{n}$ is the ordinary spatial dot product of the pointer with this matrix-valued vector:

$$\sigma_{\hat{n}} \equiv \boldsymbol{\sigma} \cdot \hat{n} = n_1\sigma_1 + n_2\sigma_2 + n_3\sigma_3. \tag{4.3}$$

This dot product is in physical three-space. It is not the inner product of two state vectors.

Substituting the Pauli matrices gives

$$\sigma_{\hat{n}} = \begin{pmatrix} n_3 & n_1 - in_2 \\ n_1 + in_2 & -n_3 \end{pmatrix}. \tag{4.4}$$

It is convenient to abbreviate

$$n_- = n_1 - in_2, \qquad n_+ = n_1 + in_2 = n_-^*, \tag{4.5}$$

so that

$$\sigma_{\hat{n}} = \begin{pmatrix} n_3 & n_- \\ n_+ & -n_3 \end{pmatrix}. \tag{4.6}$$

The real diagonal and conjugate off-diagonal entries make the operator Hermitian.

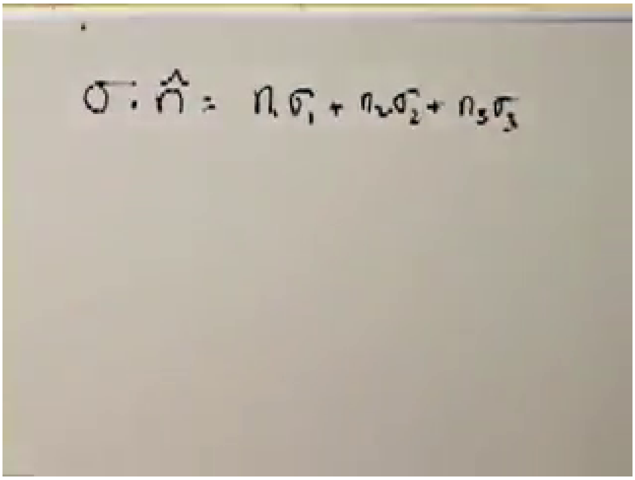

Figure 4.1: The spatial component along $\hat{n}$ becomes a linear combination of three matrix observables.

Lecture frame: 00:05:09.

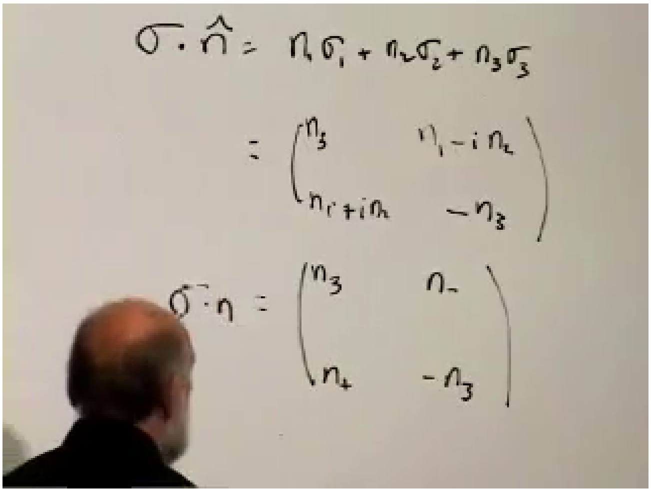

Figure 4.2: The arbitrary-axis spin observable in explicit two-by-two form.

Lecture frame: 00:07:34.

Operationally, measuring $\sigma_{\hat{n}}$ means orienting a strong magnetic field along $\hat{n}$ and observing the spin-dependent response, represented in the lecture by whether a photon is emitted. The matrix and the laboratory question are two descriptions of the same measurement.

Two small identities will carry the rest of the calculation. First,

$$\sigma_{\hat{n}}^2 = I. \tag{4.7}$$

This follows either by multiplying Equation (4.4) by itself or by using

$$\sigma_i^2 = I, \qquad \sigma_i \sigma_j = -\sigma_j \sigma_i \quad (i \neq j). \tag{4.8}$$

Every eigenvalue λ of $\sigma_{\hat{n}}$ therefore satisfies $\lambda^2 = 1$, so the only possible outcomes are

$$\lambda = \pm 1. \tag{4.9}$$

Second,

$$n_+ n_- = n_1^2 + n_2^2 = 1 - n_3^2 = (1 - n_3)(1 + n_3). \tag{4.10}$$

This is not a deep theorem, but it is the identity that keeps the eigenvector algebra under control.

4.2 From Preparation to Probability

Prepare the electron with spin $+1$ along $\hat{n}$, then measure along a different direction $\hat{m}$. The central one-spin question is

$$P(m, + \mid n, +) = P(\sigma_{\hat{m}} = +1 \mid \sigma_{\hat{n}} = +1). \tag{4.11}$$

Symmetry suggests that the answer should depend only on the angle between $\hat{n}$ and $\hat{m}$, and that interchanging the two directions should not change it. Those expectations are useful checks, not substitutes for the probability postulate.

[a] **Question from the audience.** Does quantum mechanics itself tell us that the three Pauli matrices are spin operators, or is that learned from experiment?

Response. The identification is part of the accumulated empirical structure of quantum mechanics, not the result of one isolated experiment. Repeated spin experiments reveal probabilistic regularities, and the matrix formalism was built through the work of Einstein, Born, Heisenberg, Dirac, and others to encode those regularities. The postulates are exceptionally successful guesses constrained by data.

[a] Lecture timestamp: 00:14:03.

Let

$$\sigma_{\hat{n}} \, |n,+\rangle = |n,+\rangle \,, \qquad \sigma_{\hat{m}} \, |m,+\rangle = |m,+\rangle \,. \tag{4.12}$$

The symbols $n,+$ and $m,+$ are labels inside the kets. They identify the eigenstates whose eigenvalue is $+1$.

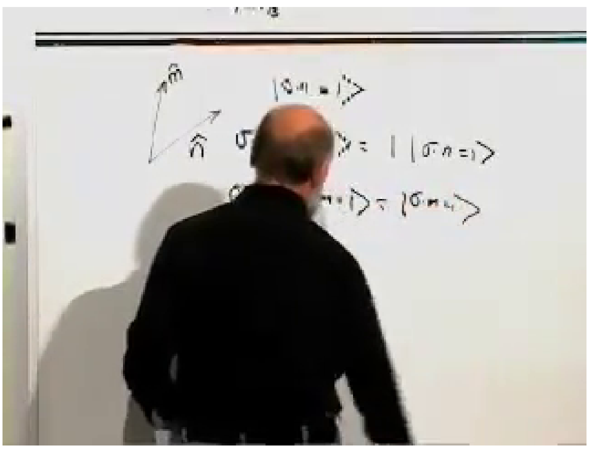

Figure 4.3: The $+1$ eigenstates for spin along $\hat{n}$ and $\hat{m}$.
Lecture frame: 00:17:36.

The first object to calculate is the probability amplitude

$$\mathcal{A}(m,+ \mid n,+) = \langle m,+ \,|\, n,+\rangle \,. \tag{4.13}$$

The probability is its absolute square:

$$P(m,+\mid n,+) = |\langle m,+\,|\,n,+\rangle|^2\,. \tag{4.14}$$

Amplitude first, probability second: that order is the essential quantum step.

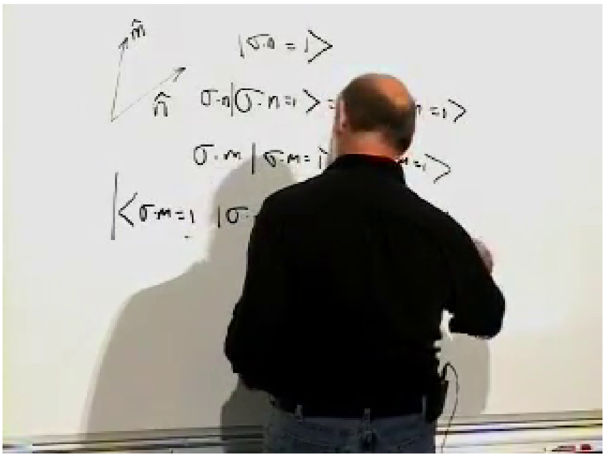

Figure 4.4: The overlap between the prepared n-state and the tested m-state is the probability amplitude.

Lecture frame: 00:19:59.

4.2.1 Solving for the Eigenstate

Write a candidate $+1$ eigenvector of $\sigma_{\hat{n}}$ as

$$\begin{pmatrix}\alpha\\ \beta\end{pmatrix}. \tag{4.15}$$

Normalization can wait, and an overall phase has no physical effect. Away from the south-pole coordinate patch, we may therefore choose $\alpha = 1$ and write $\beta = z$. The top component of

$$\begin{pmatrix} n_3 & n_- \\ n_+ & -n_3\end{pmatrix}\begin{pmatrix}1\\ z\end{pmatrix} = \begin{pmatrix}1\\ z\end{pmatrix} \tag{4.16}$$

gives

$$n_3 + n_- z = 1, \qquad z = \frac{1-n_3}{n_-}. \tag{4.17}$$

The lower component gives the same condition once Equation (4.10) is used.

The unnormalized vector has squared norm

$$\begin{aligned}\begin{pmatrix}1 & \dfrac{1-n_3}{n_+}\end{pmatrix}\begin{pmatrix}1\\ \dfrac{1-n_3}{n_-}\end{pmatrix} &= 1+\frac{(1-n_3)^2}{n_+n_-}\\ &= 1+\frac{1-n_3}{1+n_3} = \frac{2}{1+n_3}.\end{aligned} \tag{4.18}$$

Hence a normalized choice is

$$|n,+\rangle = \sqrt{\frac{1+n_3}{2}}\begin{pmatrix}1\\ \dfrac{1-n_3}{n_-}\end{pmatrix}. \tag{4.19}$$

At $n_3 = -1$, this coordinate expression is singular although the state is not; the eigenstate is simply $|d\rangle = (0,1)^T$. The singularity belongs to the phase convention, not to the spin.

Replacing n by m and conjugating gives

$$\langle m,+| = \sqrt{\frac{1+m_3}{2}}\begin{pmatrix}1 & \dfrac{1-m_3}{m_+}\end{pmatrix}. \tag{4.20}$$

The overlap looks unpleasant, and multiplying it by its conjugate looks worse. Repeated use of Equation (4.10) reduces the result to

$$\begin{aligned}P(m,+\mid n,+) &= \frac{1+n_1m_1+n_2m_2+n_3m_3}{2}\\ &= \frac{1+\hat{n}\cdot\hat{m}}{2}.\end{aligned} \tag{4.21}$$

If θ_{nm} is the angle between the two spatial directions,

$$P(m,+\mid n,+) = \frac{1+\cos\theta_{nm}}{2} = \cos^2\frac{\theta_{nm}}{2}. \tag{4.22}$$

[a] **Question from the audience.** Must the overlap vanish because the two states are orthogonal?

Response. No. The labels $m, +$ and $n, +$ refer to different measurement axes, not to different eigenvalues of one fixed observable. If $m = n$, the two states coincide and their overlap is one. They become orthogonal only when the physical directions are opposite, so that one state is the $+1$ state and the other is the -1 state of the same spin component.

[a]Lecture timestamp: 00:32:47.

4.3 What the Probability Formula Means

Equation (4.21) passes three immediate tests:

$$\begin{aligned} \hat{m} = \hat{n} \quad &\Longrightarrow \quad P = 1, \\ \hat{m} = -\hat{n} \quad &\Longrightarrow \quad P = 0, \\ \hat{m} \cdot \hat{n} = 0 \quad &\Longrightarrow \quad P = \frac{1}{2}. \end{aligned} \tag{4.23}$$

The answer depends only on the relative angle and is symmetric under $m \leftrightarrow n$. Rotating the entire apparatus changes no prediction.

This is the remarkable compromise made by spin one-half. Along every axis the observed result remains $+1$ or -1, never a fractional spin value. Smooth rotational dependence appears instead in the probabilities of those two discrete answers.

[a] **Question from the audience.** How do we know that repeated trials concern the same electron, and how is a probability measured?

Response. One may confine one electron in a cavity and monitor the conserved charge through its electric flux, reprepare that electron along $\hat{n}$, and repeat the measurement along $\hat{m}$. Equivalently, one may prepare a large ensemble of electrons identically and measure each once. The long-run ratio of plus and minus outcomes is the same in either procedure, within statistical error.

[a]Lecture timestamp: 00:39:20.

[a] **Question from the audience.** Must all members of the ensemble be measured in one collective experiment?

Response. No. Each member is prepared by the same protocol and measured separately. A million such trials form one statistical experiment only in the sense that their outcome frequencies estimate the same probability.

[a]Lecture timestamp: 00:41:29.

[a] **Question from the audience.** If one electron is reused, how many photons can it emit?

Response. There is no fixed lifetime quota. Each cycle includes external work when the magnetic field is switched on. If the spin is in the higher-energy orientation, that supplied energy can leave as a photon during relaxation. The apparatus can therefore repeat the preparation-and-emission cycle indefinitely in principle.

[a]Lecture timestamp: 00:42:23.

The formula can be obtained very quickly after rotational invariance is known: choose $\hat{n}$ as the third axis, so $|n,+\rangle = (1,0)^T$, and read the probability from the upper component of $|m,+\rangle$. But using that shortcut at the outset would assume the symmetry that the explicit calculation has just demonstrated.

[a] **Question from the audience.** Could a better choice of basis remove the ugly arbitrary-axis eigenvectors?

Response. Yes, once rotations and unitary transformations are available, one can align a coordinate axis with one of the directions and make the calculation short. At this stage the explicit calculation serves a purpose: it verifies that the chosen Pauli matrices really produce a rotationally invariant answer rather than assuming that fact.

[a]Lecture timestamp: 00:43:33.

[a] **Question from the audience.** What is the physical reason that the answer depends only on the angle?

Response. The structural answer belongs to the quantum theory of symmetry: spatial rotations are represented by unitary transformations acting on the spin states and observables. That language comes later. For now, the angle dependence is an earned result of the Pauli-matrix calculation.

[a]Lecture timestamp: 00:48:30.

4.4 Measurement Is a Filter

A measurement does more than report a number. If $\sigma_{\hat{m}}$ is measured and the result is $+1$, the electron is left in $|m,+\rangle$. The apparatus has acted as both detector and state preparer, much as a polarizer tests a photon and leaves the transmitted photon polarized along its own axis.

This state-change rule makes the problem of simultaneous measurement concrete. Two observables can have reproducible definite values together only when there are states that are eigenvectors of both.

[a] **Question from the audience.** Can one measurement determine two different spin components, such as σ_1 and σ_2, simultaneously?

Response. No. Their eigenvectors are different. A measurement of one component leaves the electron in an eigenstate of that component, but that state is not an eigenstate of the other. The first measurement therefore destroys the definiteness required by the second.

[a]Lecture timestamp: 00:51:29.

The general finite-dimensional statement is most cleanly made for Hermitian operators.

Theorem 4.1. *Two Hermitian operators A and B are simultaneously diagonalizable if and only if they commute:*

$$[A, B] \equiv AB - BA = 0. \tag{4.24}$$

Equivalently, they admit a complete common orthonormal eigenbasis.

For the easy direction, suppose

$$A\,|\alpha_i\rangle = \lambda_i\,|\alpha_i\rangle\,, \qquad B\,|\alpha_i\rangle = \mu_i\,|\alpha_i\rangle \tag{4.25}$$

for a complete basis. Then

$$\begin{aligned} AB\,|\alpha_i\rangle &= \lambda_i\mu_i\,|\alpha_i\rangle\,, \\ BA\,|\alpha_i\rangle &= \mu_i\lambda_i\,|\alpha_i\rangle\,, \end{aligned} \tag{4.26}$$

so AB and BA agree on every basis vector. Hence $AB = BA$. Conversely, if Hermitian A and B commute, B preserves each eigenspace of A; diagonalizing B within any degenerate eigenspace produces a common eigenbasis.

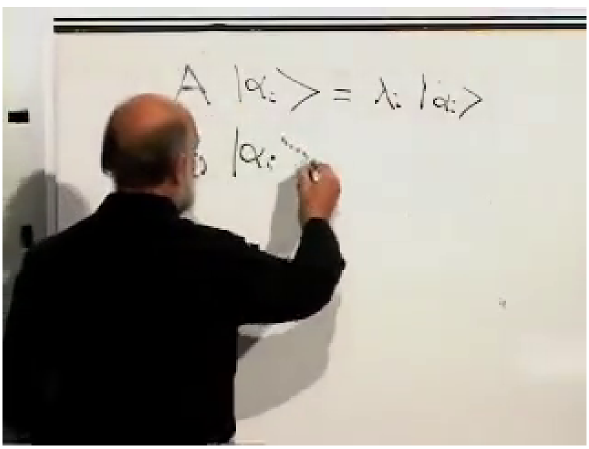

Figure 4.5: Two observables may have the same eigenvectors while assigning different eigenvalues to them.

Lecture frame: 00:55:45.

[a] **Question from the audience.** Could we apply magnetic fields along x and y at once and call that a simultaneous measurement?

Response. The fields add to one resultant direction. The apparatus then measures spin along that resultant axis, perhaps at 45°, rather than measuring σ_x and σ_y separately. The attempted construction produces a third observable, not two compatible readings.

[a]Lecture timestamp: 00:53:56.

4.5 Every Pure One-Spin State Has an Axis

Now reverse the eigenvector problem. Given an arbitrary normalized spinor

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1, \tag{4.27}$$

does some spatial direction $\hat{n}$ satisfy

$$\sigma_{\hat{n}} |\psi\rangle = |\psi\rangle? \tag{4.28}$$

Yes. The lecture first sees why by counting. The two complex components contain four real numbers. Normalization removes one, and the physically irrelevant overall phase removes another, leaving two. A unit pointer also has two independent real parameters, such as longitude and latitude.

[a] **Question from the audience.** How can the state have only two degrees of freedom when α and β are both complex?

Response. They begin with four real components. The normalization condition removes one real degree of freedom, and multiplying the whole state by $e^{i\theta}$ changes no probability, so the common phase removes another. Two physically meaningful parameters remain, matching the two angles of a unit pointer.

[a]Lecture timestamp: 01:03:52.

[a] **Question from the audience.** Do we also need to solve for the eigenvalue?

Response. Here the target eigenvalue is fixed to $+1$. We are searching only for the direction whose up-state is the given spinor. The opposite pointer would describe the same ray as a down-state with eigenvalue -1.

[a]Lecture timestamp: 01:04:20.

The count explains why the correspondence should exist. A direct construction confirms it. Define

$$\begin{aligned} n_1 &= \langle\psi|\,\sigma_1\,|\psi\rangle = 2\,\mathrm{Re}(\alpha^*\beta), \\ n_2 &= \langle\psi|\,\sigma_2\,|\psi\rangle = 2\,\mathrm{Im}(\alpha^*\beta), \\ n_3 &= \langle\psi|\,\sigma_3\,|\psi\rangle = |\alpha|^2 - |\beta|^2. \end{aligned} \tag{4.29}$$

Normalization gives $n_1^2 + n_2^2 + n_3^2 = 1$, and direct substitution gives Equation (4.28). Thus every pure one-spin state is spin-up along one spatial axis.

There is an operational version of the statement. Prepare many copies of the same unknown pure state. Measure batches along trial directions, adjust the magnet, and search for the direction along which every outcome is up. Finite data determine the direction only within uncertainty, but the ideal state has one exact axis.

Questions are part of this method, not interruptions to it. A brief classroom aside recalls ten-year-olds asking freely about string theory, extra dimensions, and whether surrounding neutrinos cause headaches. The details of a question may be wrong; the willingness to expose the point of confusion is still valuable.

4.6 Two Spins and Local Operators

Before combining two spins, record how the Pauli matrices act on the σ_3 basis:

$$\sigma_1\,|u\rangle = |d\rangle\,, \qquad\qquad \sigma_1\,|d\rangle = |u\rangle\,,$$

$$\sigma_2 |u\rangle = i |d\rangle, \qquad \sigma_2 |d\rangle = -i |u\rangle,$$
$$\sigma_3 |u\rangle = |u\rangle, \qquad \sigma_3 |d\rangle = -|d\rangle. \tag{4.30}$$

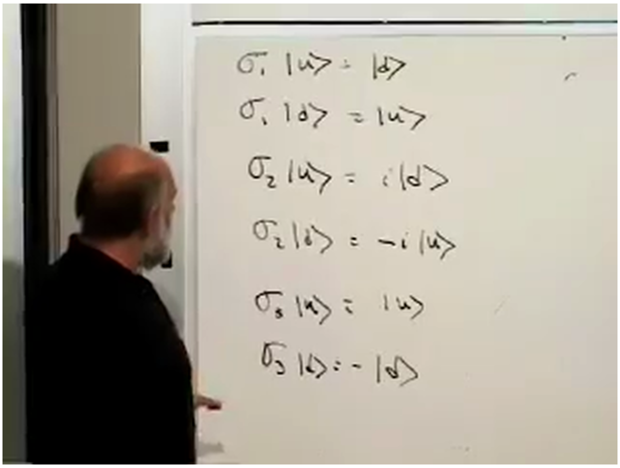

Figure 4.6: The Pauli action table repackages the matrices as rules on the up/down basis.

Lecture frame: 01:09:35.

Now imagine two electrons held at distinguishable locations so that only their spins vary. The four basis states are

$$|uu\rangle, \qquad |du\rangle, \qquad |ud\rangle, \qquad |dd\rangle. \tag{4.31}$$

The first label belongs to electron 1, the second to electron 2. These are mutually orthonormal because they represent distinct joint outcomes of the two third-component measurements.

Call the operators on electron 1 σ_i, and those on electron 2 τ_i. More explicitly,

$$\sigma_i \equiv \sigma_i \otimes I, \qquad \tau_i \equiv I \otimes \sigma_i. \tag{4.32}$$

The first set acts only on the first label, while the second set acts only on the second:

$$\sigma_1 |ud\rangle = |dd\rangle, \qquad \sigma_1 |dd\rangle = |ud\rangle,$$
$$\sigma_3 |dd\rangle = -|dd\rangle, \qquad \tau_2 |ud\rangle = -i |uu\rangle. \tag{4.33}$$

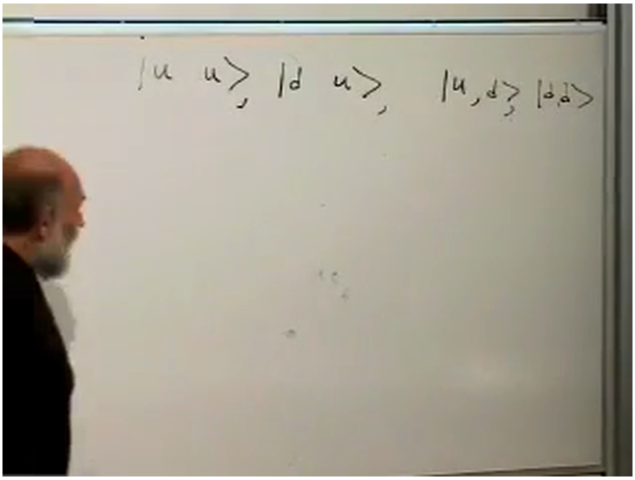

Figure 4.7: The four product-basis labels span the two-spin state space.

Lecture frame: 01:11:40.

Once an operator's action is known on the four basis states, linearity determines its action on every two-spin state.

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[a] **Question from the audience.** Why is the factor $-i$ written outside the ket? Does it act on both electrons?

Response. The symbols inside the ket are labels for a basis configuration. A complex scalar multiplies the whole state vector, so standard notation places it outside. Its origin here is the action of τ_2 on the second label; writing it outside does not mean that a separate physical operation was applied to both electrons.

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[a]Lecture timestamp: 01:17:47.

[a] **Question from the audience.** Does multiplication by i represent a rotation of the electron in ordinary space?

Response. Not by itself. It is complex scalar multiplication in state space. Spatial rotations will later be represented by particular unitary operators; an isolated factor i in this action table is not a literal real-space turn.

[a] Lecture timestamp: 01:19:44.

4.7 Six Parameters, Not Four

A general pure state of the pair is

$$|\Psi\rangle = c_{uu}\,|uu\rangle + c_{du}\,|du\rangle + c_{ud}\,|ud\rangle + c_{dd}\,|dd\rangle\,. \tag{4.34}$$

Four complex coefficients give eight real numbers. Normalization removes one and overall phase removes one:

$$8 - 1 - 1 = 6. \tag{4.35}$$

A separately prepared pure state of electron 1 needs two real parameters, and electron 2 needs two more, for only four. The two missing parameters are not an accounting error. They signal pure states of the pair that cannot be assigned separate pure states for the parts.

First identify the states that do come from separate preparations. If

$$|\psi_1\rangle = \alpha_1\,|u\rangle + \beta_1\,|d\rangle\,, \qquad |\psi_2\rangle = \alpha_2\,|u\rangle + \beta_2\,|d\rangle\,, \tag{4.36}$$

then

$$\begin{aligned} |\psi_1\rangle \otimes |\psi_2\rangle = \alpha_1\alpha_2\,|uu\rangle &+ \alpha_1\beta_2\,|ud\rangle \\ &+ \beta_1\alpha_2\,|du\rangle + \beta_1\beta_2\,|dd\rangle\,. \end{aligned} \tag{4.37}$$

This is a *product state.* Each electron has its own polarization axis, just as it would if the two magnetic preparations were performed independently.

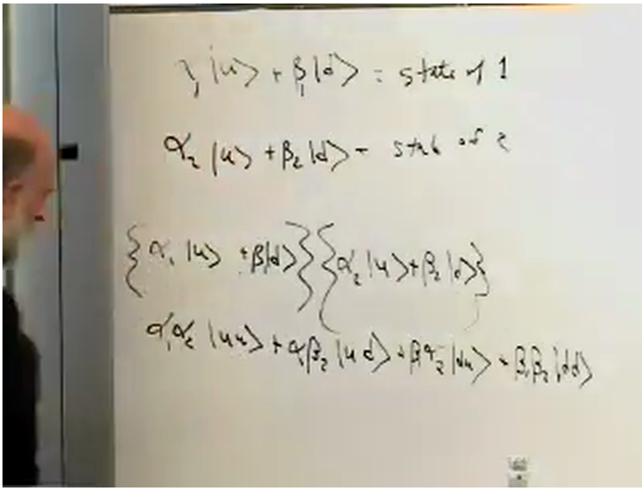

Figure 4.8: Multiplying two one-spin states produces a restricted four-term product state.

Lecture frame: 01:25:15.

If the four coefficients are arranged as

$$C = \begin{pmatrix} c_{uu} & c_{ud} \\ c_{du} & c_{dd} \end{pmatrix}, \tag{4.38}$$

the state is a product precisely when C has rank one, equivalently

$$c_{uu}c_{dd} - c_{ud}c_{du} = 0. \tag{4.39}$$

Most choices of four amplitudes do not satisfy this condition. They are entangled.

4.8 The First Entangled State

The lecture's first example is

$$|\psi_+\rangle = \frac{1}{\sqrt{2}}\left(|ud\rangle + |du\rangle\right). \tag{4.40}$$

It cannot be factored into one state for electron 1 and one for electron 2. If a third-axis measurement finds electron 2 down, electron 1 is up; if it finds electron 2 up, electron 1 is down. The pair has definite information that neither particle possesses separately.

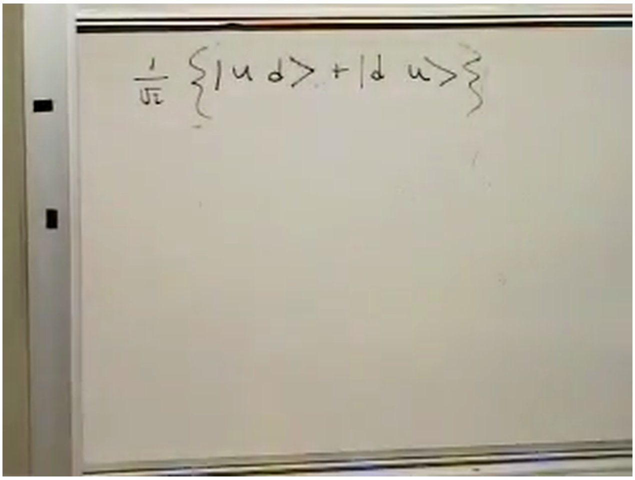

Figure 4.9: The first entangled superposition contains one up and one down but assigns neither result to a particle in advance.

Lecture frame: 01:29:30.

To see what either electron looks like alone, calculate the three local expectation values. For the first component,

$$\sigma_1 |\psi_+\rangle = \frac{1}{\sqrt{2}} \left(|dd\rangle + |uu\rangle \right), \tag{4.41}$$

which is orthogonal to $|\psi_+\rangle$. Therefore

$$\langle\psi_+| \sigma_1 |\psi_+\rangle = 0. \tag{4.42}$$

Similarly,

$$\sigma_2 |\psi_+\rangle = \frac{i}{\sqrt{2}} \left(|dd\rangle - |uu\rangle \right), \tag{4.43}$$

so

$$\langle\psi_+| \sigma_2 |\psi_+\rangle = 0. \tag{4.44}$$

For the third component,

$$\sigma_3 |\psi_+\rangle = \frac{1}{\sqrt{2}} (|ud\rangle - |du\rangle), \tag{4.45}$$

and the two diagonal contributions cancel:

$$\langle\psi_+| \sigma_3 |\psi_+\rangle = \frac{1}{2}(1 - 1) = 0. \tag{4.46}$$

Thus

$$\begin{aligned} &\langle\psi_+| \sigma_i |\psi_+\rangle = 0, \\ &\langle\psi_+| \tau_i |\psi_+\rangle = 0, \qquad i = 1, 2, 3. \end{aligned} \tag{4.47}$$

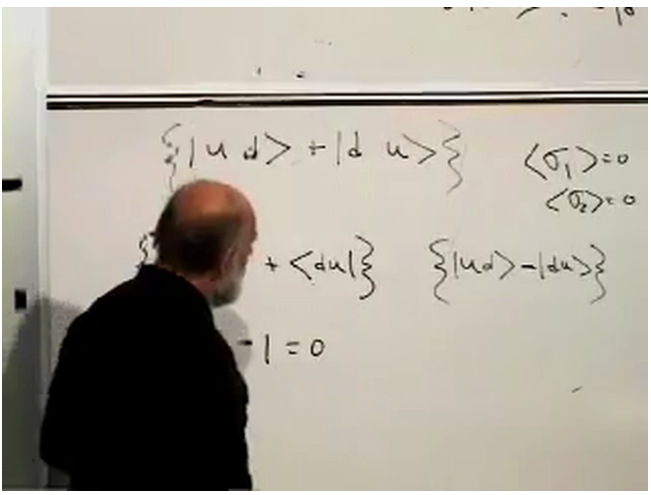

Figure 4.10: The transformed state and the cancellation $1 - 1 = 0$ establish vanishing local spin expectation values.

Lecture frame: 01:35:50.

For any direction,

$$\langle\psi_+| \sigma_{\hat{n}} |\psi_+\rangle = 0, \qquad \langle\psi_+| \tau_{\hat{n}} |\psi_+\rangle = 0. \tag{4.48}$$

Since each local measurement has only outcomes ± 1, zero expectation means

$$P(\sigma_{\hat{n}} = +1) = P(\sigma_{\hat{n}} = -1) = \frac{1}{2}, \tag{4.49}$$

and likewise for the second electron. No direction polarizes either electron by itself. This is impossible for a pure one-spin state but perfectly possible for one part of an entangled pure state.

---

[a] **Question from the audience.** Are the half-half local result and the fact that measuring one electron informs us about the other the same claim?

Response. No. First ignore electron 2: every direction measured on electron 1 alone gives a half-half distribution. That establishes the absence of any local polarization. Correlation is a separate joint statement about how the two recorded outcomes are related. Entanglement supplies both features, but one should not use one as a substitute for proving the other.

---

[a]Lecture timestamp: 01:38:23.

4.9 Correlation Before Bell

Correlation by itself is not uniquely quantum. Mix a penny and a dime, separate them without looking, and inspect one. Before inspection either object has a fifty-fifty chance of being the penny, but finding the penny here tells us that the distant object is the dime. The pair carries definite information even when either part by itself does not.

---

[a] **Question from the audience.** Does the penny-and-dime inference work only if we know the original pair contained exactly one of each?

Response. Exactly. The classical inference depends on trusted preparation. Likewise, a quantum correlation prediction applies only when the two-electron system has been prepared in the specified joint state.

---

[a]Lecture timestamp: 01:41:45.

[a] **Question from the audience.** For $|\psi_+\rangle$, is the probability that both third-component measurements are up equal to zero? Does that prove the spins are opposite along every axis?

Response. The probability of *uu* in the third-axis basis is zero, and the two third-axis outcomes are opposite. But the second conclusion is too strong for this plus-sign state. Its local spin is zero in every direction, while its joint correlation depends on the direction. Universal opposite outcomes belong to the minus-sign singlet introduced at the end of the lecture.

[a] Lecture timestamp: 01:42:09.

[a] **Question from the audience.** Could the classical pair be imagined as two coins welded so that one face is always opposite the other?

Response. That is a possible classical mechanism for fixed anti-correlation, provided the shaking does not disturb the linkage. It remains a hidden classical arrangement established at preparation. The analogy is deliberately provisional; Bell's theorem will test where every such classical account fails.

[a] Lecture timestamp: 01:43:51.

[a] **Question from the audience.** Where does the law of large numbers enter if one outcome already determines the partner's outcome?

Response. Conditional certainty is a statement about one trial once the joint state and one result are given. Repeated trials are still needed to verify experimentally that the promised correlation is reliable and to estimate expectation values and imperfect probabilities. The law of large numbers connects those repeated frequencies to the theoretical distribution.

[a] Lecture timestamp: 01:44:32.

The two observers may be separated by an enormous distance. Looking at one result changes what the local observer knows about the other result, but it does not send a controllable message to the distant observer. At this stage the situation still resembles the separated penny and dime.

---

[a] **Question from the audience.** Is the separated penny-and-dime picture the right way to think about the quantum pair?

Response. It is the right first classical comparison, not the final quantum explanation. Bell's theorem concerns the measurable difference between a pair carrying pre-existing classical properties and an entangled quantum pair. That distinction is postponed until the necessary correlations have been calculated.

---

[a]Lecture timestamp: 01:46:46.

4.10 The Singlet and Its Preparation

At the end of the blackboard calculation, replace the plus sign by a minus:

$$|\psi_-\rangle = \frac{1}{\sqrt{2}}\left(|ud\rangle - |du\rangle\right). \tag{4.50}$$

The change leaves the local expectations in Equation (4.47) equal to zero and leaves the third-axis outcomes opposite. It does change the transverse joint correlations. This minus-sign state is the spin singlet, the rotationally invariant state in which the two spins are opposite along every common measurement axis.

In the simplified physical picture used here, bringing two localized electron magnetic moments close enough to interact lets the higher-energy alignment relax by photon emission. With no preferred external axis, the lower spin state is represented by the rotationally symmetric singlet. The exact interaction and the reason the relative phase must be negative require a further calculation.

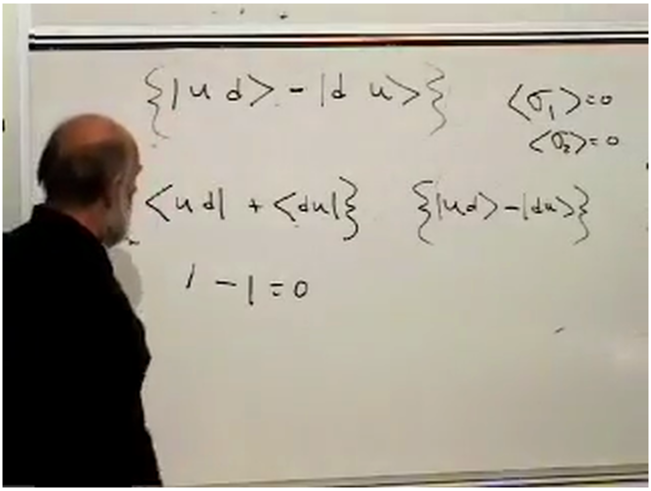

Figure 4.11: The final blackboard change replaces the symmetric superposition by the minus-sign singlet while retaining zero local spin.

Lecture frame: 01:47:25.

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[a] **Question from the audience.** Is the practical preparation simply to bring two electrons close together and wait?

Response. In this idealized model, yes: let the spin-spin interaction act and allow radiative relaxation into the lower state. The electrons need only be close enough, for long enough, that their magnetic interaction is effective. A real preparation protocol must control the orbital state and competing interactions as well.

---

[a]Lecture timestamp: 01:49:02.

[a] **Question from the audience.** Is this just the Pauli exclusion principle?

Response. No. The two electrons have been assigned distinguishable spatial locations, so their spin discussion does not yet require antisymmetrizing overlapping orbital wavefunctions. Exclusion becomes relevant when identical electrons occupy overlapping one-particle states; the present entanglement argument already works for separated sites.

[a]Lecture timestamp: 01:49:16.

[a] **Question from the audience.** Would an ordinary electron beam naturally contain such interacting pairs?

Response. Usually not. A beam is dilute, and typical electrons remain too far apart for appreciable spin-spin interaction during the available flight time. Collisions or a deliberately confining interaction can create entangled pairs, but mere membership in the same beam does not.

[a]Lecture timestamp: 01:50:31.

[a] **Question from the audience.** What distance is required for the magnetic interaction?

Response. There is no universal sharp cutoff; the decisive quantity is the interaction time scale. The lecture gives a rough microscopic estimate of order 10^{-11} cm for rapid coupling. At larger separation the energy splitting is smaller, the emitted photon has longer wavelength, and relaxation takes much longer. A sufficiently long wait can compensate in principle, while an ordinary beam provides too little time.

[a]Lecture timestamp: 01:51:45.

[a] **Question from the audience.** Why must the singlet contain a minus sign rather than a plus sign?

Response. That is not obvious from the local half-half statistics, because those statistics are identical for the two relative phases. The answer lies in how the state transforms under rotations and how the interaction selects total spin zero. The derivation belongs to the next lecture.

[a] Lecture timestamp: 01:53:10.

The course has now crossed the intended threshold. A single pure spin always has a definite up-axis. A pure pair need not assign any pure state to either member, yet it can encode exact relations between their outcomes. That extra structure is entanglement; Bell's theorem will determine why no classical penny-and-dime story can reproduce all of it.

CHAPTER 5

THE SINGLET, BELL'S INEQUALITY, AND NO-CLONING

Overview. A relative minus sign turns a merely correlated two-spin state into the rotationally invariant singlet. We identify that state by calculating its total spin, use its perfect anti-correlation to formulate a Bell experiment, and translate classical propositions into quantum projection operators. The resulting probabilities violate Bell's classical counting inequality. Questions about locality, experimental timing, and uncertainty then lead to a second consequence of linear quantum mechanics: an unknown state cannot be copied by a universal cloning machine.

5.1 What Difference Can a Sign Make?

Consider two electrons and suppress the tensor-product symbol in the basis states:

$$|ud\rangle \equiv |u\rangle_1 \otimes |d\rangle_2\,, \qquad |du\rangle \equiv |d\rangle_1 \otimes |u\rangle_2\,. \tag{5.1}$$

Two especially simple superpositions differ only by one sign,

$$|\psi_\pm\rangle = \frac{|ud\rangle \pm |du\rangle}{\sqrt{2}}. \tag{5.2}$$

The factor $1/\sqrt{2}$ makes the total probability one. In either state, a measurement along the third axis finds opposite answers: if electron 1 is up, electron 2 is down, and conversely. The sign therefore cannot be detected by that correlation alone. Nevertheless, the two states are physically different.

The calculation will repeatedly use the Pauli actions

$$\begin{aligned}
\sigma_1\,|u\rangle &= |d\rangle\,, & \sigma_1\,|d\rangle &= |u\rangle\,,\\
\sigma_2\,|u\rangle &= i\,|d\rangle\,, & \sigma_2\,|d\rangle &= -i\,|u\rangle\,,\\
\sigma_3\,|u\rangle &= |u\rangle\,, & \sigma_3\,|d\rangle &= -\,|d\rangle\,.
\end{aligned} \tag{5.3}$$

We reserve σ_i for operators acting on electron 1 and introduce an identical family τ_i acting on electron 2.

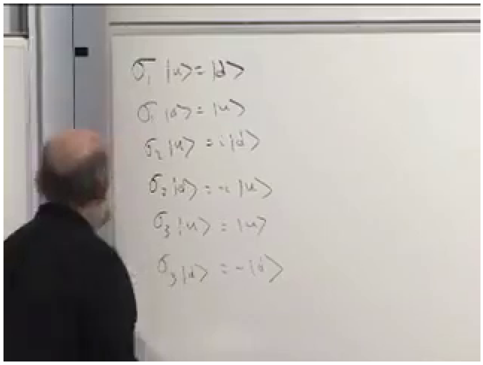

Figure 5.1: The complete Pauli-action table used throughout the comparison.

Lecture frame: 00:03:15.

Neither constituent spin has a preferred direction in either state:

$$\langle \sigma_i \rangle_{\psi_\pm} = 0, \qquad \langle \tau_i \rangle_{\psi_\pm} = 0, \qquad i = 1, 2, 3. \tag{5.4}$$

For example, the explicit check requested for the plus state gives

$$\begin{aligned} \langle \psi_+ | \, \sigma_3 \, | \psi_+ \rangle &= \frac{1}{2}\big(\langle ud| + \langle du|\big)\big(|ud\rangle - |du\rangle\big) \\ &= \frac{1}{2}(1 - 1) = 0. \end{aligned} \tag{5.5}$$

The cross terms vanish because $\langle ud \,|\, du \rangle = 0$.

[a] **Question from the audience.** Does a zero expectation value mean that a single spin can be measured to have value zero?

Response. No. A component of one spin has only the outcomes $+1$ and -1; σ_i has no zero eigenvalue. Equation (5.4) says that repeated measurements are a fair coin toss, not that any individual result is zero.

[a] Lecture timestamp: 00:07:13.

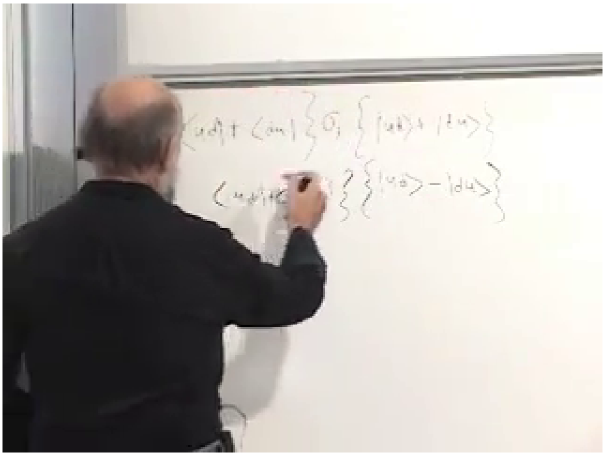

Figure 5.2: The plus-state expectation value is evaluated by acting with σ_3 and using orthogonality.

Lecture frame: 00:07:00.

The first plausible diagnostic has therefore failed. Local averages cannot see the relative sign, so we must measure a property of the pair.

5.2 The Total Spin Selects the Singlet

The pair's magnetic moment is represented by the vector sum

$$\boldsymbol{\Sigma} = \boldsymbol{\sigma} + \boldsymbol{\tau}. \tag{5.6}$$

Its third component annihilates both basis vectors:

$$\begin{aligned} (\sigma_3 + \tau_3)\,|ud\rangle &= (1-1)\,|ud\rangle = 0, \\ (\sigma_3 + \tau_3)\,|du\rangle &= (-1+1)\,|du\rangle = 0. \end{aligned} \tag{5.7}$$

Consequently,

$$(\sigma_3 + \tau_3)\,|\psi_\pm\rangle = 0. \tag{5.8}$$

Both sign choices have total third component zero on every trial.

The first component separates them. From Equation (5.3),

$$\begin{aligned}(\sigma_1 + \tau_1)\,|ud\rangle &= |dd\rangle + |uu\rangle\,,\\ (\sigma_1 + \tau_1)\,|du\rangle &= |uu\rangle + |dd\rangle\,. \end{aligned} \tag{5.9}$$

Thus

$$\begin{aligned}(\sigma_1 + \tau_1)\,|\psi_+\rangle &= \sqrt{2}(|uu\rangle + |dd\rangle),\\ (\sigma_1 + \tau_1)\,|\psi_-\rangle &= 0. \end{aligned} \tag{5.10}$$

The plus state may have zero expectation value for this observable, but it is not its zero-eigenvalue eigenstate.

---

[a] **Question from the audience.** What outcomes can the total component have when the state is not annihilated?

Response. Each constituent answer is +1 or −1, so their sum can be 2, 0, or −2. Zero expectation does not force zero on every run.

---

[a] Lecture timestamp: 00:13:02.

The second component completes the test. Its factors of i cancel in the antisymmetric combination:

$$(\sigma_2 + \tau_2)\,|\psi_-\rangle = 0. \tag{5.11}$$

All three components therefore annihilate the minus state,

$$(\boldsymbol{\sigma} + \boldsymbol{\tau}) \cdot \hat{n}\;|\psi_-\rangle = 0 \qquad \text{for every unit vector } \hat{n}. \tag{5.12}$$

The antisymmetric state $|\psi_-\rangle$ is the *singlet.* It is rotationally invariant: every axis gives the same total-spin answer, zero. The symmetric state in Equation (5.2) belongs to the triplet sector.

---

[a] **Question from the audience.** Is the opposite-spin result true only on average?

Response. For the singlet it is true on every trial. Equation (5.12) is an eigenvector equation, so measuring the same component on both particles always gives answers whose sum is the eigenvalue zero.

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[a] Lecture timestamp: 00:19:06.

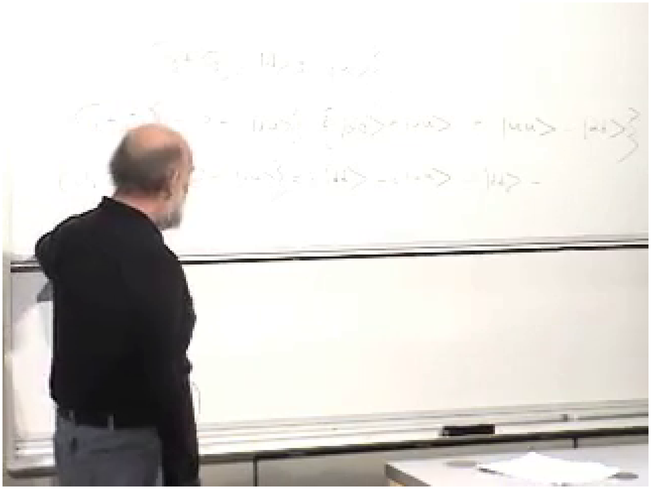

Figure 5.3: The component-by-component cancellations that identify the minus state as total spin zero.

Lecture frame: 00:16:30.

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[a] **Question from the audience.** Why is the other state called a triplet?

Response. That name comes from the three states in the total-spin-one multiplet. Its derivation is deferred; the present calculation needs only the contrast between the symmetric state and the one-dimensional singlet sector.

---

[a] Lecture timestamp: 00:19:36.

If Bob measures any component of his electron, he immediately knows what Alice would obtain by measuring the same component of hers: her answer must be the opposite. This is an Einstein–Podolsky–Rosen correlation. A pair of hidden coins can also carry opposite results, but only for one prearranged question. The singlet supplies perfect anti-correlation for whichever spatial axis is chosen.

[a] **Question from the audience.** Did Einstein, Podolsky, and Rosen first discover entanglement?

Response. No. Schrödinger discussed the phenomenon and introduced the language of entanglement. The EPR argument made the distant-correlation problem especially sharp.

[a]Lecture timestamp: 00:20:55.

[a] **Question from the audience.** Does the immediate correlation violate causality, and does it matter which observer measures first?

Response. No usable message is transmitted. Alice's local outcomes remain fifty-fifty whether or not Bob measures, in every inertial ordering. She can verify the correlation only after receiving Bob's result through an ordinary classical channel, which cannot outrun light.

[a]Lecture timestamp: 00:23:50.

5.3 Bell's Inequality Is Classical Counting

The status of Bell's result can feel unstable. On some days it appears to be the deepest feature of quantum mechanics; on others it appears to be one more inevitable consequence of the formalism. Bell himself, and Feynman as well, could express both attitudes. The inequality, however, is elementary. The surprise lies in its violation.

Take an ordinary set of objects. Each object may possess or fail to possess three properties A, B, and C. Let $N(A, \neg B)$ denote the number that have A but not B. Classical set logic implies

$$N(A, \neg B) + N(B, \neg C) \geq N(A, \neg C). \tag{5.13}$$

The proof is the set inclusion

$$A \cap C^c \subseteq (A \cap B^c) \cup (B \cap C^c). \tag{5.14}$$

Indeed, an element of $A \cap C^c$ either belongs to B or it does not. In the first case it belongs to $B \cap C^c$; in the second it belongs to $A \cap B^c$. Counting the union can only overcount the set on the left.

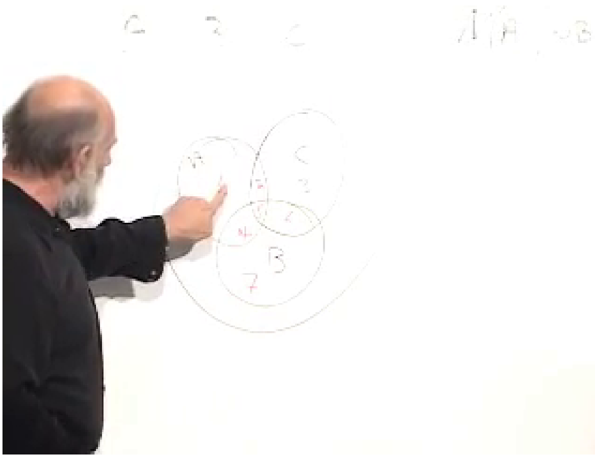

Figure 5.4: The board proof partitions three overlapping classical subsets into nonnegative regions.

Lecture frame: 00:31:15.

Equivalently, label the disjoint regions of the three-set diagram by nonnegative integers N_i. Both sides of Equation (5.13) contain the regions required by $A \cap C^c$, while the left side contains additional nonnegative regions. Nothing quantum has entered. This is a theorem about subsets and ordinary logical alternatives.

5.4 Turning the Set Statement into a Spin Experiment

Prepare many independent electron pairs in the singlet state and define three propositions about electron 1:

$$\begin{aligned} A &: \text{ spin 1 is up at } 0^\circ\text{, along the third axis,} \\ B &: \text{ spin 1 is up at } 45^\circ \text{ in the } zx\text{-plane,} \\ C &: \text{ spin 1 is up at } 90^\circ\text{, along the first axis.} \end{aligned} \tag{5.15}$$

In the singlet, the negation of a proposition about electron 1

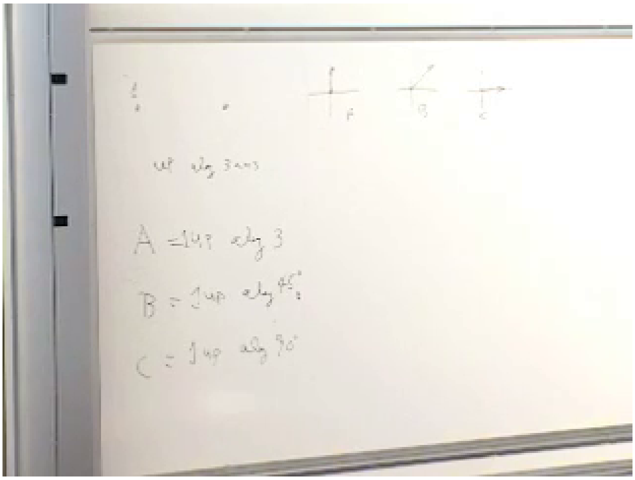

Figure 5.5: The three spin propositions are fixed at 0°, 45°, and 90° in one plane.

Lecture frame: 00:39:00.

may be rewritten as a positive proposition about electron 2. For example,

$$\begin{aligned} \neg B:\ & \text{spin 1 is down at } 45^\circ \\ & \iff \text{spin 2 is up at } 45^\circ. \end{aligned} \tag{5.16}$$

Likewise, $\neg C$ becomes the statement that electron 2 is up at 90°.

---

[a] **Question from the audience.** Should $\neg C$ make electron 2 down at 90°?

Response. No. First negate C: electron 1 is down at 90°. Perfect singlet anti-correlation then makes electron 2 up along that same axis. The two reversals must be kept separate.

---

[a]Lecture timestamp: 00:40:04.

For a large ensemble, counts and probabilities differ only by the common number of prepared pairs. Rotational invariance also gives

$$P(A, \neg B) = P(B, \neg C), \tag{5.17}$$

because the second experiment is the first rotated by 45°. Bell's classical inequality would therefore demand

$$2P(A, \neg B) \geq P(A, \neg C). \tag{5.18}$$

We now need an efficient way to calculate these probabilities.

5.5 Propositions Become Projectors

In an ordinary three-dimensional vector space, a projector removes every component perpendicular to a chosen subspace. A one-dimensional projector drops a vector orthogonally onto a line; a two-dimensional projector drops it onto a plane. If P projects onto a specified subspace,

$$P\,|\alpha\rangle = |\alpha_P\rangle\,. \tag{5.19}$$

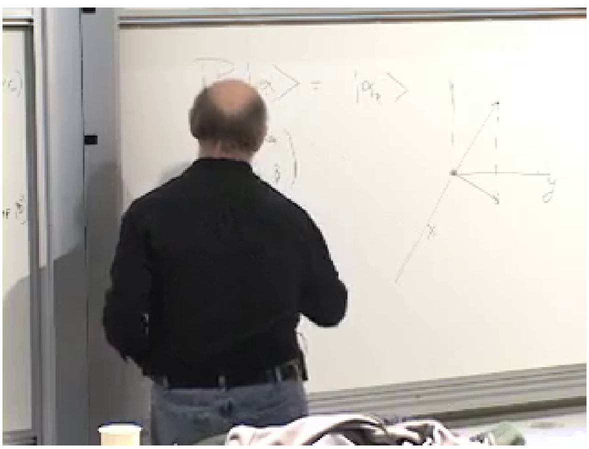

Figure 5.6: A geometric projection and its operator notation are developed side by side.

Lecture frame: 00:49:53.

In the one-spin Hilbert space, the proposition that $\sigma_3 = +1$ corresponds to the one-dimensional subspace spanned by $|u\rangle$. Its projector acts as

$$P_{\sigma_3=+1}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \begin{pmatrix}\alpha\\ 0\end{pmatrix}, \qquad P_{\sigma_3=-1}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \begin{pmatrix}0\\ \beta\end{pmatrix}. \tag{5.20}$$

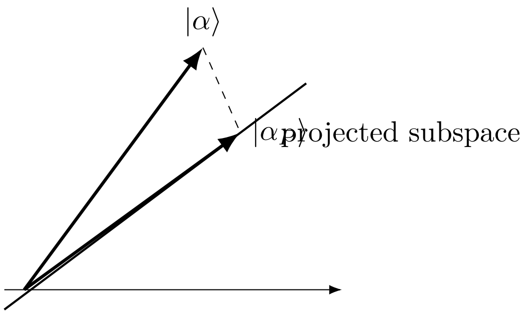

Figure 5.7: A clean reconstruction of the projection geometry.

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[a] **Question from the audience.** Does the first projector report how high the vector reaches on a vertical axis?

Response. No. It simply discards the orthogonal coefficient β. The components are Hilbert-space amplitudes, not ordinary spatial coordinates of the spin.

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[a] Lecture timestamp: 00:49:55.

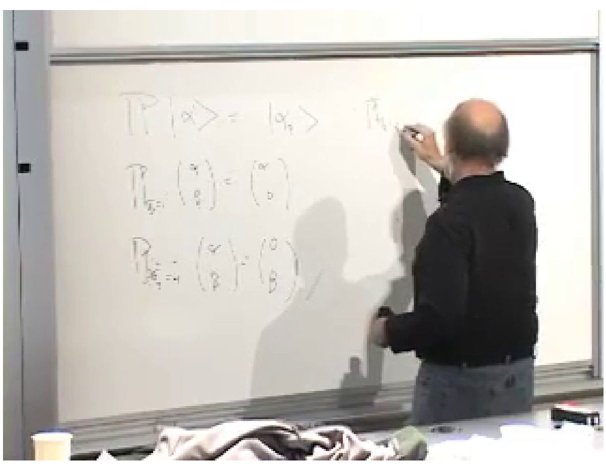

Figure 5.8: Projectors first act on a general spinor and are then written as explicit matrices.

Lecture frame: 00:51:40.

The matrix form is

$$P_{\sigma_3=+1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \frac{I + \sigma_3}{2}. \tag{5.21}$$

The same construction works along any unit vector $\hat{n}$:

$$P_{\boldsymbol{\sigma}\cdot\hat{n}=+1} = \frac{I + \boldsymbol{\sigma}\cdot\hat{n}}{2}. \tag{5.22}$$

---

[a] **Question from the audience.** What is the projector onto $\sigma_1 = +1$?

Response. Replace σ_3 by σ_1 in Equation (5.21): $P_{\sigma_1=+1} = (I + \sigma_1)/2$.

---

[a]Lecture timestamp: 00:53:09.

The Born rule now has a useful proposition form:

$$P(\mathcal{A} \text{ is true in } |\psi\rangle) = \langle\psi| P_{\mathcal{A}} |\psi\rangle . \tag{5.23}$$

For a normalized spinor $|\psi\rangle = (\alpha, \beta)^T$, Equation (5.23) with $P_{\sigma_3=+1}$ gives $|\alpha|^2$, exactly as it should.

This language exposes the difference between classical and quantum propositions. In classical logic, a property corresponds to a subset of a set. In quantum mechanics, a property corresponds to a subspace of Hilbert space, or equivalently to its projector. Subsets and vector subspaces do not obey the same lattice operations; Bell's contradiction grows from that difference.

5.5.1 The Joint Bell Projector

For $A \wedge \neg B$, electron 1 must be up along the third axis and electron 2 must be up along

$$\hat{n}_{45} = \frac{\hat{x} + \hat{z}}{\sqrt{2}}. \tag{5.24}$$

The relevant component of the second spin is

$$\boldsymbol{\tau} \cdot \hat{n}_{45} = \frac{\tau_1 + \tau_3}{\sqrt{2}}. \tag{5.25}$$

The joint projector is therefore

$$P_{A \wedge \neg B} = \frac{I + \sigma_3}{2} \frac{I + (\tau_1 + \tau_3)/\sqrt{2}}{2}. \tag{5.26}$$

---

[a] **Question from the audience.** Why is there a factor $1/\sqrt{2}$ in the 45° spin component?

Response. The direction must be a unit vector. Its first- and third-axis components are both $1/\sqrt{2}$, so the corresponding observable is $(\tau_1 + \tau_3)/\sqrt{2}$.

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[a]Lecture timestamp: 00:59:18.

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[a] **Question from the audience.** Is the entire expression $I + \boldsymbol{\tau} \cdot \hat{n}_{45}$ divided by two?

Response. Yes. The projector is $\left(I + \boldsymbol{\tau} \cdot \hat{n}_{45}\right)/2$; the factor $1/2$ multiplies the whole numerator.

---

[a]Lecture timestamp: 01:00:52.

The two factors in Equation (5.26) act on different tensor factors and commute. Applying both first removes every component with electron 1 down and then every surviving component with electron 2 down along $\hat{n}_{45}$.

5.6 The Quantum Numbers Reverse the Inequality

Let

$$|\psi_s\rangle = \frac{|ud\rangle - |du\rangle}{\sqrt{2}} \tag{5.27}$$

denote the singlet. Begin with

$$P(A, \neg B) = \langle \psi_s | \, P_{A \wedge \neg B} \, | \psi_s \rangle \, . \tag{5.28}$$

The factor $(I + \sigma_3)/2$ kills $|du\rangle$ and leaves $|ud\rangle/\sqrt{2}$. The second factor then gives

$$\begin{aligned}\frac{I + (\tau_1 + \tau_3)/\sqrt{2}}{2}\,|ud\rangle \\ &= \frac{1}{2}\left[|ud\rangle + \frac{|uu\rangle - |ud\rangle}{\sqrt{2}}\right] \\ &= \frac{1}{2}\Bigg[\left(1 - \frac{1}{\sqrt{2}}\right)|ud\rangle \\ &\qquad + \frac{1}{\sqrt{2}}\,|uu\rangle\Bigg]. \qquad (5.29)\end{aligned}$$

The $|uu\rangle$ term has zero overlap with the singlet bra, and the surviving normalization factors yield

$$P(A, \neg B) = \frac{1}{4}\left(1 - \frac{1}{\sqrt{2}}\right). \qquad (5.30)$$

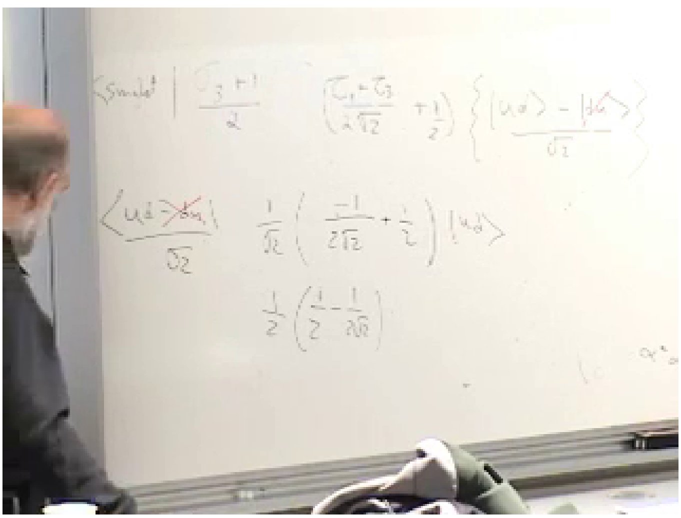

Figure 5.9: The projector calculation keeps the $|ud\rangle$ overlap and discards the orthogonal $|uu\rangle$ contribution.

Lecture frame: 01:08:20.

Rotational invariance gives the same result for the next pair of directions:

$$P(B, \neg C) = \frac{1}{4}\left(1 - \frac{1}{\sqrt{2}}\right). \tag{5.31}$$

For $A \wedge \neg C$, the second axis is the first spatial axis, so

$$P_{A\wedge\neg C} = \frac{I + \sigma_3}{2}\frac{I + \tau_1}{2}. \tag{5.32}$$

The first factor again leaves $|ud\rangle / \sqrt{2}$. The τ_1 term produces $|uu\rangle$, which is orthogonal to the singlet bra, while the identity term survives. Hence

$$P(A, \neg C) = \frac{1}{4}. \tag{5.33}$$

Combining the three results,

$$\underbrace{P(A, \neg B) + P(B, \neg C)}_{\frac{1}{2}(1-1/\sqrt{2})\approx 0.146} < \underbrace{P(A, \neg C)}_{\frac{1}{4}=0.25}. \tag{5.34}$$

Equation (5.34) reverses Equation (5.18). The conclusion is not that quantum mechanics contradicts itself. It is that the singlet correlations cannot arise from a local classical model in which all three spin propositions possess ordinary set-theoretic truth values independent of the distant measurement choice. A classical simulation can reproduce the correlations only by abandoning at least one Bell assumption, most conspicuously locality.

5.7 What the Bell Experiment Tests

Quantum mechanics already predicts the violation. An experiment must additionally close the classical escape route in which one detector informs the other what setting or outcome it encountered. The two wings should be far enough apart, and the choices and detections close enough in time, that a light-speed signal cannot travel between them. Aspect's experiment made that timing and isolation operationally convincing.

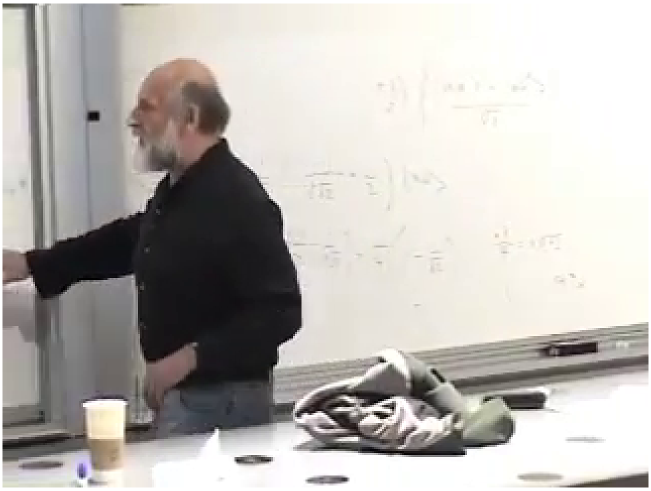

Figure 5.10: The completed board calculation displays the smaller Bell left-hand side and the larger right-hand side.

Lecture frame: 01:12:15.

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[a] **Question from the audience.** If quantum mechanics already gives the answer, did the Bell experiments teach us anything?

Response. They tested whether nature follows the quantum prediction in precisely this correlation experiment and, crucially, made the two measurement wings independent on the relevant light-travel time. The experimental control and timing were themselves substantial achievements.

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[a]Lecture timestamp: 01:13:15.

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[a] **Question from the audience.** What did Bell add after Einstein's discussion of spooky action?

Response. Bell supplied a quantitative inequality. He identified a feasible correlation experiment whose quantum prediction rules out a local classical underlying account, rather than leaving the objection at the level of philosophical unease. Bell remained ambivalent about whether this was inevitable or profound and became increasingly willing to question standard quantum mechanics.

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[a]Lecture timestamp: 01:18:19.

The particles may be miles apart while still sharing one joint state, but this remains a quantum experiment on small systems. Environmental interactions tend to destroy the required phase relations. Air molecules, stray photons, and thermal disturbances can entangle with the pair, so separation must be carried out in a cold, well-isolated environment.

---

[a] **Question from the audience.** Does a common state spread over a large distance make this a macroscopic quantum object?

Response. It is better described as small quantum systems far apart. Their separation can be large, but preserving their joint wave function becomes increasingly delicate because uncontrolled interaction with the environment destroys the useful coherence. Two ordinary bowling balls do not provide the same experiment.

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[a]Lecture timestamp: 01:19:53.

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[a] **Question from the audience.** Can measuring one particle reveal complementary information about the other and thereby evade the uncertainty principle?

Response. One may measure, for example, the position of one particle and the momentum of the other. That is not a simultaneous measurement of noncommuting observables on one particle. The uncertainty restriction on a single system has not been evaded.

---

[a]Lecture timestamp: 01:21:23.

5.8 When Does a Product Mean And?

The Bell calculation multiplied two projectors. This represents a conjunction only for compatible propositions.

[a] **Question from the audience.** Why did the product of two projectors represent the logical word and?

Response. Because the two projectors commute. One acts on electron 1 and the other on electron 2, so their order is irrelevant and both properties can be tested together. For a single electron, $(I+\sigma_3)/2$ and $(I+\sigma_1)/2$ do not commute; spin up along both axes is not a jointly definite proposition of this kind.

[a]Lecture timestamp: 01:22:39.

This is another place where quantum propositions depart from ordinary subsets. A classical intersection always exists. The quantum analogue represented by a product projector has the intended logical meaning only when the relevant subspaces are compatible.

The same discussion points toward measurement itself. A measuring device has a pointer state. If an up electron drives the pointer one way and a down electron drives it another, then a superposed electron becomes entangled with two pointer positions. Measurement may therefore be studied as entanglement between system and apparatus. That problem is postponed in favor of a shorter theorem that uses only linearity.

[a] **Question from the audience.** Is wave-function collapse usefully viewed as one measurement constraining the joint state?

Response. That question belongs to the measurement process. The apparatus and measured system become entangled: one spin value correlates with one pointer position and the other value with another. The detailed measurement discussion is deferred.

[a]Lecture timestamp: 01:25:09.

5.9 Linearity Forbids a Universal Xerox Machine

Only one fact about time evolution is needed. If a physical process sends

$$|a\rangle \longrightarrow |a'\rangle\,, \qquad |b\rangle \longrightarrow |b'\rangle\,, \tag{5.35}$$

then linearity requires

$$\alpha \left| a \right\rangle + \beta \left| b \right\rangle \longrightarrow \alpha \left| a' \right\rangle + \beta \left| b' \right\rangle . \tag{5.36}$$

An ordinary copying machine can reproduce classical information with controllably small errors. The quantum demand is stronger: one fixed device must duplicate the complete amplitudes of any state presented to it. This is why no-cloning is also called the no-quantum-Xerox principle.

Imagine a machine with one input port and two output ports. If it clones the third-axis basis, it must perform

$$\left| u \right\rangle \longrightarrow \left| uu \right\rangle , \qquad \left| d \right\rangle \longrightarrow \left| dd \right\rangle . \tag{5.37}$$

The machine can supply energy, mass, or charge to its outputs; conservation bookkeeping is not the obstruction. The contradiction concerns the state dependence of the transformation.

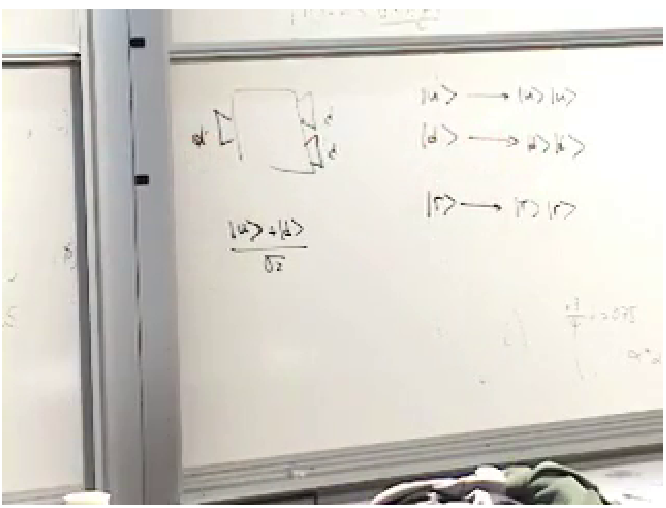

Figure 5.11: A hypothetical one-input, two-output machine is required to duplicate both basis states.

Lecture frame: 01:33:45.

Now input a spin pointing right,

$$|r\rangle = \frac{|u\rangle + |d\rangle}{\sqrt{2}}. \tag{5.38}$$

Equations (5.36) and (5.37) force the output

$$|r\rangle \longrightarrow \frac{|uu\rangle + |dd\rangle}{\sqrt{2}}. \tag{5.39}$$

Exact cloning instead requires two independent copies:

$$\begin{aligned} |rr\rangle &= |r\rangle \otimes |r\rangle \\ &= \frac{1}{2}\left(|uu\rangle + |ud\rangle + |du\rangle + |dd\rangle\right). \end{aligned} \tag{5.40}$$

The mixed terms in Equation (5.40) are absent from Equation (5.39). The two outputs are different.

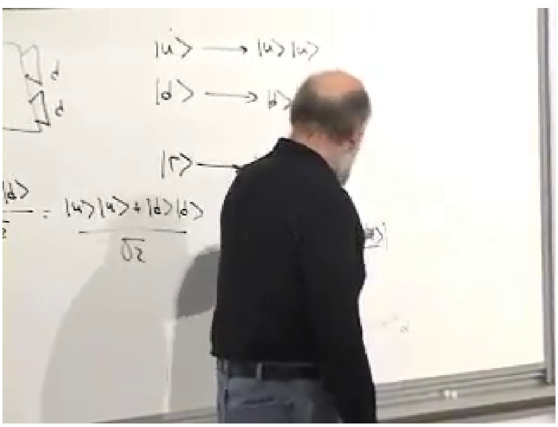

Figure 5.12: Linearity produces only the parallel-spin terms, whereas two true right-pointing copies also contain both mixed terms.

Lecture frame: 01:35:20.

Theorem 5.1 (No-cloning)**.** *No single physical machine can exactly clone every unknown quantum state.*

Proof. Assume one machine clones the orthogonal basis states as in Equation (5.37). Linearity then fixes its output on their superposition to be Equation (5.39). Exact cloning of the same input demands Equation (5.40). Since the two state vectors differ, the assumed universal machine cannot exist. □

The reason is structural. Sending $|\psi\rangle$ to $|\psi\rangle \otimes |\psi\rangle$ is quadratic in the amplitudes, whereas every allowed evolution of the complete quantum state is linear.

---

[a] **Question from the audience.** Does the proof assume that the two outputs are independent?

Response. Yes, because two clones must be independently usable copies. Independent systems combine by the tensor product, which is why two right-pointing copies are $|r\rangle \otimes |r\rangle$. Correlated outputs that cannot be used separately are not two clones.

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[a]Lecture timestamp: 01:35:49.

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[a] **Question from the audience.** A laser seems to copy a photon's state through stimulated emission. Where does the argument fail?

Response. The atom and the electromagnetic modes also participate. Their state contributes to the output and appears as noise from the standpoint of universal cloning. Approximate cloning is possible, but perfect state-independent cloning is not.

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[a]Lecture timestamp: 01:37:20.

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[a] **Question from the audience.** Can the quantum information be transferred even though it cannot be duplicated?

Response. Yes. Transfer is a different operation: the original need not retain the state. No-cloning forbids ending with two independent perfect copies, not moving the state from one carrier to another.

---

[a]Lecture timestamp: 01:40:24.

[a] **Question from the audience.** Must the input state be unknown?

Response. If a state is known in advance, one can prepare more systems in that state, or build a device adapted to a chosen orthogonal basis. What is impossible is one machine that accepts an arbitrary input and duplicates it without prior knowledge of which state arrived.

[a] Lecture timestamp: 01:41:03.

[a] **Question from the audience.** Which of the two right-state expressions is the actual machine output?

Response. Linearity forces Equation (5.39); cloning demands Equation (5.40). Their disagreement is precisely the contradiction.

[a] Lecture timestamp: 01:42:35.

[a] **Question from the audience.** Can the extra information carried by entangled two-particle states be quantified?

Response. A pure state of one spin has two real physical parameters. Two separately prepared spins supply four, whereas a general pure state in the four-dimensional two-spin Hilbert space has six after normalization and overall phase are removed. The additional states include entangled states, and their degree of entanglement can be quantified by entanglement entropy.

[a] Lecture timestamp: 01:43:40.

The sign in Equation (5.2) has now done three jobs. It selected the unique total-spin-zero state, supplied correlations that violate a classical set inequality, and exposed the difference between a linear quantum evolution and the nonlinear act of making two copies. Each result comes from treating the joint state as a vector in its full tensor-product space rather than as a list of pre-existing local properties.

CHAPTER 6

SUBSPACES, PROJECTION OPERATORS, AND MEASUREMENT AS ENTANGLEMENT

Bell's theorem has left two questions on the table. What remains of entanglement when one member of a pair is disturbed, absorbed, or measured? And what, precisely, turns an ordinary interaction into a measurement? The route to an answer runs through elementary linear algebra. Properties are subspaces, probabilities are expectation values of projectors, and a measuring apparatus is another quantum system with which the object becomes entangled.

6.1 Opening Motivation: Entanglement, Locality, and Measurement

Suppose a singlet has been separated into two distant parts and one part is measured. Must quantum field theory describe a virtual exchange between them? Nothing detectable happens at the far side, and no usable message arrives there. One may rotate one spin locally with a magnetic field, changing the detailed form of the joint state, while the amount of entanglement remains fixed under that local operation. Quantitative measures of entanglement will come later; the immediate point is that two systems which no longer interact cannot change their shared entanglement merely through an operation confined to one side.

[a] **Question from the audience.** Does a measurement on one member of a spacelike-separated singlet require some virtual exchange with the other?

Response. No detectable exchange occurs at the other side. A local operation may rotate one factor and alter the form in which the joint state is written, but it cannot by itself send a message or change the entanglement shared across the separation.

[a]Lecture timestamp: 00:00:22.

That forces a broader notion of what counts as the partner. If one member of an entangled pair falls into a black hole, the other does not cease to be entangled; it becomes entangled with the black-hole degrees of freedom and, after evaporation, with the emitted radiation. The same idea needs no black hole. Drop one member of an entangled pair into a bathtub and let it mix violently with the water. Its individual identity may be lost in the many-body state, yet the distant particle remains entangled with the bathtub as a whole and eventually with the escaping vapor.

This is not only a story about persistence. It is also a story about reversibility. If interaction can create entanglement, then interaction can, in principle, undo it. Disentanglement is possible, but only when the systems come back into contact and interact again. Anything that can happen one way can happen the other way.

[a] **Question from the audience.** If both spins of an entangled pair are measured, do the electrons remain entangled? Can one measurement already disentangle them?

Response. The measurements can remove their mutual entanglement, and one measurement can be enough. To see what that statement means rather than merely assert collapse, we must describe observation as an interaction between the system and a recording apparatus.

[a]Lecture timestamp: 00:04:45.

That question sets the target: connect observation, collapse of the wave packet, and entanglement. Before approaching it, however, we need a cleaner account of subspaces and projection operators, using no heavier machinery than the argument requires.

6.2 Dimension, Linear Dependence, and Orthonormal Bases

We therefore stay, for the moment, with finite-dimensional vector spaces equipped with bras, kets, and an inner product. Susskind begins as elementarily as possible. He does not define dimension by counting components in a column vector. He defines it abstractly: the dimension of a vector space is the maximum number of linearly independent vectors that can be found in it.

Thus a collection $\{|v_i\rangle\}$ is linearly independent when the relation

$$\sum_i c_i|v_i\rangle = 0 \tag{6.1}$$

forces all the coefficients c_i to vanish. Equivalently, none of the vectors can be written as a linear combination of the others.

It is crucial here that orthogonality is not the same thing as independence. In three dimensions one may have three vectors that are not orthogonal at all and yet are linearly independent. What matters is whether one of them can be built from the other two. On the other hand, in a two-dimensional plane any third vector in that plane must be linearly dependent on the first two.

This is the point of the sketch. If the space is two-dimensional, the maximum number of linearly independent vectors is two. If the space is three-dimensional, it is three. More generally, if the space has dimension d, then, given an inner product, one can find d mutually orthogonal unit vectors spanning it. Those are basis vectors. Susskind explicitly says he will not prove that theorem here. He only wants the conclusion: once

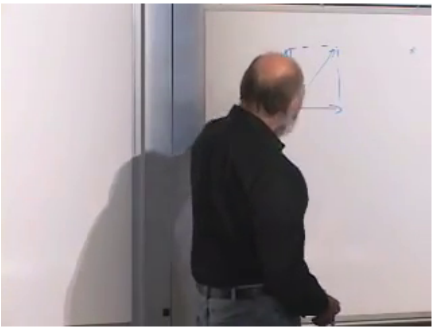

Figure 6.1: Three vectors drawn on the board. The lecture uses this sketch to separate the idea of orthogonality from the idea of linear independence, and then to emphasize that in a plane a third vector is dependent on the first two.

Lecture frame: 00:09:35.

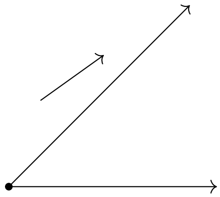

Figure 6.2: A minimal reconstruction of the board sketch. The exact coefficient relation is not fixed by the frame alone; the geometric point is that in a two-dimensional plane a third vector is not independent.

such a basis exists, every vector in the space can be expanded in it.

---

[a] **Question from the audience.** Is the number of spatial degrees of freedom the same as the dimension of the state space?

Response. No. A particle moving in a room has three spatial coordinates but an enormous number of mutually orthogonal quantum states. Configuration-space dimension and Hilbert-space dimension are different notions; finite-dimensional examples are being used only to expose the logic cleanly.

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[a]Lecture timestamp: 00:10:58.

6.3 Basis Expansion, Dyads, and the Resolution of Identity

Now choose an orthonormal basis $\{|n\rangle\}$, with $n = 1, \dots, d$. The blackboard at this point is especially clear and compact:

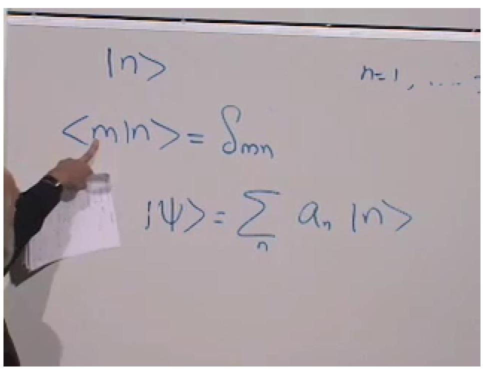

Figure 6.3: Orthonormal basis and expansion of an arbitrary state.

Lecture frame: 00:14:15.

We write

$$\langle m|n\rangle = \delta_{mn}, \tag{6.2}$$

$$|\psi\rangle = \sum_n a_n |n\rangle. \tag{6.3}$$

Here δ_{mn} is the Kronecker delta: it is 1 when $m = n$ and 0 otherwise.

Worked derivation. The coefficients are recovered by projecting onto the basis. Multiply by $\langle m|$ on the left:

$$\langle m|\psi\rangle = \sum_n a_n \langle m|n\rangle \tag{6.4}$$

$$= \sum_n a_n \delta_{mn} \tag{6.5}$$

$$= a_m. \tag{6.6}$$

So the coefficients are simply

$$a_m = \langle m|\psi\rangle. \tag{6.7}$$

The coefficient is an ordinary complex number, so it may be placed on either side of the ket without changing anything.

If we substitute the coefficient back into the expansion, we get

$$|\psi\rangle = \sum_n |n\rangle\langle n|\psi\rangle. \tag{6.8}$$

This suggests that the object $|n\rangle\langle n|$, and more generally $|a\rangle\langle b|$, ought to be read as an operator. That is exactly the next move. A dyad acts by

$$\big(|a\rangle\langle b|\big)|\psi\rangle = \langle b|\psi\rangle\,|a\rangle. \tag{6.9}$$

The bra extracts a complex number; the ket supplies the resulting direction. This is Susskind's low-tech route from vectors to operators.

Now read the previous expansion operatorially. Since

$$|\psi\rangle = \left(\sum_n |n\rangle\langle n|\right)|\psi\rangle \tag{6.10}$$

for every $|\psi\rangle$, the operator in parentheses must be the identity:

$$\sum_n |n\rangle\langle n| = I. \tag{6.11}$$

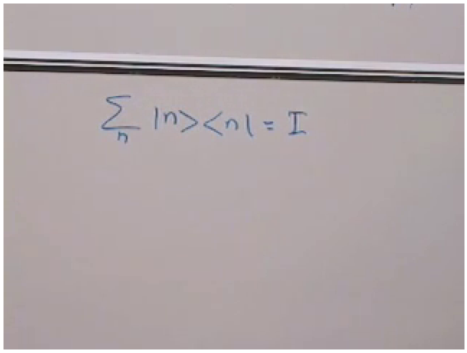

Figure 6.4: The completed blackboard resolution of identity.
Lecture frame: 00:18:30.

This is the resolution of the identity.

---

[a] **Question from the audience.** Must the vectors in the resolution of identity be a basis?

Response. Yes. The derivation used both normalization and mutual orthogonality, and the sum must run over a complete basis. An arbitrary collection of vectors does not resolve the identity.

---

[a]Lecture timestamp: 00:18:39.

The identity operator is itself the projector onto the whole space. Every vector is an eigenvector of I, and every eigenvalue is $+1$.

---

[a] **Question from the audience.** Does the resolution of identity say that every vector has eigenvalue one?

Response. Yes, for the identity operator. It is exceptional in that every vector is its eigenvector and $I|\psi\rangle = |\psi\rangle$ for every state.

---

[a]Lecture timestamp: 00:19:14.

6.4 Properties as Subspaces and Projection Operators

With that machinery in hand, the lecture makes its central conceptual correction. A property such as $k = \lambda$ is not, in general, one vector. It is a whole subspace.

Let k be a Hermitian observable and suppose

$$k|A\rangle = \lambda|A\rangle, \qquad k|B\rangle = \lambda|B\rangle, \tag{6.12}$$

with $|A\rangle$ and $|B\rangle$ linearly independent. Then any linear combination has the same eigenvalue:

$$\begin{aligned} k\big(\alpha|A\rangle + \beta|B\rangle\big) &= \alpha k|A\rangle + \beta k|B\rangle & (6.13)\\ &= \alpha\lambda|A\rangle + \beta\lambda|B\rangle & (6.14)\\ &= \lambda\big(\alpha|A\rangle + \beta|B\rangle\big). & (6.15)\end{aligned}$$

So the eigenspace for λ is closed under addition and scalar multiplication. It is a vector subspace.

The lecture's first concrete example is a two-spin system. The states

$$|ud\rangle, \qquad |uu\rangle \tag{6.16}$$

have different values of the second spin, but the same value of the first: the first spin is up in both cases. So they span the subspace in which $\sigma_3 = +1$ for the first spin.

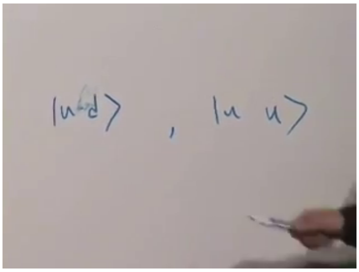

Figure 6.5: Two two-spin states with first spin up.

Lecture frame: 00:23:27.

That is the point worth preserving: a property is not necessarily one state. It is a whole collection of states with the same eigenvalue.

To avoid the notational confusion that appears briefly on the blackboard, we shall keep capital letters $A, B, \ldots$ for particular eigenvectors and use lower-case $a, b, \ldots$ as summation indices running over an orthonormal basis of the eigensubspace. If $\{|a\rangle\}$ is such a basis for the subspace $k = \lambda$, then the projection operator onto that subspace is

$$\Pi_{k=\lambda} = \sum_a |a\rangle\langle a|. \tag{6.17}$$

This is just the identity operator restricted to the subspace. If $|\psi\rangle$ already lies in that subspace, the projector returns it unchanged. If $|\psi\rangle$ lies partly outside, the projector discards the orthogonal remainder and keeps only the component inside.

6.5 Probability Postulate, Compatible Properties, and the Bell Recap

Once properties have been reinterpreted as subspaces, the probability postulate takes its cleanest form. If $|\psi\rangle$ is normalized, then the probability that it has the property $k = \lambda$ is

$$P(k = \lambda) = \langle\psi|\Pi_{k=\lambda}|\psi\rangle. \tag{6.18}$$

Insert the dyadic form of the projector:

$$\begin{aligned} P(k = \lambda) &= \left\langle \psi \middle| \sum_a |a\rangle\langle a| \middle| \psi \right\rangle && (6.19) \\ &= \sum_a \langle\psi|a\rangle\langle a|\psi\rangle && (6.20) \\ &= \sum_a |\langle a|\psi\rangle|^2. && (6.21) \end{aligned}$$

This is exactly the lecture's interpretation: the total probability for the property is the sum of the probabilities of all the orthogonal ways in which the property can occur.

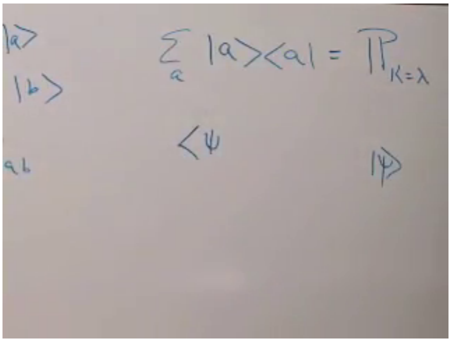

Figure 6.6: Projector onto the $k = \lambda$ eigensubspace. The blackboard layout already prepares the next move: sandwich the projector between $\langle\psi|$ and $|\psi\rangle$.

Lecture frame: 00:31:09.

For the first spin being up in a two-spin system, the relevant subspace is spanned by $|uu\rangle$ and $|ud\rangle$, so

$$\Pi_{\sigma_3=+1} = |uu\rangle\langle uu| + |ud\rangle\langle ud|. \tag{6.22}$$

In the lecture's shorthand this is also written as

$$\Pi_{\sigma_3=+1} = \frac{1+\sigma_3}{2}, \tag{6.23}$$

with the identity on the second spin suppressed.

---

[a] **Question from the audience.** What is the difference between an observable, a property, and a projector?

Response. The observable is the Hermitian operator k. The property $k = \lambda$ is represented by its eigensubspace and means that measuring k would certainly yield λ. The projector $\Pi_{k=\lambda}$ isolates that subspace, and its expectation value gives the probability that the property is present.

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[a]Lecture timestamp: 00:33:04.

[a] **Question from the audience.** Must the degenerate eigenvectors A and B already be orthogonal, and are they a basis for the whole space?

Response. The original eigenvectors need not be orthogonal, but an orthonormal basis can always be chosen within their eigensubspace. It is a basis for that subspace only; additional orthogonal directions are needed to span the full space. Projecting onto each subspace basis vector and adding gives the projection onto the subspace.

[a]Lecture timestamp: 00:34:07.

[a] **Question from the audience.** Does the symbol a in $k|A\rangle = \lambda|A\rangle$ mean the same thing as the a summed in the projector?

Response. No. The first denotes a particular eigenvector, whereas the index under the summation runs over every vector in an orthonormal basis of the eigensubspace. Distinct notation prevents a fixed label from being mistaken for a dummy summation index.

[a]Lecture timestamp: 00:35:33.

[a] **Question from the audience.** Are the inner products in the projector formula probabilities or probability amplitudes?

Response. Each $\langle a|\psi\rangle$ is an amplitude. Multiplication by its complex conjugate gives a probability, and summing those probabilities over the orthogonal ways of realizing $k = \lambda$ gives the total probability for the property.

[a]Lecture timestamp: 00:37:15.

The lecture then asks how to combine properties. Suppose we have two compatible properties, represented by commuting projectors P_k and P_ℓ. In the lecture's language, the AND statement is represented by the product:

$$\Pi_{k=\lambda \wedge \ell=\mu} = P_k P_\ell = P_\ell P_k. \tag{6.24}$$

The logic is simple. $P_k|\psi\rangle$ lies in the $k = \lambda$ subspace. Because the operators commute, $P_\ell P_k|\psi\rangle = P_k P_\ell|\psi\rangle$, so the same vector lies in the $\ell = \mu$ subspace as well. Thus the product projects onto the simultaneous property. The corresponding probability is

$$P(k = \lambda, \ell = \mu) = \langle \psi | P_k P_\ell | \psi \rangle. \tag{6.25}$$

The first step in that argument is worth displaying on its own. For any state,

$$k\big(P_k|\psi\rangle\big) = \lambda P_k|\psi\rangle, \tag{6.26}$$

so $P_k|\psi\rangle$ lies in the $k = \lambda$ eigensubspace. If P_k and P_ℓ commute, the product can be written with either projector on the left and therefore belongs to both subspaces.

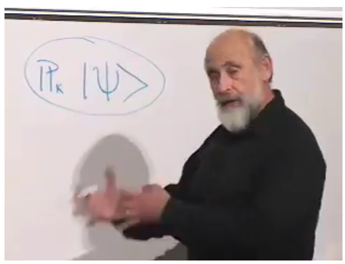

Figure 6.7: The projected state as a vector in the eigensubspace.

Lecture frame: 00:41:57.

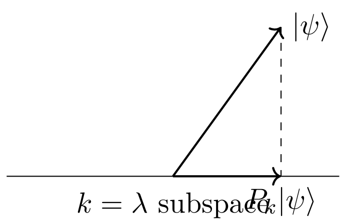

Figure 6.8: Projecting an arbitrary vector into the $k = \lambda$ eigensubspace.

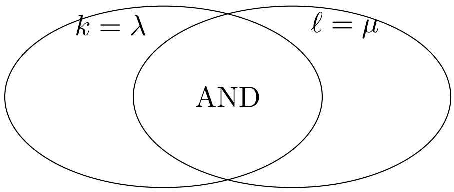

Figure 6.9: A reconstruction of the intersecting-subspace picture behind the AND statement for commuting projectors.

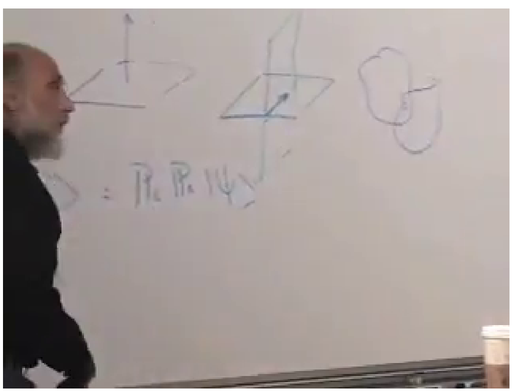

Figure 6.10: The board argument that commuting projectors select the simultaneous eigensubspace.

Lecture frame: 00:45:20.

[a] **Question from the audience.** Is the simultaneous property the same as the intersection in classical logic?

Response. Yes. For commuting projectors, $P_k P_\ell$ projects onto the intersection of the two eigensubspaces. If the intersection contains only the zero vector, no nonzero state has both properties.

[a]Lecture timestamp: 00:44:42.

For completeness, the projector onto the compatible OR subspace is

$$P_{k\vee\ell} = P_k + P_\ell - P_k P_\ell. \tag{6.27}$$

When the alternatives are mutually exclusive, $P_k P_\ell = 0$ and this reduces to the sum $P_k + P_\ell$. This subtraction is essential whenever the two subspaces overlap, because their intersection must not be counted twice.

Compatibility itself is then tied back to physical examples. In a two-spin system, any component σ_i of the first spin commutes with any component τ_j of the second:

$$[\sigma_i, \tau_j] = 0. \tag{6.28}$$

By contrast, different components of the same spin do not commute:

$$[\sigma_1, \sigma_3] \neq 0. \tag{6.29}$$

That is why one may measure one component of each of two separated spins, but not simultaneously measure two different components of the same spin.

[a] **Question from the audience.** What are concrete examples of compatible observables?

Response. Any operator acting on the first spin commutes with every operator acting on the second, because they act on different tensor factors. Position components x, y, z are another compatible family. By contrast, two different components of the same spin, or position and its conjugate momentum, do not commute.

[a]Lecture timestamp: 00:46:12.

[a] **Question from the audience.** Does a subsystem have to mean a different particle?

Response. No. It means an independently specified set of degrees of freedom. Distinct particles give the simplest tensor factors, but different degrees of freedom of one particle can also be treated separately when their operator structure permits it.

[a]Lecture timestamp: 00:47:26.

[a] **Question from the audience.** If so many operators on different subsystems commute, what is a simple incompatible pair?

Response. Two components of the same spin provide the standard example: σ_1 and σ_3 do not commute, so no apparatus can assign both sharp values simultaneously. The nonzero matrix commutator is the algebraic test.

[a]Lecture timestamp: 00:48:37.

Now Susskind revisits Bell's theorem. He does not redo last lecture's arithmetic. He rebuilds its logic in the new language. The classical inequality, in probabilistic form, is

$$P(A \wedge \neg B) + P(B \wedge \neg C) \geq P(A \wedge \neg C). \tag{6.30}$$

This is a theorem of classical set logic, not of quantum mechanics. In the singlet state, spins are anti-aligned along any common axis, so a NOT statement for the first spin can be traded for the corresponding positive statement for the second. That lets one rewrite the inequality as a statement about joint properties of a two-spin system.

Suppressing identity operators on untouched factors, the single-property projectors appearing in the lecture are

$$\frac{1+\sigma_3}{2}, \qquad \frac{1+\frac{\tau_1+\tau_3}{\sqrt{2}}}{2}, \qquad \frac{1+\frac{\sigma_1+\sigma_3}{\sqrt{2}}}{2}, \qquad \frac{1+\tau_1}{2}. \tag{6.31}$$

Therefore the three relevant joint projectors are

$$\Pi_1 = \frac{1+\sigma_3}{2}\,\frac{1+\frac{\tau_1+\tau_3}{\sqrt{2}}}{2}, \tag{6.32}$$

$$\Pi_2 = \frac{1+\frac{\sigma_1+\sigma_3}{\sqrt{2}}}{2}\,\frac{1+\tau_1}{2}, \tag{6.33}$$

$$\Pi_3 = \frac{1+\sigma_3}{2}\,\frac{1+\tau_1}{2}. \tag{6.34}$$

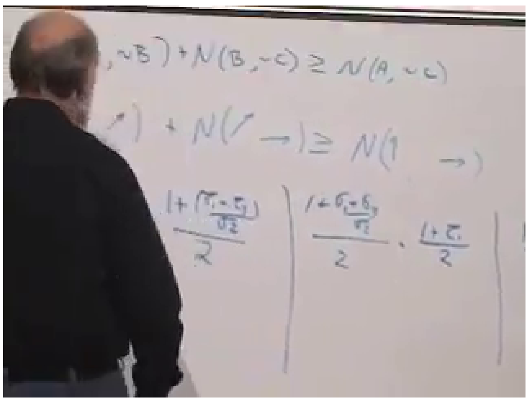

Figure 6.11: The three analyzer choices written as products of one-spin projectors.

Lecture frame: 01:01:45.

The lecture speaks of the singlet in the familiar unnormalized blackboard form

$$|\Psi_{\text{singlet}}\rangle \propto |ud\rangle - |du\rangle. \tag{6.35}$$

The proportionality sign is enough here. What matters for the logic is the state itself and, in particular, the relative sign, not the overall normalization. Evaluating $\langle\Pi_1\rangle + \langle\Pi_2\rangle$ against $\langle\Pi_3\rangle$ in the singlet state yields the violation obtained in the previous lecture. The important conclusion is the one Susskind emphasizes verbally: quantum mechanics is not classical set theory in disguise.

[a] **Question from the audience.** If spin-up and spin-down are orthogonal, does the minus sign mean that one ket is the negative of the other?

Response. No. $|u\rangle$ and $|d\rangle$ are distinct orthogonal vectors. A relative sign belongs to a superposition such as $|ud\rangle - |du\rangle$ and can change its physical properties; only an overall sign multiplying the entire state is irrelevant.

[a] Lecture timestamp: 00:52:31.

[a] **Question from the audience.** Is Bell's inequality violated for every quantum state and every choice of properties?

Response. No. Many states and many choices of axes satisfy a particular Bell inequality. Violation depends on both the state and the selected observables. Varying the intermediate and final angles changes whether this test exposes the nonclassical correlations.

[a] Lecture timestamp: 00:53:06.

[a] **Question from the audience.** What can be concluded when a particular state violates, or happens to satisfy, this Bell inequality?

Response. A violation proves that the observed statistics cannot arise from the classical set logic used in the derivation. Satisfaction of one chosen inequality does not by itself make the state classical, because another choice of properties may reveal a violation.

[a] Lecture timestamp: 01:02:47.

6.6 Locality After Bell

The projector calculation is not merely a symbolic game. Imagine repeatedly preparing correlated pairs in the singlet ground state, separating the two members, and dividing the identically prepared ensemble into three large groups. The first group measures the 0° and 45° settings, the second the 45° and 90° settings, and the third the 0° and 90° settings. Frequencies in the three groups estimate the three probabilities in the inequality.

[a] **Question from the audience.** What laboratory arrangement do the three Bell probabilities actually describe?

Response. A source produces many identically prepared entangled pairs and sends the two members to separated analyzers. Split the pairs into three ensembles and use the corresponding pair of analyzer directions on each ensemble. Counting the joint outcomes estimates each probability. A charged particle and antiparticle could be separated by a field; practical Bell tests often use photons instead.

[a]Lecture timestamp: 01:04:30.

[a] **Question from the audience.** The classical set proof used one population. How can three separately measured ensembles test it?

Response. The inequality is a statement about probabilities as well as counts. If all three samples are drawn from the same preparation procedure, their relative frequencies estimate the three probability distributions of one common ensemble. Identical preparation is the indispensable assumption.

[a]Lecture timestamp: 01:08:00.

That Bell recap leads straight into the physical unease still hanging over the room. If the correlations are real, why can they not be used to send a signal?

[a] **Question from the audience.** Why can the entangled correlations not be used to send a faster-than-light message?

Response. Each wing separately still sees an uncontrollable random sequence. The correlation becomes visible only when the two records are compared through an ordinary classical channel. A decisive Bell test nevertheless arranges the choices and detections so that no light signal could pass between the wings in time, thereby excluding a local communication mechanism for producing the joint statistics.

[a]Lecture timestamp: 01:09:41.

If enough time elapses for a light signal to travel between apparatuses, a correlation does not by itself contradict relativity; one might merely have discovered an unnoticed ordinary influence. The sharp experiment closes that escape route. In Bell's language, a hidden-variable account reproducing the correlations must be nonlocal, but nonlocal dependence is not the same thing as a controllable superluminal message.

---

[a] **Question from the audience.** Is angular momentum conservation essential to this argument?

Response. No. Spin is a convenient two-state model, not a sacred ingredient. Any pair of two-level systems with the required entangled state and alternative measurement bases can display the same logical structure.

---

[a] Lecture timestamp: 01:12:18.

This discussion does not close the lecture; it clears the ground for the next pivot. Bell has taught us that quantum mechanics is not classical logic, but the next question is different. What is a measurement, operationally and mathematically, and how does it enter the story of interference?

6.7 Finite-State Two-Slit, Interference, and Measurement as Entanglement

The lecture now changes model. Susskind turns to the two-slit experiment, but he immediately simplifies it into a finite-state problem so that the same vector-space technology can be used without bringing in the full machinery of continuum wave mechanics. He even says, in effect, that this is still the two-slit experiment, only stripped down to a small set of basis states.

There is a source state $|0\rangle$, two slit states $|1\rangle$ and $|-1\rangle$, and a set of screen states $|n\rangle$. These $|n\rangle$ are orthonormal because distinct screen positions are distinct configurations. The lecture explicitly chooses not to track normalization carefully here. Unitarity is postponed. What matters at this stage is linearity.

By symmetry the source evolves to an equal-amplitude

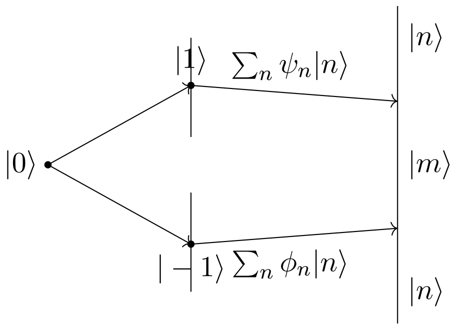

Figure 6.12: A finite-state version of the two-slit experiment: the source state $|0\rangle$, the two slit states $|1\rangle$ and $|-1\rangle$, and the screen basis $|n\rangle$.

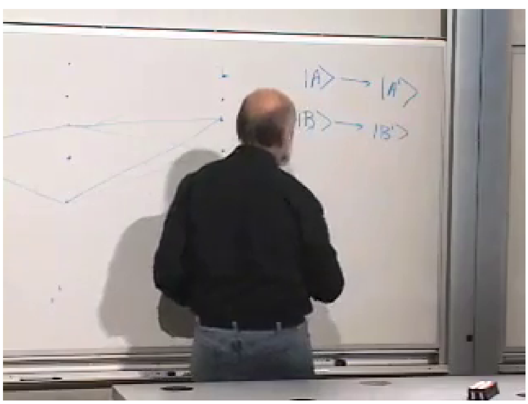

Figure 6.13: The finite position lattice beside the linear evolution rule $A \to A'$, $B \to B'$.

Lecture frame: 01:18:00.

superposition of the two slit states:

$$|0\rangle \longrightarrow |1\rangle + |-1\rangle. \tag{6.36}$$

Now let the upper slit state evolve to

$$|1\rangle \longrightarrow \sum_n \psi_n |n\rangle, \tag{6.37}$$

and the lower slit state to

$$|-1\rangle \longrightarrow \sum_n \phi_n |n\rangle. \tag{6.38}$$

The arrows here mean time evolution, not equality of states at the same instant. By linearity,

$$|1\rangle + |-1\rangle \longrightarrow \sum_n (\psi_n + \phi_n)|n\rangle. \tag{6.39}$$

That is the key rule. Amplitudes add before probabilities are formed.

---

[a] **Question from the audience.** How can the slit state $|1\rangle$ turn into states also labelled by integers n? Are these kets at the same place and time?

Response. The labels refer to different time slices. $|1\rangle$ denotes the electron at the upper slit, whereas the $|n\rangle$ on the right denote possible later positions on the screen. The arrow is time evolution, not an equality between simultaneous configurations.

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[a]Lecture timestamp: 01:21:59.

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[a] **Question from the audience.** Should the amplitudes be normalized, and do they carry physical units?

Response. A complete evolution must preserve total probability; that is unitarity. Here the common normalization is suppressed because only the addition and cancellation of path amplitudes matter. Any scale factors can be restored when the state is normalized.

---

[a]Lecture timestamp: 01:25:30.

The probability of arrival at the screen site m is therefore

$$P_m = |\psi_m + \phi_m|^2 \tag{6.40}$$
$$= |\psi_m|^2 + |\phi_m|^2 + \psi_m^* \phi_m + \phi_m^* \psi_m. \tag{6.41}$$

The first two terms are what classical reasoning would have produced. The last two are the interference terms.

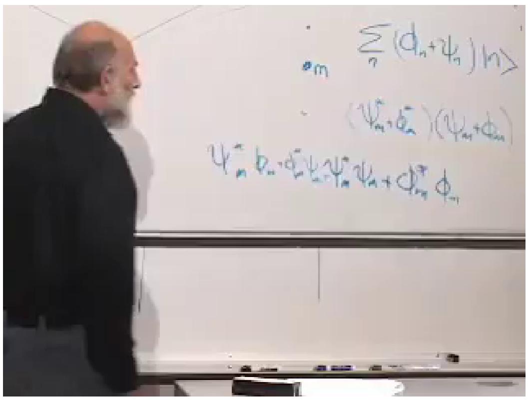

Figure 6.14: The screen amplitude and its four-term probability expansion on the board.

Lecture frame: 01:29:20.

At the symmetric point on the screen, call it $m = 0$, symmetry suggests $\psi_0 = \phi_0$. Then

$$P_0 = |\psi_0 + \phi_0|^2 = |2\psi_0|^2 = 4|\psi_0|^2. \tag{6.42}$$

Both slits open produce four times the one-slit probability at that point, not twice. This is constructive interference.

Now insert a phase shift so that $\phi_0 = -\psi_0$. Then

$$P_0 = |\psi_0 - \psi_0|^2 = 0. \tag{6.43}$$

Each slit by itself permits arrival at that point, but opening both together can make arrival there impossible. This is destructive interference. Susskind explicitly dwells on how odd this is: for his money it is as strange as Bell.

---

[a] **Question from the audience.** What if the phase shifter or slit choice is changed only after the electron is already in flight? Does the other slit need time to learn about it?

Response. The answer depends on the precise spacetime arrangement and on whether the electron encounters the altered part of the apparatus. Quantum mechanics evolves the state through the actual setup; probabilities still come from adding the applicable amplitudes. A delayed-choice proposal must be specified sharply before locality claims can be assessed.

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[a] Lecture timestamp: 01:33:14.

Now we introduce the apparatus. There is one additional two-state system, a spin initially in the state $|d\rangle$. If the electron goes through the upper slit, it flips that spin to $|u\rangle$. If it goes through the lower slit, it leaves it unchanged. The initial composite state is $|0, d\rangle$, and the first update rule becomes

$$|0, d\rangle \longrightarrow |1, u\rangle + |-1, d\rangle. \tag{6.44}$$

That one line is the whole meaning of measurement in the lecture. The position of the electron has become entangled with the record stored in the apparatus.

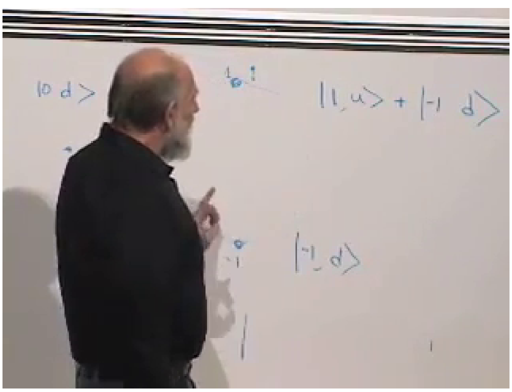

Figure 6.15: The detector spin correlates its up and down records with the two slit alternatives.

Lecture frame: 01:44:10.

[a] **Question from the audience.** Must there be a detector at both slits?

Response. One reliable detector is enough. If it records passage through the upper slit, an unchanged record identifies the lower alternative, provided every transmitted electron must pass through one of the two openings. A second detector is possible but adds another subsystem without changing the principle.

[a]Lecture timestamp: 01:40:20.

[a] **Question from the audience.** Why not say that the electron passes through both slits?

Response. Without a path record, the state is a coherent superposition and no retrospective path fact is available. With a complete which-path measurement, the apparatus records one alternative or the other on each run. What persists through both alternatives is the state amplitude, not two separately observed electron trajectories.

[a]Lecture timestamp: 01:40:51.

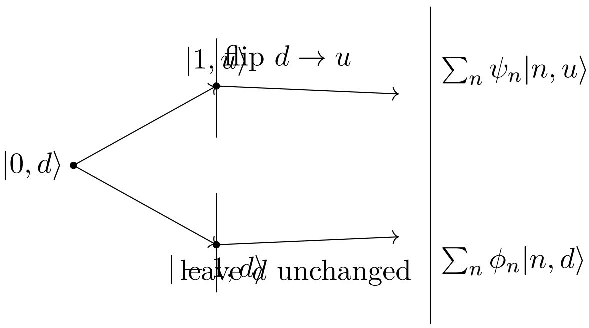

Figure 6.16: Measurement as entanglement with a detector spin. The upper path flips the apparatus spin; the lower path leaves it untouched.

Assume, as the lecture does, that nothing further happens to the spin during the journey to the screen. Then

$$|1,u\rangle \longrightarrow \sum_n \psi_n |n,u\rangle, \tag{6.45}$$

$$|-1,d\rangle \longrightarrow \sum_n \phi_n |n,d\rangle. \tag{6.46}$$

Therefore the final state with both slits open is

$$|\Psi_{\text{final}}\rangle = \sum_n \psi_n |n,u\rangle + \sum_n \phi_n |n,d\rangle. \tag{6.47}$$

This is not a superposition of states of one electron alone. It is an entangled state of electron and apparatus.

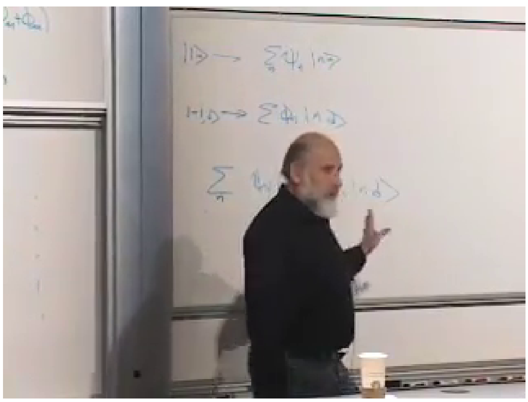

Figure 6.17: The two path amplitudes reach the screen carrying orthogonal apparatus records.

Lecture frame: 01:47:35.

The cleanest way to compute the arrival probability at the site m is to use the projector onto that site on the electron subsystem:

$$\Pi_m = |m\rangle\langle m| \otimes I_{\text{spin}}. \tag{6.48}$$

Applying Π_m to the final state gives

$$\Pi_m |\Psi_{\text{final}}\rangle = |m\rangle\big(\psi_m |u\rangle + \phi_m |d\rangle\big). \tag{6.49}$$

So the probability of arrival at m is

$$\begin{aligned} P_m &= \langle \Psi_{\text{final}} | \Pi_m | \Psi_{\text{final}} \rangle && (6.50) \\ &= \left(\psi_m^* \langle u| + \phi_m^* \langle d|\right)\left(\psi_m |u\rangle + \phi_m |d\rangle\right) && (6.51) \\ &= |\psi_m|^2 + |\phi_m|^2 + \psi_m^* \phi_m \langle u|d\rangle + \phi_m^* \psi_m \langle d|u\rangle. && (6.52) \end{aligned}$$

But the record states are orthogonal:

$$\langle u|d\rangle = 0. \tag{6.53}$$

Hence the cross terms vanish:

$$P_m = |\psi_m|^2 + |\phi_m|^2. \tag{6.54}$$

That is the lecture's mathematical punchline. Once which-path information is recorded, interference disappears and probabilities add in the classical way.

---

[a] **Question from the audience.** What is the difference between the superposition $|1\rangle + |-1\rangle$ and the state $|1,u\rangle + |-1,d\rangle$?

Response. The first concerns one system in a linear combination of two positions; there is nothing else with which it could be entangled. The second concerns two systems and correlates path with detector spin. Reading the spin then reveals information about the path.

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[a]Lecture timestamp: 01:49:09.

6.8 Environmental Records and Partial Measurement

A deliberately built detector is not the only recorder. An atom near a slit, residual gas, a thermal surface, or any other degree of freedom can become correlated with the path. If its final states are $|E_1\rangle$ and $|E_2\rangle$, the path state has the form

$$|\Psi\rangle = \sum_n \left(\psi_n |n\rangle |E_1\rangle + \phi_n |n\rangle |E_2\rangle\right). \tag{6.55}$$

The interference terms are now multiplied by the overlap $\langle E_1|E_2\rangle$. Orthogonal records make the paths perfectly distinguishable and erase interference completely. Identical records leave $\langle E_1|E_2\rangle = 1$ and preserve it. Intermediate overlap gives partial visibility.

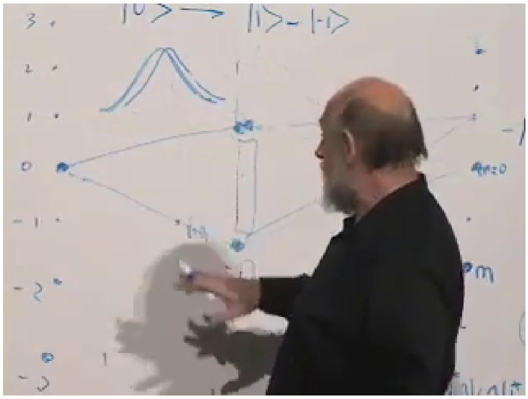

Figure 6.18: Overlapping and separated environmental wave packets illustrate partial and complete path records.

Lecture frame: 01:53:40.

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[a] **Question from the audience.** Would an ordinary atom near the slit become entangled with the passing electron even if nobody deliberately observes it?

Response. Yes. Knowledge is not the operative verb; physical correlation is. If the atom or surrounding gas is disturbed enough that its later state distinguishes the path, it records which-way information and suppresses interference. This is why such experiments require isolation, low temperatures, and often weakly interacting particles such as photons.

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[a] Lecture timestamp: 01:50:48.

Cooling places environmental degrees of freedom near their ground states and reduces accidental records. An energy gap helps as well: if the incident particle cannot supply the minimum excitation energy, the environment may be unable to retain path information. Conversely, charged particles can disturb nearby matter through their electric fields, making electron interference especially demanding.

[a] **Question from the audience.** Can an interference pattern really be obtained with single electrons?

Response. Each electron makes only one localized hit. Send the particles through one at a time, with enough separation that they cannot interact, and collect many repetitions. The accumulated frequency distribution displays the interference pattern; one event alone is only a blip.

[a] Lecture timestamp: 01:54:49.

[a] **Question from the audience.** Does measurement destroy entanglement or create it?

Response. The which-path interaction creates entanglement between system and apparatus, and that entanglement destroys interference between the alternatives. A later measurement can correlate the apparatus with yet another system and may condition the remaining state, so one must always say which two subsystems and which interference are being discussed.

[a] Lecture timestamp: 01:55:59.

[a] **Question from the audience.** Can the interaction be weakened so that the apparatus obtains only partial path information?

Response. Yes. A weak interaction may flip the detector only with some amplitude, leaving nonorthogonal record states. Then the paths are only partly distinguishable and the interference is only partly reduced. Stronger measurement means greater distinguishability, stronger entanglement with the record, and lower fringe visibility.

[a] Lecture timestamp: 01:57:44.

The collapse relevant here is therefore concrete: it is the loss of the cross terms when path information has been deposited in another system. With no record, amplitudes add. With perfectly distinguishable records, their inner products vanish and probabilities add. Between those extremes, measurement strength and interference visibility vary continuously.

CHAPTER 7

MEASUREMENT, DENSITY MATRICES, AND ENTROPY

The two-slit experiment contains the measurement problem in its smallest useful form. When two alternatives remain coherent, their amplitudes interfere. When another system acquires a record of the alternative taken, the object and the recorder become entangled, and the interference changes. We shall make that mechanism exact before introducing the trace, the density matrix, and entropy—the language needed to quantify entanglement.

7.1 Rebuilding the Two-Slit Experiment

Consider a discrete model with a source at site 0, two openings A and B, and a screen whose possible arrival sites are represented by orthonormal states $\{|m\rangle\}$. The particle may be an electron, a photon, or any other quantum object. We call it an electron, and we ignore its own spin so that position is the only degree of freedom under discussion. The rest of the barrier is closed: an electron reaching any site other than A or B is reflected.

The structural principle is linearity. If two states evolve as

$$|A\rangle \longrightarrow |A'\rangle, \qquad |B\rangle \longrightarrow |B'\rangle, \tag{7.1}$$

then every superposition evolves with the same coefficients:

$$\alpha|A\rangle + \beta|B\rangle \longrightarrow \alpha|A'\rangle + \beta|B'\rangle. \tag{7.2}$$

In a symmetric arrangement the source feeds the two openings with equal amplitude. Suppressing the common normalization, as on the board,

$$|0\rangle \longrightarrow |A\rangle + |B\rangle. \tag{7.3}$$

The normalized state would be $(|A\rangle + |B\rangle)/\sqrt{2}$; the omitted factor plays no role in the interference argument.

Now study each route separately. Propagation from the openings to the screen has the form

$$|A\rangle \longrightarrow \sum_m \psi_m |m\rangle, \tag{7.4}$$

$$|B\rangle \longrightarrow \sum_m \phi_m |m\rangle. \tag{7.5}$$

The complex number ψ_m is the amplitude to arrive at screen site m after starting at A, and ϕ_m is the corresponding amplitude after starting at B.

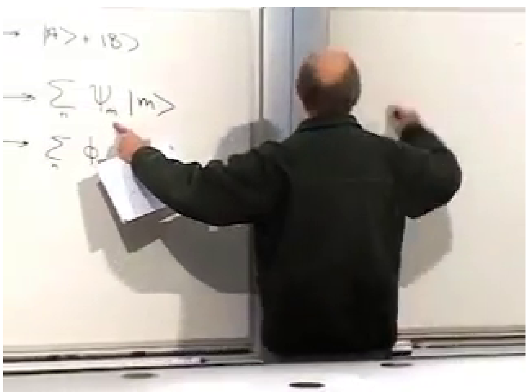

Figure 7.1: The separate A- and B-route expansions in the screen basis.

Lecture frame: 00:05:39.

If B is closed, the probability at m is

$$P_A(m) = |\psi_m|^2. \tag{7.6}$$

If A is closed,

$$P_B(m) = |\phi_m|^2. \tag{7.7}$$

Classical probability would suggest adding these two answers when both openings are available. Quantum mechanics instead adds the amplitudes first:

$$|A\rangle + |B\rangle \longrightarrow \sum_m (\psi_m + \phi_m)|m\rangle. \tag{7.8}$$

At a fixed screen site,

$$P(m) = |\psi_m + \phi_m|^2 \tag{7.9}$$
$$= |\psi_m|^2 + |\phi_m|^2 + \psi_m^* \phi_m + \phi_m^* \psi_m. \tag{7.10}$$

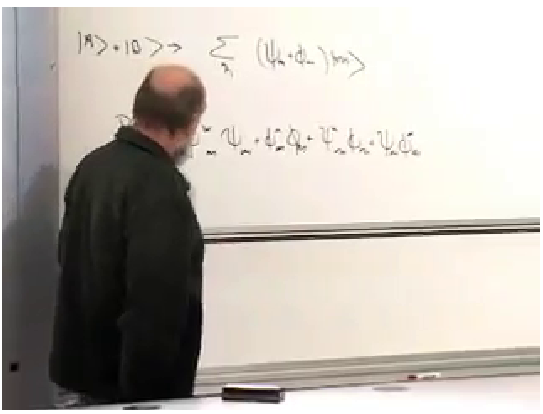

Figure 7.2: The four-term screen probability. The last two terms are the interference terms.

Lecture frame: 00:09:21.

At a symmetry point, $\psi_m = \phi_m$, and the two alternatives reinforce one another. At a point where their phases differ by π, $\psi_m = -\phi_m$, the probability vanishes. Either route alone can reach that point, yet opening both routes can prevent the electron from arriving there. The question is now unavoidable: what changes when something records which opening was used?

7.2 A Detector Becomes Entangled

Introduce an auxiliary two-state system with states $|u\rangle$ and $|d\rangle$. It is not the electron's spin in this model; it is a minimal external apparatus. Initially the detector is untriggered:

$$|0, d\rangle. \tag{7.11}$$

Arrange the interaction so that passage through A flips the detector, whereas passage through B leaves it unchanged:

$$|0, d\rangle \longrightarrow |A, u\rangle + |B, d\rangle. \tag{7.12}$$

After the electron leaves the openings, it no longer interacts with the auxiliary spin. The two branches therefore propagate as

$$|A, u\rangle \longrightarrow \sum_m \psi_m |m, u\rangle, \tag{7.13}$$

$$|B, d\rangle \longrightarrow \sum_m \phi_m |m, d\rangle, \tag{7.14}$$

and the final state is

$$|\Psi\rangle = \sum_m \left(\psi_m |m, u\rangle + \phi_m |m, d\rangle\right). \tag{7.15}$$

The route and the detector record are now entangled. Reading the detector would reveal which opening was used.

---

[a] **Question from the audience.** Has the wave packet already collapsed when the spin flips?

Response. For the electron alone, one may use collapse language: the path alternatives no longer interfere. For the complete electron–detector system, however, neither branch has been discarded. The state is still $|A, u\rangle + |B, d\rangle$. Including the apparatus replaces the statement *the electron collapsed* by the precise statement that system–apparatus entanglement was established.

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[a] Lecture timestamp: 00:13:24.

The auxiliary spin is only a convenient pointer. A fired or unfired device, an emitted or absent photon, or any other pair of distinguishable records can play the same role.

[a] **Question from the audience.** Does passing through the metal plate itself always leave a path record?

Response. Not necessarily. What matters is whether the interaction leaves the plate in distinguishable final states. If the plate has an excitation gap and the electron lacks enough energy to excite it, no lasting internal mark need be left. Mere interaction is not enough; the two alternatives must become correlated with distinguishable records.

[a]Lecture timestamp: 00:15:17.

[a] **Question from the audience.** Could the plate's recoil reveal whether the electron passed through the upper or lower opening?

Response. Only if the opposite momentum kicks produce sufficiently distinguishable plate states. A well-localized plate already has a broad momentum wave function. Small upward and downward shifts can overlap almost completely, so the recoil states are not orthogonal and do not provide an unambiguous route record. This is where the uncertainty principle protects interference.

[a]Lecture timestamp: 00:16:29.

A partial record is possible. If the detector is kicked only slightly, the two final record states need not be orthogonal. Interference is then reduced rather than eliminated. For the formal calculation we first take the limiting case of a complete measurement:

$$\langle u|d\rangle = 0. \tag{7.16}$$

7.3 The Projector Calculation

We now calculate the probability that the electron arrives at a fixed screen site m, without asking for the detector outcome. In the combined Hilbert space, the property *electron at m* is a two-dimensional subspace spanned by

$$|m, u\rangle, \qquad |m, d\rangle. \tag{7.17}$$

The corresponding projector is

$$\Pi_m = |m, u\rangle\langle m, u| + |m, d\rangle\langle m, d|. \tag{7.18}$$

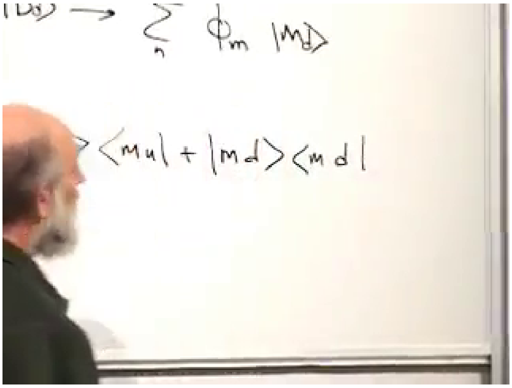

Figure 7.3: The projector onto a fixed screen position, summed over the two orthogonal detector records.

Lecture frame: 00:22:04.

The probability postulate gives

$$P(m) = \langle \Psi | \Pi_m | \Psi \rangle. \tag{7.19}$$

Writing out the sums,

$$\begin{aligned} P(m) = \sum_{n,n'} &\left(\psi_n^* \langle n, u| + \phi_n^* \langle n, d|\right) \\ &\times \Pi_m \\ &\times \left(\psi_{n'} |n', u\rangle + \phi_{n'} |n', d\rangle\right). \end{aligned} \tag{7.20}$$

The position inner products require $n = m = n'$. The detector inner products then decide which terms remain:

$$\langle u|u\rangle = 1, \qquad \langle d|d\rangle = 1, \qquad \langle u|d\rangle = \langle d|u\rangle = 0. \tag{7.21}$$

Consequently,

$$P(m) = |\psi_m|^2 + |\phi_m|^2. \tag{7.22}$$

The cross terms have not been removed by decree. They vanish because the two route amplitudes now multiply orthogonal vectors of the enlarged state space. To interfere, alternatives must agree not only at the screen endpoint but in every degree of freedom that carries a record.

7.4 What Counts as a Measurement?

The clean spin model supplies a general rule: measurement occurs when an interaction establishes a correlation between alternatives of the system and distinguishable states of an apparatus. The following variations test that rule.

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[a] **Question from the audience.** What if the spin is placed symmetrically and flips for passage through either opening?

Response. Then both branches carry the same record, $|A,u\rangle + |B,u\rangle = (|A\rangle + |B\rangle) \otimes |u\rangle$. The detector state factors out and reveals no route information, so the interference pattern remains.

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[a]Lecture timestamp: 00:26:53.

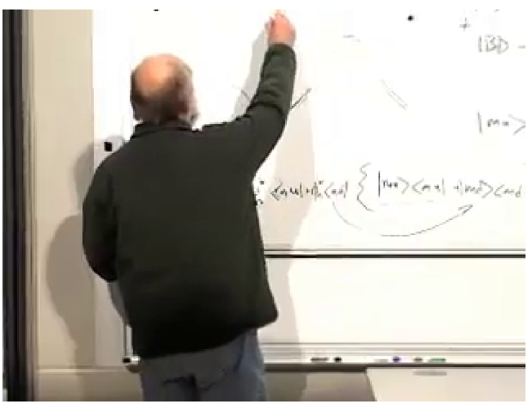

Figure 7.4: The board variation in which a symmetrically placed detector responds identically to both routes.

Lecture frame: 00:27:14.

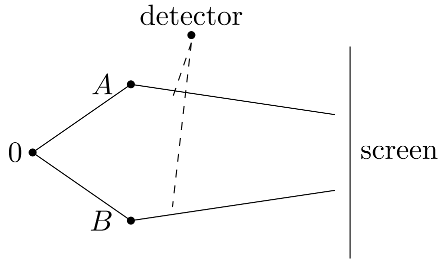

Figure 7.5: A clean reconstruction of the symmetric detector placement: both routes produce the same detector state.

[a] **Question from the audience.** Must someone actually read the auxiliary spin before interference disappears?

Response. No. The electron has already correlated its route with the spin. One may reset the detector, repeat the experiment one electron at a time, and inspect only the distribution of screen hits. The fringes are absent even if nobody reads the spin; the physical record, not human awareness of it, matters.

[a]Lecture timestamp: 00:27:55.

[a] **Question from the audience.** Would a sufficiently light recoiling plate destroy interference?

Response. Yes, if the recoil shifts the plate into orthogonal momentum states for the two routes. If the shifted wave packets still overlap, the destruction is only partial. Mass by itself is not the criterion; distinguishability of the final plate states is.

[a]Lecture timestamp: 00:29:36.

Radiation provides another imperfect recorder. An accelerated charged particle may emit a photon. Radiation emitted near A need not be the same state as radiation emitted near B, so the electromagnetic field can retain route information. For a slowly deflected electron the emission probability may be small; the lecture uses an illustrative estimate of roughly one event in a hundred. Stronger acceleration makes radiation and therefore loss of contrast more likely. The important point is qualitative: even an unobserved photon can carry away coherence.

[a] **Question from the audience.** How does one infer the interference probability in a real experiment with imperfect contrast?

Response. One repeats the experiment many times and estimates the screen distribution from relative frequencies. Finite samples and environmental records leave residual counts in nominally dark regions. The inferred probability distribution improves with the number of trials, while decohering processes reduce the physical fringe contrast.

[a]Lecture timestamp: 00:32:00.

[a] **Question from the audience.** Does any possibility of determining the route destroy all interference?

Response. Complete distinguishability destroys it completely. Partial information produces partial loss. A detector that rotates only slightly leaves two nonorthogonal record states, so one route becomes more likely than the other when the detector is read, but not certain; correspondingly, the fringes are blurred rather than erased.

[a]Lecture timestamp: 00:33:13.

Let $|E_A\rangle$ and $|E_B\rangle$ denote arbitrary record states. The combined state at the screen can be written

$$|\Psi\rangle = \sum_m \left(\psi_m |m\rangle |E_A\rangle + \phi_m |m\rangle |E_B\rangle\right). \tag{7.23}$$

Ignoring the record system gives

$$\begin{aligned} P(m) &= |\psi_m|^2 + |\phi_m|^2 \quad &(7.24)\\ &\quad + \psi_m^* \phi_m \langle E_A|E_B\rangle + \phi_m^* \psi_m \langle E_B|E_A\rangle. \quad &(7.25) \end{aligned}$$

Thus the overlap $\langle E_A|E_B\rangle$ is the coherence factor. Identical records preserve full interference; orthogonal records remove it; intermediate overlap gives intermediate visibility.

[a] **Question from the audience.** If photons are emitted only occasionally, what happens when one keeps only the photon-emission events?

Response. That selected subensemble generally has poorer interference because a photon carries route information. The loss need not be complete: photon wave packets emitted at the two openings may themselves overlap, so even the radiative records may fail to be perfectly orthogonal.

[a]Lecture timestamp: 00:34:12.

Environmental bombardment scales this mechanism up. Air molecules, ambient photons, and surrounding matter continuously acquire records of macroscopic positions. A person or the Moon does not remain visibly spread into a coherent position superposition: mass slows free spreading, and continual environmental entanglement suppresses observable interference between macroscopically separated alternatives.

[a] **Question from the audience.** Is this already a theory of measurement, rather than merely another use of the phrase *wave-packet collapse*?

Response. It is the mechanism in a deliberately exact example. In an electron-only description, collapse means discarding the off-diagonal $\psi^*\phi$ terms. In the composite description no ad hoc deletion is needed: entanglement with orthogonal apparatus states makes those terms unavailable to measurements on the electron alone. More complicated examples should be traced back to this structure.

[a]Lecture timestamp: 00:36:02.

[a] **Question from the audience.** Would a gravitational field that bends the route naturally change the radiation argument or the interference rule?

Response. The detailed radiation may change, and quantum mechanics does not assign a single sharp trajectory before a path measurement. The general conclusion does not change: only a degree of freedom that retains distinguishable route information suppresses interference. A concrete apparatus must be specified before a more detailed answer is meaningful.

[a] Lecture timestamp: 00:38:04.

[a] **Question from the audience.** Is this interference different from the interference of classical electromagnetic waves?

Response. For many photons, the observed pattern agrees with ordinary wave interference of the electromagnetic field. The single-particle experiment shows the deeper quantum statement: photons and electrons arrive as individual events, while their quantum amplitudes interfere. For electrons the relevant wave is the electron wave function, not an electromagnetic wave.

[a] Lecture timestamp: 00:39:44.

[a] **Question from the audience.** What happens to portions of the initial wave aimed between A and B?

Response. In this deliberately discrete model the rest of the barrier is blocked. Those portions are reflected and do not contribute to the transmitted screen amplitude. The model isolates the two transmitted alternatives without claiming that a real beam has only two momentum components.

[a] Lecture timestamp: 00:40:45.

7.5 The Cat, the Screen, and the Movable Cut

The same mathematics applies to Schrödinger's cat. Let the gun have states $|U\rangle$ and $|F\rangle$ for unfired and fired, and let the cat have states $|L\rangle$ and $|D\rangle$ for live and dead. Starting from a live cat and an unfired gun, the interaction may produce

$$|\Psi_{\text{cat}}\rangle = \alpha|L,U\rangle + \beta|D,F\rangle. \tag{7.26}$$

The cat alone is not in the coherent state $\alpha|L\rangle + \beta|D\rangle$. It is entangled with a gun that has recorded the alternative. Looking at the gun can reveal the cat's branch, and the orthogonal gun records remove ordinary live–dead interference in the cat subsystem.

---

[a] **Question from the audience.** Is the cat itself in a superposition of alive and dead?

Response. Not in the simple isolated sense. The composite cat–gun system is in a superposition of correlated alternatives, live with unfired and dead with fired. The cat subsystem is entangled with a record-bearing gun, exactly as the electron route is entangled with the auxiliary spin.

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[a]Lecture timestamp: 00:41:18.

The final screen also has to be treated as an apparatus. An electron arriving at m may blacken a photographic grain, make a scintillator emit a photon, or flip one detector in a row of local two-state devices. The mark becomes correlated with the arrival position.

---

[a] **Question from the audience.** Why may we speak of probabilities at the final screen rather than carry only state vectors forever?

Response. The probability postulate assigns $\langle \Pi_m \rangle$ to the property *arrival at m*. To complete the physical experiment, the screen then leaves a position record. The projector gives the probability; blackening, scintillation, or a local detector flip realizes the measurement whose statistics are being collected.

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[a]Lecture timestamp: 00:43:38.

[a] **Question from the audience.** Why discuss a screen measurement but not a probability for passage through A or B?

Response. One may assign and test path probabilities only by installing a path recorder. Without such a record the alternatives remain coherent and must be added as amplitudes. At the screen, a position-sensitive apparatus is assumed. The difference is not a privileged time but whether a corresponding measurement interaction has been arranged.

[a]Lecture timestamp: 00:45:00.

The boundary between system and apparatus can be moved. Include an observer with record states $|O_u\rangle$ and $|O_d\rangle$; then

$$|A,u\rangle + |B,d\rangle \longrightarrow |A,u,O_u\rangle + |B,d,O_d\rangle. \tag{7.27}$$

Another observer may in turn become correlated with the first. Smoke in the cat's box, the gun, the cat, Schrödinger, and Wigner can all be included in one longer entanglement chain. Wherever the cut is drawn, the correlations must remain consistent: if Wigner later asks Schrödinger what he saw, their records agree.

[a] **Question from the audience.** Where exactly should the system end and the measuring observer begin?

Response. Quantum mechanics does not select a unique convenient cut. One may collapse the electron-only description, include the detector and collapse a larger description, or include an observer and enlarge the entangled state again. The physical requirement is consistency of all accessible probabilities and records, not a unique bookkeeping boundary.

[a]Lecture timestamp: 00:47:24.

[a] **Question from the audience.** Does the uncertainty principle enter because measuring the path disturbs the event?

Response. Yes. The recoil example makes the connection explicit: localization of the plate produces momentum uncertainty, which can prevent small path-dependent kicks from becoming distinguishable. More generally, a quantum measurement is not complete until system–apparatus entanglement has changed the state, and that change can alter later interference.

[a] Lecture timestamp: 00:49:47.

This marks a sharp contrast with the classical ideal. Classically one imagines making a measurement arbitrarily gentle, extracting information with negligible disturbance. Here extracting path information requires establishing a correlation with an apparatus. That is a genuine change of the joint state, and it changes the answer to a later interference experiment.

[a] **Question from the audience.** What happens if the separation between A and B becomes very large?

Response. There is no universal new rule. In this source-and-barrier geometry the amplitudes to reach very distant openings may simply become small, so most particles hit the blocked region. Apart from such apparatus-dependent changes in ψ_m and ϕ_m, separation alone does not decide whether coherence survives.

[a] Lecture timestamp: 00:52:06.

[a] **Question from the audience.** Is the linearity of quantum evolution fundamentally exact, and why does nature use it?

Response. Linearity is an empirical foundation of quantum mechanics. Small nonlinear modifications are notoriously difficult to make logically coherent and tend to spoil the structure. The deeper reason that physical propositions follow the logic of subspaces rather than classical sets is not supplied here; the lecture treats it as a fact of nature.

[a] Lecture timestamp: 00:53:04.

[a] **Question from the audience.** Could a later observation reveal the earlier path and thereby change whether interference occurred?

Response. If a later observation can reveal the path, some physical degree of freedom must have preserved a path-dependent record. That record already affects coherence; nobody has to read it. Causality must still be respected, and an abstract delayed-choice claim cannot be settled without specifying the complete spacetime arrangement and apparatus.

[a]Lecture timestamp: 00:54:53.

[a] **Question from the audience.** What if a bubble chamber is placed immediately after the openings?

Response. A track that distinguishes the two emerging routes is an unambiguous mark, so it destroys their interference. A thin path-recording layer is enough. If a downstream device does not retain enough information to infer the earlier opening, then it does not automatically serve as a which-path detector.

[a]Lecture timestamp: 00:56:47.

[a] **Question from the audience.** If the world is already highly entangled, how can an interference experiment ever begin with disentangled systems?

Response. The preparation must isolate, reset, or condition the relevant degrees of freedom well enough that the alternatives start with the same environmental record. A prior bubble-chamber track cannot simply be ignored if it remains correlated with the path. Interference requires a product-like initial relation between the controlled system and the relevant recorder.

[a]Lecture timestamp: 00:59:29.

[a] **Question from the audience.** Where does practical disentanglement come from after a measurement?

Response. Condition on a definite observed outcome. If the spin is found up, the observer retains the branch $|A, u\rangle$ and discards the alternative from the working description. That conditional product state can be used as a new starting point. The full observer-inclusive state may remain entangled, but relative to the recorded outcome the selected preparation is definite.

[a] Lecture timestamp: 01:01:04.

This nested hierarchy does not resolve every interpretive question, but it makes the operational consistency visible. System, detector, observer, and observer of observer may all be included. No matter where the description is cut, correctly correlated records do not later contradict one another.

7.6 From Entanglement to Entropy

The discussion has gone farther into measurement than planned, but it has exposed the next question: how can the degree of entanglement be measured? Two spins in the product state

$$|u\rangle \otimes |u\rangle \tag{7.28}$$

are not entangled. Each subsystem has its own definite state, and observing one supplies no new information about the other. Entangled states do not factor this way. A standard quantitative diagnostic will be entanglement entropy, but we first need the ordinary notion of entropy and then its quantum form.

7.7 Classical Entropy as Missing Information

Begin with a classical system having a finite set of N possible states. Entropy here is a property of the system together with our knowledge of it. Suppose all we know is that the system occupies one of n equally probable states. Then

$$S = \log n. \tag{7.29}$$

If $n = 1$, the state is known and $S = 0$. If $n = N$, nothing is known beyond the state space itself and the entropy is maximal, $S_{\max} = \log N$.

The logarithm counts independent yes–no choices additively. For M classical spins there are 2^M configurations, so

$$\log(2^M) = M \log 2. \tag{7.30}$$

Up to the choice of logarithm base, entropy therefore counts the amount of binary information missing from the description.

For a general probability distribution p_i, the appropriate definition is

$$S = -\sum_i p_i \log p_i. \tag{7.31}$$

---

[a] **Question from the audience.** Why is there a minus sign in the entropy formula?

Response. Every nonzero probability satisfies $0 < p_i \leq 1$, so $\log p_i \leq 0$. The minus sign makes entropy nonnegative. At $p_i = 0$, the contribution is defined by the limit $p_i \log p_i \to 0$.

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[a] Lecture timestamp: 01:14:17.

For the equal-probability distribution

$$p_i = \begin{cases} 1/n, & i \in \mathcal{N}, \\ 0, & i \notin \mathcal{N}, \end{cases} \qquad |\mathcal{N}| = n, \tag{7.32}$$

the general formula becomes

$$S = -n\left(\frac{1}{n}\right)\log\left(\frac{1}{n}\right) \tag{7.33}$$

$$= \log n. \tag{7.34}$$

It therefore reproduces the elementary counting definition. A distribution concentrated with certainty on one state has $S = 0$.

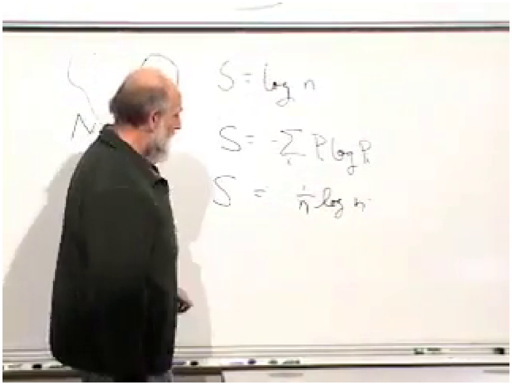

Figure 7.6: The board collects $S = \log n$, the Shannon formula, and the equal-probability check.

Lecture frame: 01:16:08.

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[a] **Question from the audience.** What is the largest entropy a system with N states can have, and what is known in that case?

Response. The maximum is $\log N$, attained by the uniform distribution $p_i = 1/N$. One still knows what the system is and therefore knows its state space; what is unknown is which of those N states it occupies.

---

[a]Lecture timestamp: 01:17:35.

The difference

$$I = S_{\max} - S \tag{7.35}$$

is the available information in this counting sense. Entropy and information are opposite sides of the same bookkeeping: entropy records information that could distinguish the state but is not available to us.

7.8 Trace: the Quantum Replacement for a Sum

For an operator M and any complete orthonormal basis $\{|i\rangle\}$, define

$$\operatorname{Tr} M = \sum_i \langle i|M|i\rangle. \tag{7.36}$$

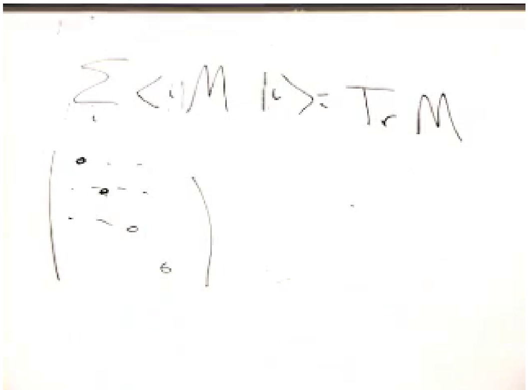

Figure 7.7: The trace written as the sum of diagonal matrix elements.

Lecture frame: 01:21:20.

Although the definition is written using a basis, the result is basis independent. For a Hermitian operator one may choose an eigenbasis, in which

$$\mathrm{Tr}\, M = \sum_i \lambda_i. \tag{7.37}$$

The eigenvalues, and hence their sum, do not depend on the coordinates used to represent the operator.

---

[a] **Question from the audience.** Does putting a matrix in diagonal form imply that it has an inverse?

Response. No. Diagonal form displays the eigenvalues; invertibility is a separate condition. A diagonal matrix is invertible precisely when none of its diagonal eigenvalues is zero. Hermitian matrices are diagonalizable even when some eigenvalues vanish.

---

[a]Lecture timestamp: 01:23:04.

After the break, return briefly to classical probability. A classical observable is a function f_i of the configuration, and its average is

$$\langle f\rangle_{\mathrm{cl}} = \sum_i p_i f_i. \tag{7.38}$$

In the quantum formulas, the trace will replace the sum over alternatives, an operator will replace the classical function, and a density matrix will replace the probability distribution.

7.9 Statistical Mixtures and the Density Matrix

Suppose someone prepares a system in one state of an orthonormal basis but does not tell us which state was chosen. They provide only preparation probabilities—for example, a spin is prepared up with probability 3/4 and down with probability 1/4. This is not the coherent superposition

$$\sqrt{\frac{3}{4}}|u\rangle + \sqrt{\frac{1}{4}}|d\rangle, \tag{7.39}$$

because no relative phase has been prepared. It is a statistical mixture of definite states. Quantum probabilities remain even when a pure state is known; here there is an additional uncertainty about which pure state was prepared.

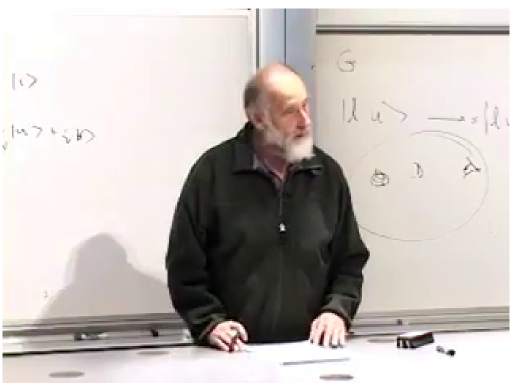

Figure 7.8: The conceptual transition from a specified ket to an ensemble description.

Lecture frame: 01:29:43.

If state $|i\rangle$ is prepared with probability ρ_i, the density operator is

$$\rho = \sum_i \rho_i |i\rangle\langle i|. \tag{7.40}$$

In the preparation basis,

$$\rho = \begin{pmatrix} \rho_1 & 0 & 0 & \cdots \\ 0 & \rho_2 & 0 & \cdots \\ 0 & 0 & \rho_3 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{7.41}$$

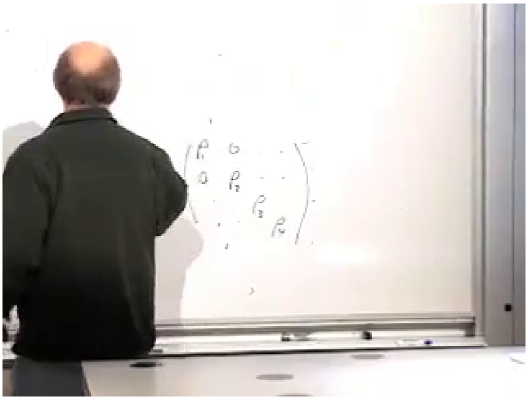

Figure 7.9: The diagonal density matrix and its normalization condition.

Lecture frame: 01:32:12.

Its eigenvalues are real and nonnegative, and normalization of probabilities becomes

$$\operatorname{Tr} \rho = \sum_i \rho_i = 1. \tag{7.42}$$

Let F be a Hermitian observable. The ensemble expectation value is

$$\langle F \rangle = \operatorname{Tr}(F\rho). \tag{7.43}$$

To recover the ordinary weighted average, evaluate the trace in the basis where ρ is diagonal and insert the resolution of

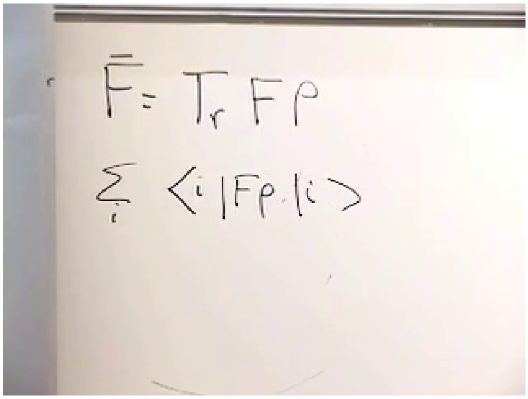

Figure 7.10: The trace expectation value $\langle F \rangle = \mathrm{Tr}(F\rho)$.

Lecture frame: 01:35:05.

identity:

$$\mathrm{Tr}(F\rho) = \sum_i \langle i|F\rho|i\rangle \tag{7.44}$$

$$= \sum_{i,j} \langle i|F|j\rangle\langle j|\rho|i\rangle \tag{7.45}$$

$$= \sum_i \rho_i \langle i|F|i\rangle. \tag{7.46}$$

Thus each pure-state expectation value is weighted by the probability that the corresponding state was prepared.

The trace has the cyclic property

$$\mathrm{Tr}(AB) = \mathrm{Tr}(BA), \tag{7.47}$$

even if A and B do not commute. Hence $\mathrm{Tr}(F\rho) = \mathrm{Tr}(\rho F)$.

7.10 Quantum Entropy and the Deferred Payoff

The entropy of a density matrix is

$$S(\rho) = -\,\mathrm{Tr}(\rho \log \rho). \tag{7.48}$$

For a Hermitian positive operator, functions such as $\log\rho$ are defined through its eigenvalues. If

$$\rho = \sum_i \rho_i |i\rangle\langle i|, \tag{7.49}$$

then

$$\log\rho = \sum_{\rho_i>0} (\log\rho_i)|i\rangle\langle i|, \tag{7.50}$$

and therefore

$$S(\rho) = -\sum_i \rho_i \log\rho_i. \tag{7.51}$$

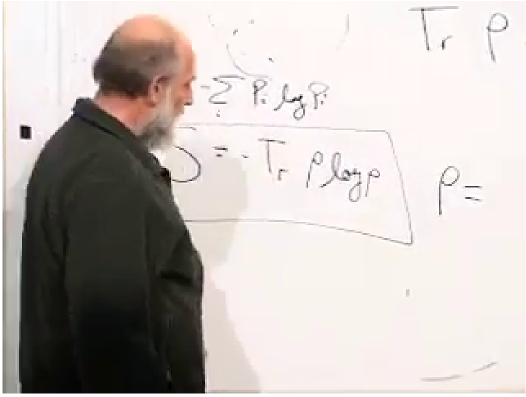

Figure 7.11: The classical entropy formula recast as $S = -\operatorname{Tr}(\rho\log\rho)$.

Lecture frame: 01:41:25.

A known pure state has density operator

$$\rho_\psi = |\psi\rangle\langle\psi|, \tag{7.52}$$

with eigenvalues $1, 0, \ldots$, so $S(\rho_\psi) = 0$. This does not mean that every measurement on $|\psi\rangle$ has a certain outcome. It means there is no additional statistical ignorance about the prepared state. At the opposite extreme,

$$\rho = \frac{I}{d} \qquad \Longrightarrow \qquad S(\rho) = \log d, \tag{7.53}$$

where $d = \dim \mathcal{H}$. The maximally mixed state assigns equal weight to every orthogonal direction.

The connection to entanglement is now close but not yet worked out. Suppose a composite system AB is in a joint state and all experiments are confined to A. There must be an object that reproduces every expectation value obtainable on A while B is ignored. That object is a density matrix for A. If the joint state factorizes, the density matrix of A is pure and has zero entropy. If A and B are entangled, the density matrix of A is generally mixed and has nonzero entropy. Constructing that reduced density matrix and evaluating its entropy is the task reserved for the next lecture.

CHAPTER 8

REDUCED STATES, ENTANGLEMENT ENTROPY, AND UNITARY EVOLUTION

A density matrix first appears as the natural description of incomplete preparation: a quantum system has been prepared, but we do not know exactly how. Entanglement gives the same mathematical object a deeper role. Even when the state of a composite system is known completely, a subsystem generally has no state vector of its own. Its reduced density matrix contains everything that can be predicted locally. We shall derive that statement in components, test it on two spin states, and use entropy to distinguish product states from entangled ones. Then we shall turn to a second question: how a quantum state changes with time. Preservation of inner products will lead to unitarity, a Hermitian generator, and the Schrödinger equation.

8.1 Density Matrices and Incomplete Preparation

Suppose an electron has been placed in a strong magnetic field and allowed to align with it. Someone else chose the field direction and prepared the electron, but did not tell us the direction. The spin may therefore be in a perfectly definite pure state even though we do not know which state it is. We may know a probability distribution over possible preparations, or we may know nothing at all. A density matrix ρ is the quantum description of precisely this kind of incomplete information.

The classical analogue is a probability density on phase space or a probability distribution over discrete configurations. The quantum object is an operator with three essential properties:

$$\rho^\dagger = \rho, \qquad \mathrm{Tr}\,\rho = 1, \qquad \lambda_i \geq 0, \tag{8.1}$$

where the λ_i are the eigenvalues of ρ. Hermiticity makes the eigenvalues real and supplies an orthonormal eigenbasis. Positivity makes them admissible probabilities, and normalization gives

$$\sum_{i=1}^{n} \lambda_i = 1. \tag{8.2}$$

In the eigenbasis of ρ, λ_i is the probability assigned to the i-th eigenstate.

Complete ignorance is the case in which no direction in Hilbert space is preferred:

$$\rho_{\max} = \frac{1}{n} I. \tag{8.3}$$

Every eigenvalue is $1/n$. At the opposite extreme, complete knowledge is represented by a normalized state $|\psi\rangle$, with

$$\rho_\psi = |\psi\rangle \langle\psi| \,. \tag{8.4}$$

This is the projector onto the one-dimensional subspace spanned by $|\psi\rangle$.

---

[a] **Question from the audience.** Can a preparation be described as a fifty-fifty choice between two states that are not orthogonal, for example two spin directions separated by 45°?

Response. Such a preparation procedure is possible, but those two weights are not generally the eigenvalues of the resulting density matrix. One first forms the weighted sum of projectors and then diagonalizes it. The clean probability interpretation of the eigenvalues belongs to the orthonormal eigenbasis of ρ, not necessarily to the nonorthogonal states named by the preparer.

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[a] Lecture timestamp: 00:06:47.

[a] **Question from the audience.** What distinguishes a pure state from a mixed state?

Response. A pure-state density matrix has one nonzero eigenvalue, equal to 1. A mixed state has more than one nonzero eigenvalue. Older literature used the German word *Gemisch*; the modern term is simply *mixed state*.

[a]Lecture timestamp: 00:07:47.

Let us verify the spectrum of the pure-state projector. Normalization gives

$$\rho_\psi \left|\psi\right\rangle = \left|\psi\right\rangle \left\langle\psi \,|\, \psi\right\rangle = \left|\psi\right\rangle, \tag{8.5}$$

so $|\psi\rangle$ is an eigenvector with eigenvalue 1. If $|\phi\rangle$ is orthogonal to $|\psi\rangle$, then

$$\rho_\psi \left|\phi\right\rangle = \left|\psi\right\rangle \left\langle\psi \,|\, \phi\right\rangle = 0. \tag{8.6}$$

Every direction orthogonal to $|\psi\rangle$ therefore has eigenvalue zero.

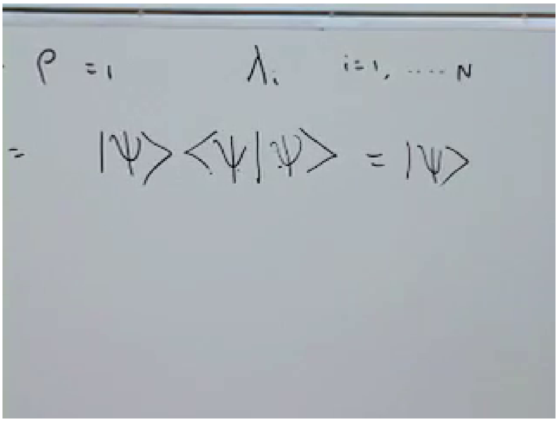

Figure 8.1: The pure-state density matrix acts as a projector: $\rho\,|\psi\rangle = |\psi\rangle$.

Lecture frame: 00:09:32.

8.2 Expectation Values from the Trace

The operational rule for a density matrix is

$$\overline{M} = \mathrm{Tr}(\rho M), \tag{8.7}$$

where M is an observable. For two factors the cyclic property of the trace gives

$$\mathrm{Tr}(\rho M) = \mathrm{Tr}(M\rho), \tag{8.8}$$

whether or not ρ and M commute.

In any orthonormal basis $\{|i\rangle\}$,

$$\mathrm{Tr}\, X = \sum_i \langle i| X |i\rangle . \tag{8.9}$$

The numerical value does not depend on the basis. For a pure state,

$$\begin{align} \overline{M} &= \mathrm{Tr}(|\psi\rangle \langle\psi| M) \tag{8.10} \\ &= \sum_i \langle i \mid \psi\rangle \langle\psi| M |i\rangle \tag{8.11} \\ &= \langle\psi| M \left(\sum_i |i\rangle \langle i| \right) |\psi\rangle \tag{8.12} \\ &= \langle\psi| M |\psi\rangle . \tag{8.13} \end{align}$$

The resolution of the identity,

$$\sum_i |i\rangle \langle i| = I, \tag{8.14}$$

turns the density-matrix rule back into the familiar pure-state expectation value.

[a] **Question from the audience.** Should the quantity on the left be $\overline{M}$, the average of M?

Response. Yes. The trace is not the operator M itself; it is the scalar expectation value $\overline{M}$.

[a]Lecture timestamp: 00:14:38.

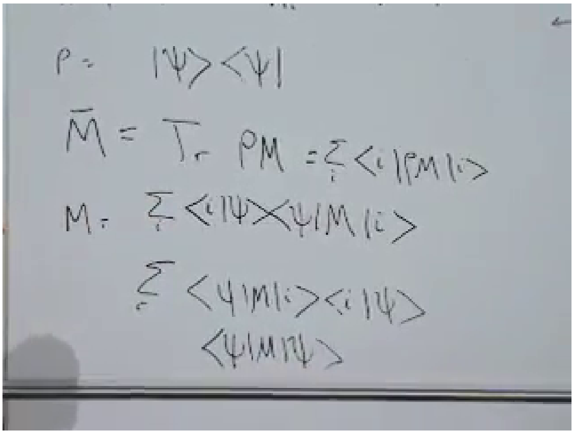

Figure 8.2: The trace calculation for a pure-state projector and the resolution of the identity.

Lecture frame: 00:14:25.

For a general density matrix, choose its eigenbasis:

$$\rho \left|j\right\rangle = \lambda_j \left|j\right\rangle . \tag{8.15}$$

Insert a complete set of states between ρ and M:

$$\overline{M} = \sum_i \left\langle i\right| \rho M \left|i\right\rangle \tag{8.16}$$

$$= \sum_{i,j} \left\langle i\right| \rho \left|j\right\rangle \left\langle j\right| M \left|i\right\rangle \tag{8.17}$$

$$= \sum_{i,j} \lambda_j \delta_{ij} \left\langle j\right| M \left|i\right\rangle \tag{8.18}$$

$$= \sum_j \lambda_j \left\langle j\right| M \left|j\right\rangle . \tag{8.19}$$

This is an ordinary weighted average: λ_j is the probability of the j-th eigenstate, and $\left\langle j\right| M \left|j\right\rangle$ is the expectation of M in that state.

[a] **Question from the audience.** Why do both labels in the last matrix element become j, and which index is still being summed?

Response. Because ρ is diagonal in this basis, $\langle i|\,\rho\,|j\rangle = \lambda_j \delta_{ij}$, so only terms with $i = j$ survive. The original double sum collapses to one sum, which may be called a sum over j. The name of a dummy summation index has no physical significance.

[a]Lecture timestamp: 00:18:12.

8.3 Entropy as Missing Information

For a classical probability distribution $\{p_i\}$, define

$$S = -\sum_i p_i \log p_i, \tag{8.20}$$

with $0 \log 0$ understood as its limiting value 0. If one probability equals 1 and all others vanish, then $S = 0$: the state is known. If all n possibilities are equally likely,

$$p_i = \frac{1}{n}, \qquad S = -n\left(\frac{1}{n}\log\frac{1}{n}\right) = \log n. \tag{8.21}$$

Thus entropy grows as the probability distribution spreads over more significant possibilities. Roughly, e^S is the effective number of comparably weighted states when natural logarithms are used.

For a density matrix the same definition is applied to its eigenvalues:

$$S(\rho) = -\sum_i \lambda_i \log \lambda_i = -\operatorname{Tr}(\rho \log \rho). \tag{8.22}$$

The first expression is the one needed immediately; the trace form emphasizes that the answer is independent of basis.

8.4 A Composite System and a Local Experiment

Consider two subsystems, A and B. A basis vector is written $|a, b\rangle$. If A has N basis states and B has n, the composite

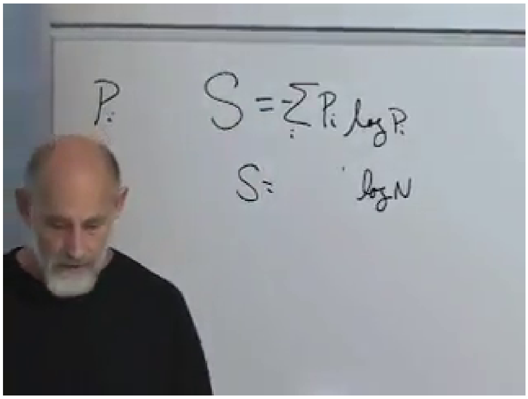

Figure 8.3: Entropy for a probability distribution and the equal-probability value $S = \log n$.

Lecture frame: 00:21:08.

system has Nn basis states. The two systems need not be alike: one could be a two-state spin and the other a three-state atom.

The most general pure state of the pair is

$$|\Psi\rangle = \sum_{a,b} \psi(a,b)\,|a,b\rangle\,, \qquad \sum_{a,b} |\psi(a,b)|^2 = 1. \tag{8.23}$$

Now let Alice measure an observable M_A that acts only on subsystem A. On the composite Hilbert space the operator is $M_A \otimes I_B$. Its matrix elements are

$$\langle a', b'|\, M_A \otimes I_B \,|a, b\rangle = \langle a'|\, M_A \,|a\rangle\; \delta_{b'b}. \tag{8.24}$$

The B label is passive.

Write the expectation value in full:

$$\langle M_A\rangle = \langle \Psi|\, M_A \otimes I_B \,|\Psi\rangle \tag{8.25}$$

$$= \sum_{a',b'} \sum_{a,b} \psi^*(a',b')\psi(a,b)\,\langle a'|\, M_A \,|a\rangle\, \delta_{b'b} \tag{8.26}$$

$$= \sum_{a',a,b} \psi^*(a',b)\psi(a,b)\,\langle a'|\, M_A \,|a\rangle\,. \tag{8.27}$$

[a] **Question from the audience.** Why are the labels on the bra primed while those on the ket are not?

Response. The bra and ket are independent sums before the matrix element is evaluated, so they require separate dummy labels. Primes distinguish the bra indices from the ket indices. The operator and the inner products then determine which of those labels must coincide.

[a]Lecture timestamp: 00:29:12.

Hold a and a' fixed and perform the sum over b. The result is a matrix on subsystem A:

$$\boxed{\rho_A(a,a') = \sum_b \psi(a,b)\psi^*(a',b)}. \tag{8.28}$$

This is the reduced density matrix of A, or equivalently the partial trace of the pure-state projector over B:

$$\rho_A = \mathrm{Tr}_B \left|\Psi\right\rangle \left\langle\Psi\right|. \tag{8.29}$$

The local expectation value becomes

$$\langle M_A \rangle = \sum_{a,a'} \rho_A(a,a') \left\langle a'\right| M_A \left|a\right\rangle = \mathrm{Tr}_A(\rho_A M_A). \tag{8.30}$$

[a] **Question from the audience.** Is the local observable simply the identity on subsystem B?

Response. Exactly. $M_A \otimes I_B$ may change the A coordinate, but it neither changes nor otherwise depends on b. That is why its B-space matrix element is $\delta_{b'b}$.

[a]Lecture timestamp: 00:35:39.

Every prediction for experiments confined to A is therefore contained in ρ_A. Bob need not measure his subsystem. Merely declining to use the B degrees of freedom forces Alice's description into density-matrix form.

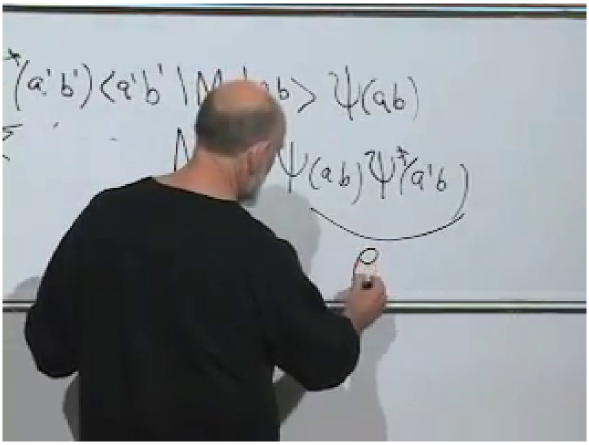

Figure 8.4: The B index is summed away, leaving a matrix in the two A indices.

Lecture frame: 00:33:50.

8.5 A Pure Whole Can Have Mixed Parts

The combined system is in the pure state $|\Psi\rangle$, yet ρ_A will generally have several nonzero eigenvalues. Complete knowledge of the whole does not imply a state vector for each part. The cat and the gun may have one entangled state vector; the cat alone is then described by a reduced density matrix.

---

[a] **Question from the audience.** Can $\mathrm{Tr}(M\rho)$ be replaced by $(\mathrm{Tr}\, M)(\mathrm{Tr}\, \rho)$?

Response. No. The trace of a product is generally not the product of the traces. Cyclicity permits $\mathrm{Tr}(M\rho) = \mathrm{Tr}(\rho M)$, but it does not factorize the trace.

---

[a] Lecture timestamp: 00:38:25.

[a] **Question from the audience.** Did we begin with only the basis state $|a, b\rangle$, or with the most general pure state?

Response. With the most general pure state of the composite system: $|\Psi\rangle = \sum_{a,b} \psi(a, b) |a, b\rangle$. A single $|a, b\rangle$ is only one basis vector. The surprise is that even this completely specified composite vector can give a mixed reduced state.

[a]Lecture timestamp: 00:38:49.

The contrast with classical mechanics is sharp. Complete classical information about a composite system fixes each of its coordinates and hence each subsystem. A classical probability distribution may certainly be marginalized, but if the original distribution is concentrated at one phase-space point, every marginal is also concentrated at one definite value. Quantum pure states need not have that property.

[a] **Question from the audience.** How is summing over B different from integrating a classical position–velocity distribution over velocity?

Response. The operations look similar, but their pure-state limits differ. A completely known classical phase-space point fixes both position and velocity, so either marginal remains definite. A completely known entangled quantum state can give a nontrivial mixed state for either subsystem. The partial trace therefore reveals a quantum failure of the classical implication from a pure whole to pure parts.

[a]Lecture timestamp: 00:40:46.

8.6 Product States: the Exceptional Case

When does ρ_A remain pure? The answer is that the wave function must factorize:

$$\psi(a, b) = \phi(a)\chi(b). \tag{8.31}$$

Assume ϕ and χ are separately normalized. Substitution into the reduced density matrix gives

$$\rho_A(a, a') = \sum_b \phi(a)\chi(b)\phi^*(a')\chi^*(b) \tag{8.32}$$

$$= \phi(a)\phi^*(a') \sum_b |\chi(b)|^2 \tag{8.33}$$

$$= \phi(a)\phi^*(a'). \tag{8.34}$$

Thus

$$\rho_A = |\phi\rangle \langle\phi| \,. \tag{8.35}$$

It has one eigenvalue 1, all remaining eigenvalues 0, and zero entropy.

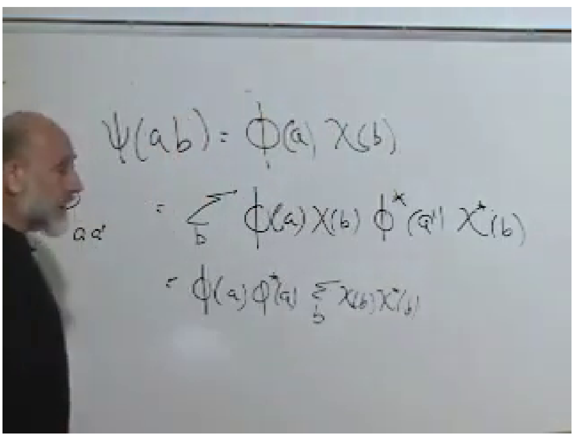

Figure 8.5: For a factorized wave function, the B-sum reduces to the normalization of χ.

Lecture frame: 00:46:35.

[a] **Question from the audience.** Is $\phi(a)$ itself the A-system state?

Response. It is the wave-function component of that state. More explicitly, $|\phi\rangle = \sum_a \phi(a)\,|a\rangle$, and similarly $|\chi\rangle = \sum_b \chi(b)\,|b\rangle$. Factorization means $|\Psi\rangle = |\phi\rangle \otimes |\chi\rangle$, a product state with no entanglement between A and B.

[a] Lecture timestamp: 00:44:17.

For every local observable,

$$\langle M_A \rangle = \mathrm{Tr}(\rho_A M_A) = \langle \phi | \, M_A \, | \phi \rangle \, . \tag{8.36}$$

Subsystem A behaves exactly as a pure state $|\phi\rangle$. For a bipartite pure state, the entropy of ρ_A therefore tests product structure: zero means a product state, while nonzero entropy means entanglement. Values near zero indicate weak entanglement; values near the maximum indicate a state far from any product across this bipartition.

8.7 The Singlet: a Pure Whole and Maximal Local Ignorance

Take the two-spin singlet

$$\left| \Psi^- \right\rangle = \frac{1}{\sqrt{2}} \left(|{\uparrow\downarrow}\rangle - |{\downarrow\uparrow}\rangle \right) . \tag{8.37}$$

Its nonzero wave-function components are

$$\psi(\uparrow,\downarrow) = \frac{1}{\sqrt{2}}, \qquad \psi(\downarrow,\uparrow) = -\frac{1}{\sqrt{2}}, \tag{8.38}$$

while $\psi(\uparrow,\uparrow) = \psi(\downarrow,\downarrow) = 0$.

Trace over spin 2. The diagonal entries are

$$\rho_1(\uparrow,\uparrow) = |\psi(\uparrow,\uparrow)|^2 + |\psi(\uparrow,\downarrow)|^2 = \frac{1}{2}, \tag{8.39}$$

$$\rho_1(\downarrow,\downarrow) = |\psi(\downarrow,\uparrow)|^2 + |\psi(\downarrow,\downarrow)|^2 = \frac{1}{2}. \tag{8.40}$$

The off-diagonal entry is

$$\begin{aligned} \rho_1(\uparrow,\downarrow) &= \psi(\uparrow,\uparrow)\psi^*(\downarrow,\uparrow) + \psi(\uparrow,\downarrow)\psi^*(\downarrow,\downarrow) & (8.41) \\ &= 0, & (8.42) \end{aligned}$$

and its conjugate also vanishes. Hence

$$\rho_1 = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \frac{1}{2} I. \tag{8.43}$$

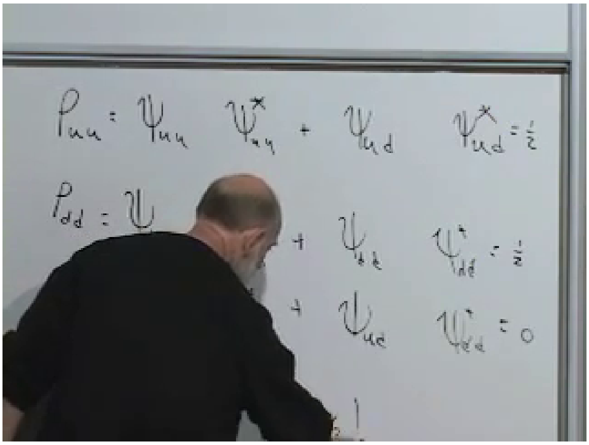

Figure 8.6: The four entries of the reduced density matrix for one spin in the singlet.

Lecture frame: 00:55:35.

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[a] **Question from the audience.** What is the entropy of the single-spin reduced state?

Response. Its two eigenvalues are both 1/2, so

$$S_1 = -2\left(\frac{1}{2}\log\frac{1}{2}\right) = \log 2.$$

This is the maximum entropy available to a two-state subsystem. Equal *eigenvalues*, not equal matrix entries in an arbitrary basis, characterize the maximally mixed state.

---

[a]Lecture timestamp: 00:55:58.

The spin expectation value vanishes along every axis:

$$\langle \boldsymbol{\sigma} \cdot \hat{\boldsymbol{n}} \rangle = \mathrm{Tr}(\rho_1\, \boldsymbol{\sigma} \cdot \hat{\boldsymbol{n}}) = \frac{1}{2}\,\mathrm{Tr}\,(\boldsymbol{\sigma} \cdot \hat{\boldsymbol{n}}) = 0. \tag{8.44}$$

[a] **Question from the audience.** Why is the trace of $\boldsymbol{\sigma}\cdot\hat{\boldsymbol{n}}$ zero for every direction?

Response. Each Pauli matrix is traceless. Since $\boldsymbol{\sigma}\cdot\hat{\boldsymbol{n}} = n_1\sigma_1 + n_2\sigma_2 + n_3\sigma_3$, linearity of the trace gives zero for every unit vector $\hat{\boldsymbol{n}}$. The spin is therefore equally likely to be found parallel or antiparallel to any chosen axis.

[a]Lecture timestamp: 00:58:37.

The whole pair has a definite pure state and entropy zero. Either spin by itself is maximally mixed. This is not ignorance caused by a defective preparation record; it is the local state implied by entanglement.

8.8 An Equal-Amplitude Product State

Now consider

$$|\Phi\rangle = \frac{1}{2}\left(|\uparrow\uparrow\rangle + |\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle + |\downarrow\downarrow\rangle\right). \tag{8.45}$$

All four amplitudes equal 1/2, so normalization follows from four contributions of 1/4. Tracing over spin 2 gives

$$\rho_1 = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}. \tag{8.46}$$

Its trace is 1, but

$$\det \rho_1 = \frac{1}{4} - \frac{1}{4} = 0. \tag{8.47}$$

[a] **Question from the audience.** How do trace and determinant determine the two eigenvalues here?

Response. For a 2×2 matrix, the sum of the eigenvalues is the trace and their product is the determinant. Thus $\lambda_1 + \lambda_2 = 1$ and $\lambda_1\lambda_2 = 0$, which forces the pair $\{1, 0\}$. The reduced state is pure and its entropy is zero.

[a]Lecture timestamp: 01:02:23.

[a] **Question from the audience.** What are the two unentangled spins actually doing?

Response. They are both aligned along the positive x-axis. The answer can be recognized by factorization,

$$|\Phi\rangle = \frac{|\uparrow\rangle + |\downarrow\rangle}{\sqrt{2}} \otimes \frac{|\uparrow\rangle + |\downarrow\rangle}{\sqrt{2}},$$

or established experimentally by computing the Pauli expectation values.

[a]Lecture timestamp: 01:03:18.

For the z-component,

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \mathrm{Tr}(\rho_1\sigma_3) = 0. \tag{8.48}$$

For the x-component,

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{8.49}$$

and

$$\rho_1\sigma_1 = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \tag{8.50}$$

$$\mathrm{Tr}(\rho_1\sigma_1) = 1. \tag{8.51}$$

Thus $\langle\sigma_1\rangle = 1$ while $\langle\sigma_2\rangle = \langle\sigma_3\rangle = 0$. Perturbing the amplitudes away from exact factorization generally changes the eigenvalues from $\{1, 0\}$ to two nonzero values. The resulting small entropy measures a small amount of entanglement.

8.9 Whose Entropy Is It?

The entropy just computed belongs to a subsystem. In the singlet example,

$$S(\rho_{12}) = 0, \qquad S(\rho_1) = S(\rho_2) = \log 2. \tag{8.52}$$

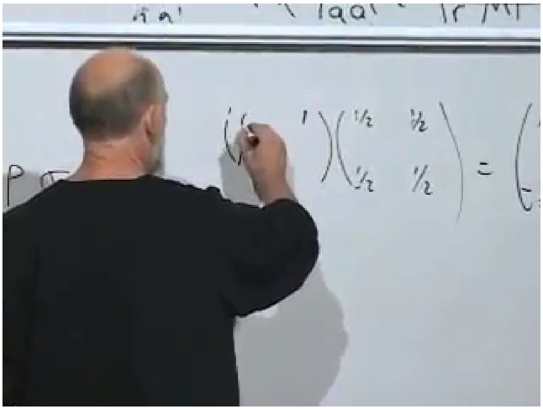

Figure 8.7: The σ_1 matrix calculation for the equal-amplitude reduced state.

Lecture frame: 01:05:13.

The whole is pure while each part is mixed. Entanglement entropy therefore does not obey the naive additive rule $S_{12} = S_1 + S_2$. For a bipartite pure state the two reduced density matrices have the same nonzero eigenvalues, so their entropies are equal even though the entropy of the whole vanishes.

---

[a] **Question from the audience.** Is every density matrix ultimately an entanglement density matrix?

Response. Operationally, one may describe ordinary uncertainty without tracking what carries the missing record. If the preparation apparatus, environment, or observer is included, that uncertainty can often be viewed as entanglement with unobserved degrees of freedom. Mathematically the idea is exact: every density matrix admits a purification on a larger Hilbert space, although that purification does not uniquely identify a physical preparation history.

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[a]Lecture timestamp: 01:09:03.

8.10 From Discrete Classical Motion to Quantum Motion

We now leave entanglement temporarily and ask how states change with time. For a classical system with finitely many configurations, reversible evolution is a permutation. Each state has one successor and one predecessor; the motion decomposes into cycles.

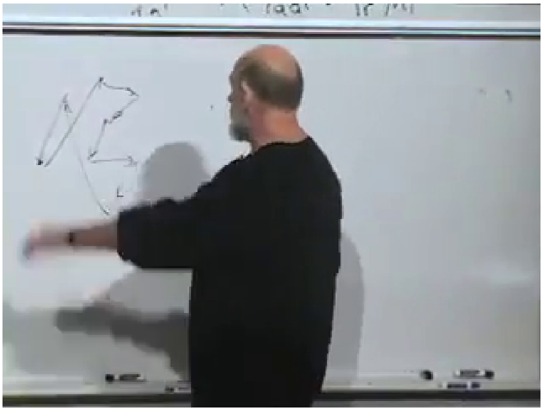

Figure 8.8: Classical reversible evolution represented by arrows among discrete configurations.

Lecture frame: 01:11:40.

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[a] **Question from the audience.** Why must the time steps also be discrete if the configurations are discrete?

Response. A strictly classical two-state model contains no intermediate configurations through which a continuous path could pass. Turning a physical coin slowly introduces its orientation as an additional continuous coordinate, so it is no longer a two-configuration model. A quantum two-state system is different: its Hilbert space contains a continuous family of superpositions even though an orthonormal basis has only two members.

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[a]Lecture timestamp: 01:11:51.

A spin-$\frac{1}{2}$ state can rotate continuously from $|\uparrow\rangle$ toward $|\downarrow\rangle$ through superpositions. Finite dimension does not imply a finite set of state vectors. This allows continuous time evolution in a discrete quantum system.

Reversible microscopic dynamics also preserves distinctions. Two different classical initial conditions do not merge into one microscopic state, because that would erase which initial condition occurred. In Hamiltonian mechanics this invertibility is accompanied by Liouville's theorem: phase-space volume is preserved.

---

[a] **Question from the audience.** What about nonlinear bifurcations in which several initial conditions approach the same steady state?

Response. Attractors occur in effective dissipative descriptions, where environmental degrees of freedom have been discarded. Fundamental microscopic Hamiltonian evolution remains invertible; information may be transferred to unobserved variables, but distinct complete states do not literally become one complete state.

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[a]Lecture timestamp: 01:15:26.

The quantum statement is unitarity. Orthogonal initial states evolve into orthogonal final states, and more generally every inner product is preserved.

---

[a] **Question from the audience.** Must the two states be subjected to exactly the same influence? What happens to initial z-up and z-down states in an x-directed magnetic field?

Response. The same evolution operator must act on both states. In the transverse field the two spins precess, but their Bloch vectors remain opposite and the state vectors remain orthogonal. Applying different external histories would amount to comparing different evolution operators.

---

[a]Lecture timestamp: 01:16:50.

8.11 Linearity and Unitarity

The first principle of quantum time evolution is linearity:

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle\,. \tag{8.53}$$

The operator depends on the elapsed time and satisfies

$$U(0) = I. \tag{8.54}$$

Like the linearity of quantum mechanics itself, this rule is justified by its empirical success.

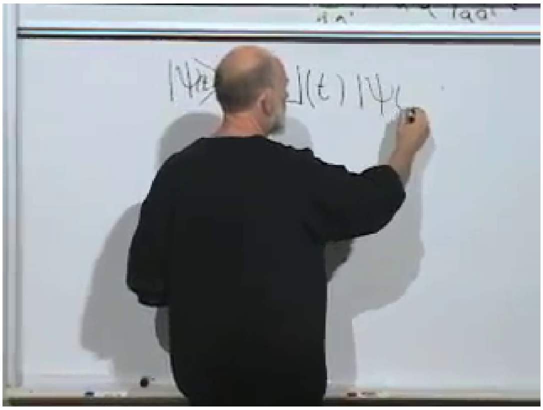

Figure 8.9: The state at time t is obtained by a linear operator $U(t)$ acting on the initial state.

Lecture frame: 01:19:00.

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[a] **Question from the audience.** What must $U(t)$ be at $t = 0$?

Response. It must be the identity. At zero elapsed time the state is unchanged for every possible initial vector.

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[a]Lecture timestamp: 01:19:01.

The second principle is preservation of inner products:

$$\langle\phi(t)\,|\,\psi(t)\rangle = \langle\phi(0)\,|\,\psi(0)\rangle\,. \tag{8.55}$$

When $\phi = \psi$, this preserves normalization and hence total probability. When the initial states are orthogonal, it preserves perfect distinguishability. For arbitrary states it preserves their entire geometric relation.

The bra evolves according to

$$\langle \phi(t)| = \langle \phi(0)|\, U^{\dagger}(t), \tag{8.56}$$

so

$$\begin{aligned} \langle \phi(t) \,|\, \psi(t) \rangle &= \langle \phi(0)|\, U^{\dagger}(t) U(t)\, |\psi(0)\rangle && (8.57) \\ &= \langle \phi(0) \,|\, \psi(0) \rangle \,. && (8.58) \end{aligned}$$

Since this equality holds for every pair of initial vectors,

$$\boxed{U^{\dagger}(t) U(t) = I}. \tag{8.59}$$

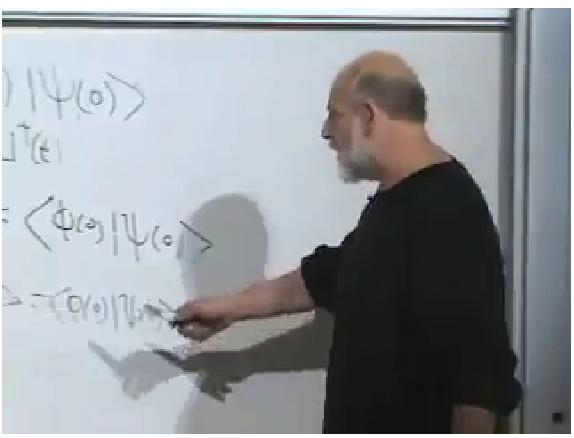

Figure 8.10: Ket evolution, bra evolution, and the inner-product condition leading to $U^{\dagger}U = I$.

Lecture frame: 01:23:30.

[a] **Question from the audience.** Why does the bra acquire $U^\dagger$ rather than U?

Response. Taking the Hermitian conjugate reverses the order of factors and replaces each operator by its Hermitian conjugate:

$$\big(U\,|\phi\rangle\big)^\dagger = \langle\phi|\,U^\dagger.$$

That is the dual action required for consistent inner products.

[a]Lecture timestamp: 01:22:10.

[a] **Question from the audience.** If U were just a complex number, what would $U^\dagger U = 1$ mean?

Response. It would mean $|U| = 1$, so $U = e^{i\theta}$ lies on the unit circle. A unitary matrix is the operator analogue of a pure phase: it preserves lengths and inner products.

[a]Lecture timestamp: 01:24:34.

8.12 Infinitesimal Evolution and the Hamiltonian

Let ϵ be a very small time interval. Since $U(0) = I$, write the first-order expansion

$$U(\epsilon) = I - i\epsilon h + O(\epsilon^2). \tag{8.60}$$

The minus sign and the factor of i are convenient conventions. Taking the Hermitian conjugate gives

$$U^\dagger(\epsilon) = I + i\epsilon h^\dagger + O(\epsilon^2). \tag{8.61}$$

Unitarity then requires

$$\begin{aligned} I &= U^\dagger(\epsilon)U(\epsilon) && (8.62)\\ &= \left(I + i\epsilon h^\dagger\right)(I - i\epsilon h) + O(\epsilon^2) && (8.63)\\ &= I + i\epsilon(h^\dagger - h) + O(\epsilon^2). && (8.64)\end{aligned}$$

The first-order term must vanish:

$$\boxed{h^\dagger = h}. \tag{8.65}$$

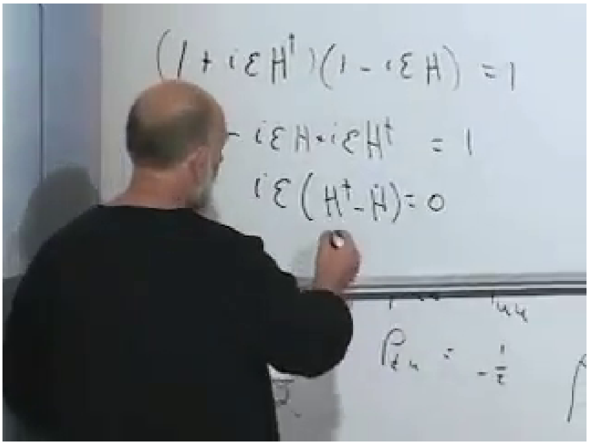

Figure 8.11: First-order unitarity forces the infinitesimal generator to be Hermitian.

Lecture frame: 01:29:35.

The infinitesimal generator is Hermitian.

In the temporary convention above, h has units of angular frequency. Restoring physical units, define the energy operator

$$H = \hbar h. \tag{8.66}$$

This Hermitian operator is the Hamiltonian, and its eigenvalues are energy levels.

Apply the infinitesimal operator to a state:

$$|\psi(\epsilon)\rangle = \left(I - \frac{i\epsilon}{\hbar}H\right)|\psi(0)\rangle + O(\epsilon^2). \tag{8.67}$$

Subtract the initial state, divide by ϵ, and take the limit:

$$\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = -\frac{i}{\hbar}H|\psi(t)\rangle. \tag{8.68}$$

Equivalently,

$$\boxed{i\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = H|\psi(t)\rangle}. \tag{8.69}$$

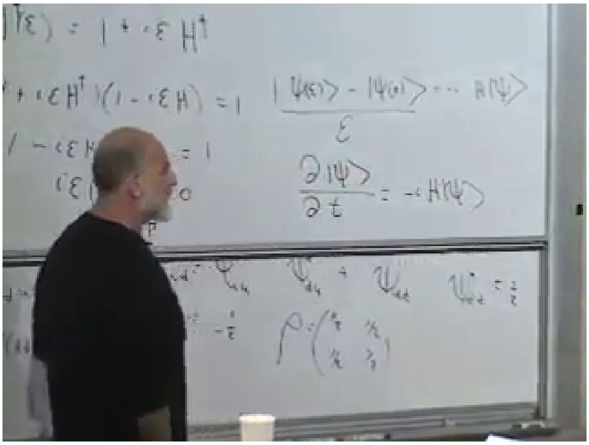

Figure 8.12: The infinitesimal difference quotient becomes the time derivative of the state.

Lecture frame: 01:32:18.

This is the general Schrödinger equation.

The Hamiltonian depends on the physical setup. It may change with time if an external magnetic field or another control changes. When the setup is time independent, H is a fixed Hermitian operator.

---

[a] **Question from the audience.** Does preservation of inner products imply that initially unentangled systems must remain unentangled?

Response. No. Unitarity preserves inner products between complete state vectors, not the product structure of a chosen partition. An interaction Hamiltonian can carry a product state into an entangled state. If unitaries could not do this, measurements could not correlate a system with an apparatus.

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[a]Lecture timestamp: 01:34:30.

8.13 The Units of Entropy

[a] **Question from the audience.** Does entanglement have a physical unit such as joules?

Response. Entanglement entropy is an information measure, not an energy. With natural logarithms it is measured in nats; with base-two logarithms it is measured in bits. Both are dimensionless counting conventions.

[a]Lecture timestamp: 01:35:08.

Thermodynamic notation can obscure this point because the Boltzmann constant converts between temperature measured in kelvins and energy:

$$\langle K \rangle_{\text{monatomic}} = \frac{3}{2} k_{\mathrm{B}} T. \tag{8.70}$$

The reversible heat relation is

$$\delta Q_{\text{rev}} = T \, \mathrm{d}S_{\text{th}}. \tag{8.71}$$

If the information entropy is

$$S_{\text{info}} = -\sum_i p_i \log p_i, \tag{8.72}$$

then the conventional thermodynamic entropy is

$$S_{\text{th}} = k_{\mathrm{B}} S_{\text{info}}. \tag{8.73}$$

Setting $k_{\mathrm{B}} = 1$ measures temperature in energy units and leaves entropy dimensionless. Boltzmann's constant is then recognized as a conversion factor rather than a new kind of entanglement unit. The historical aside connecting Boltzmann's statistical ideas with Einstein's 1905 work on Brownian motion does not alter the information-theoretic definition.

8.14 Energy Eigenstates and Phase

[a] **Question from the audience.** Why is the Hermitian generator H identified with energy?

Response. Unitarity alone proves only that the generator is Hermitian. Its identification with energy is an additional physical statement: energy generates time translations. For a time-independent Hamiltonian, the same operator that governs evolution is conserved, and its eigenvalues are the stationary energy levels. Concrete Hamiltonians supply the experimental content of that identification.

[a]Lecture timestamp: 01:39:54.

The relation between Planck's constants and frequency is

$$E = h\nu = \hbar\omega, \qquad \omega = 2\pi\nu, \qquad h = 2\pi\hbar. \tag{8.74}$$

[a] **Question from the audience.** Should the evolution equation contain h, $\hbar$, or an extra factor of 2π?

Response. Using angular frequency ω, the equation is $E = \hbar\omega$, and the Schrödinger equation contains $\hbar$. Using ordinary frequency ν, the equivalent relation is $E = h\nu$. The factor 2π is entirely accounted for by $\omega = 2\pi\nu$ and $h = 2\pi\hbar$.

[a]Lecture timestamp: 01:40:30.

Suppose the initial state is an energy eigenvector:

$$H\,|\psi(0)\rangle = E\,|\psi(0)\rangle\,. \tag{8.75}$$

[a] **Question from the audience.** If the state remains an energy eigenvector, does that mean it does not change?

Response. It may acquire a time-dependent complex factor and remain an eigenvector with the same eigenvalue. Unitarity restricts that factor to a phase. Thus the ray is stationary even while the ket rotates in phase.

[a]Lecture timestamp: 01:43:18.

Write

$$|\psi(t)\rangle = f(t)\,|\psi(0)\rangle\,. \tag{8.76}$$

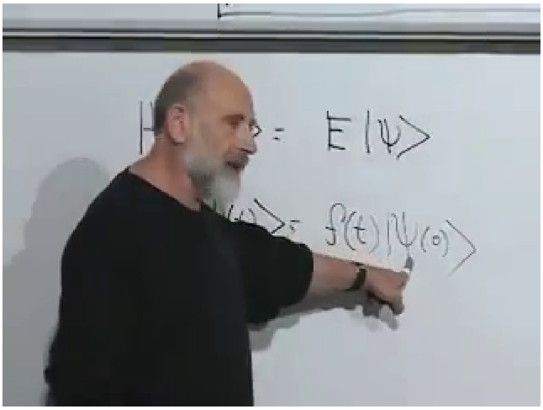

Figure 8.13: An energy eigenstate remains along the same Hilbert-space direction up to a scalar factor $f(t)$.
Lecture frame: 01:43:54.

Substitution into the Schrödinger equation gives

$$i\hbar\frac{\mathrm{d}f}{\mathrm{d}t}\,|\psi(0)\rangle = f(t)H\,|\psi(0)\rangle \tag{8.77}$$

$$= Ef(t)\,|\psi(0)\rangle\,. \tag{8.78}$$

Cancel the common vector:

$$\frac{\mathrm{d}f}{\mathrm{d}t} = -\frac{iE}{\hbar}f(t). \tag{8.79}$$

With $f(0) = 1$,

$$f(t) = e^{-iEt/\hbar}, \qquad |\psi(t)\rangle = e^{-iEt/\hbar}\,|\psi(0)\rangle\,. \tag{8.80}$$

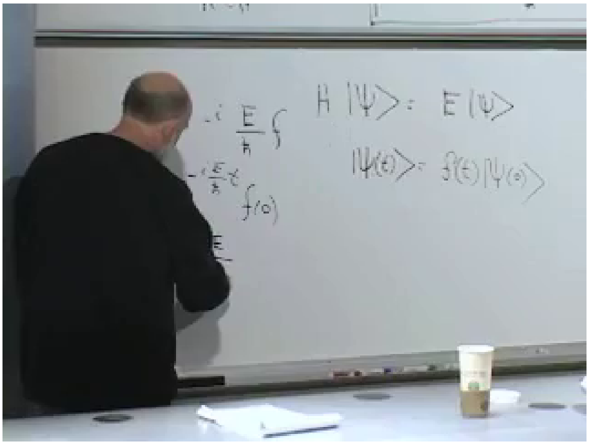

Figure 8.14: The energy eigenvalue equation and the resulting phase evolution.

Lecture frame: 01:46:10.

---

[a] **Question from the audience.** What is the coefficient of t in the phase $e^{-iEt/\hbar}$?

Response. It is the angular frequency,

$$\omega = \frac{E}{\hbar}.$$

Equivalently $E = \hbar\omega$. A real exponential coefficient would describe a growth or decay rate; the imaginary exponent here describes unitary phase rotation.

---

[a] Lecture timestamp: 01:45:39.

This phase-frequency relation and the Schrödinger equation are two forms of the same law of time evolution. The equation is general; a free particle, a particle in a potential, and a spin in a magnetic field differ through their Hamiltonians. Choosing and solving those Hamiltonians is the next step.

CHAPTER 9

TIME EVOLUTION, CONSERVATION, AND COUPLED SPINS

A quantum state is not a fixed point in Hilbert space. As time passes it moves, yet it must move without losing normalization, probabilities, or the distinguishability encoded in inner products. Those requirements nearly determine the form of quantum dynamics. Linearity gives a time-evolution operator, probability conservation makes that operator unitary, and an infinitesimal unitary transformation is generated by a Hermitian operator. That operator is the Hamiltonian. We shall derive the Schrödinger equation from these statements, see how energy eigenstates accumulate phase, and then turn the formalism into two concrete calculations: precession of one spin in a magnetic field and oscillation between two states of a coupled spin pair.

9.1 Linear Motion in Hilbert Space

Let the state at time t be $|\psi(t)\rangle$. We assume that its relation to the state at time zero is linear:

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle\,. \tag{9.1}$$

The linearity is a postulate, not a theorem derived from something more elementary. It is an experimentally successful part of quantum mechanics, and attempts to replace it generally endanger either probability or conservation laws. In a chosen basis, $U(t)$ is simply a time-dependent matrix.

The dual vector obeys the conjugate rule. If

$$|\phi(t)\rangle = U(t)\,|\phi(0)\rangle\,, \tag{9.2}$$

then

$$\langle\phi(t)| = \langle\phi(0)|\,U^\dagger(t). \tag{9.3}$$

For a matrix, the dagger means complex conjugation followed by transposition. The bra therefore carries the same evolution law as the ket, expressed in the dual space.

9.1.1 Probability Conservation Implies Unitarity

A normalized state represents total probability one:

$$\langle \psi(t) \,|\, \psi(t) \rangle = 1. \tag{9.4}$$

Time evolution must preserve this norm. In fact, a linear map that preserves the norm of every vector also preserves every inner product. Geometrically, preserving the lengths of two vectors and their difference preserves the triangle formed by them and hence their angle. In a complex vector space the same conclusion follows by also considering combinations with a factor of i.

Thus, for arbitrary $|\phi\rangle$ and $|\psi\rangle$,

$$\langle \phi(t) \,|\, \psi(t) \rangle = \langle \phi(0) \,|\, \psi(0) \rangle \,. \tag{9.5}$$

Substituting the evolution laws gives

$$\langle \phi(0) |\, U^\dagger(t) U(t) \,|\psi(0)\rangle = \langle \phi(0) \,|\, \psi(0) \rangle \,. \tag{9.6}$$

Since this equation holds for every pair of vectors,

$$\boxed{U^\dagger(t) U(t) = I}. \tag{9.7}$$

Such an operator is unitary. The scalar analogue is a complex number of unit magnitude,

$$e^{i\theta} \left(e^{i\theta}\right)^* = 1. \tag{9.8}$$

A unitary operator is the matrix generalization of a pure phase: it can rotate a state through Hilbert space, but it cannot stretch or shrink it.

9.2 The Infinitesimal Generator

Finite evolution is built from very small time steps. Let the elapsed time be ϵ, small enough that terms of order ϵ^2 may be ignored. At zero elapsed time nothing happens, so

$$U(0) = I. \tag{9.9}$$

The leading correction must be proportional to ϵ. We write it in the conventional form

$$\boxed{U(\epsilon) = I - \frac{i\epsilon}{\hbar} H}. \tag{9.10}$$

Only the factor of ϵ is forced at this stage. The i, $\hbar$, and the name H are chosen so that the object left at the end has the standard normalization and is Hermitian.

---

[a] **Question from the audience.** Why is Planck's constant such a remarkably small number in familiar units?

Response. The numerical size reflects the units chosen by human beings. Chemists selected macroscopic amounts that fit in hand-sized vessels, while atoms are vastly smaller than people. In units adapted to those macroscopic scales, an atomic quantum of action must look tiny. This is not a separate dynamical mystery; it is largely the historical and human scale built into the units.

---

[a] Lecture timestamp: 00:12:46.

Taking the Hermitian conjugate,

$$U^\dagger(\epsilon) = I + \frac{i\epsilon}{\hbar} H^\dagger. \tag{9.11}$$

Now multiply and retain only first-order terms:

$$U^\dagger(\epsilon)U(\epsilon) = \left(I + \frac{i\epsilon}{\hbar} H^\dagger\right)\left(I - \frac{i\epsilon}{\hbar} H\right) \tag{9.12}$$

$$= I + \frac{i\epsilon}{\hbar}\left(H^\dagger - H\right) + O(\epsilon^2). \tag{9.13}$$

Unitarity requires this product to equal I, and therefore

$$\boxed{H^\dagger = H}. \tag{9.14}$$

The infinitesimal generator of time evolution is Hermitian. Since Hermitian operators represent observables, every dynamical system possesses an observable that generates its motion in time. This is the Hamiltonian. We use H for the operator and reserve E for its possible measured values, the eigenvalues.

---

[a] **Question from the audience.** Why should this generator be called energy? Is energy not $p^2/2m$, or the capacity to do work?

Response. Those are useful descriptions in special classical settings, not a general definition. The expression $p^2/2m$ is only the kinetic energy of a nonrelativistic free particle, and it does not define such things as energy stored in an electromagnetic field. For an abstract system the first universal feature to demand of energy is conservation. We shall shortly prove that the expectation value of H is conserved, and its eigenstate phases will also satisfy $E = \hbar\omega$. Those two facts identify the generator with energy.

---

[a]Lecture timestamp: 00:18:34.

9.3 From a Small Step to the Schrödinger Equation

The infinitesimal update is already a differential equation in disguise. At an arbitrary origin of time,

$$\frac{d}{dt}\left|\psi(t)\right\rangle = \lim_{\epsilon \to 0} \frac{\left|\psi(t+\epsilon)\right\rangle - \left|\psi(t)\right\rangle}{\epsilon}. \tag{9.15}$$

The board first writes this at $t = 0$; shifting the origin gives the general form above.

Use

$$\left|\psi(t+\epsilon)\right\rangle = \left(I - \frac{i\epsilon}{\hbar}H\right)\left|\psi(t)\right\rangle. \tag{9.16}$$

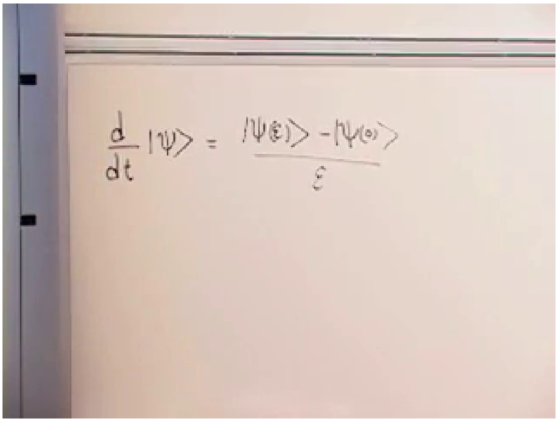

Figure 9.1: The difference quotient that turns infinitesimal evolution into a differential equation.

Lecture frame: 00:22:15.

Then

$$\frac{d}{dt}|\psi(t)\rangle = \lim_{\epsilon \to 0} \frac{\left(I - \frac{i\epsilon}{\hbar} H\right)|\psi(t)\rangle - |\psi(t)\rangle}{\epsilon} \tag{9.17}$$

$$= -\frac{i}{\hbar} H |\psi(t)\rangle . \tag{9.18}$$

Equivalently,

$$\boxed{i\hbar \frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle}. \tag{9.19}$$

This is the Schrödinger equation. Once H and the state at one instant are known, it determines the state at later times. Probabilities for later measurements then follow from the ordinary measurement rules.

9.4 Energy Eigenstates and Rotating Phase

The simplest solution begins with an eigenvector of the Hamiltonian:

$$H|\psi_E\rangle = E|\psi_E\rangle . \tag{9.20}$$

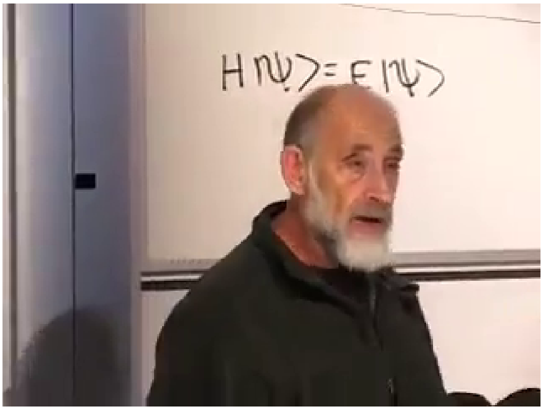

Figure 9.2: The energy-eigenvalue equation on the blackboard.
Lecture frame: 00:28:22.

Since H sends $|\psi_E\rangle$ back into the same one-dimensional direction, time evolution cannot mix it with another eigenvector. It can only multiply it by a scalar:

$$|\psi_E(t)\rangle = f(t)\,|\psi_E\rangle\,. \tag{9.21}$$

Substitution into the Schrödinger equation gives

$$\begin{aligned} i\hbar\,\dot{f}(t)\,|\psi_E\rangle &= f(t)H\,|\psi_E\rangle & (9.22)\\ &= Ef(t)\,|\psi_E\rangle\,. & (9.23) \end{aligned}$$

Therefore

$$\dot{f}(t) = -\frac{iE}{\hbar}f(t), \qquad f(t) = e^{-iEt/\hbar}, \tag{9.24}$$

and

$$\boxed{|\psi_E(t)\rangle = e^{-iEt/\hbar}\,|\psi_E(0)\rangle}. \tag{9.25}$$

An energy eigenstate is stationary in the sense that its ray in Hilbert space does not move. Its ket nevertheless accumulates phase.

The exponential may be written

$$e^{-iEt/\hbar} = \cos\!\left(\frac{Et}{\hbar}\right) - i\sin\!\left(\frac{Et}{\hbar}\right). \tag{9.26}$$

The minus sign and the absence of an extra i inside the sine were corrected during the blackboard calculation. The angular frequency is

$$\omega = \frac{E}{\hbar}, \qquad \boxed{E = \hbar\omega}. \tag{9.27}$$

Equivalently, using ordinary frequency $\nu = \omega/(2\pi)$, one has $E = h\nu$.

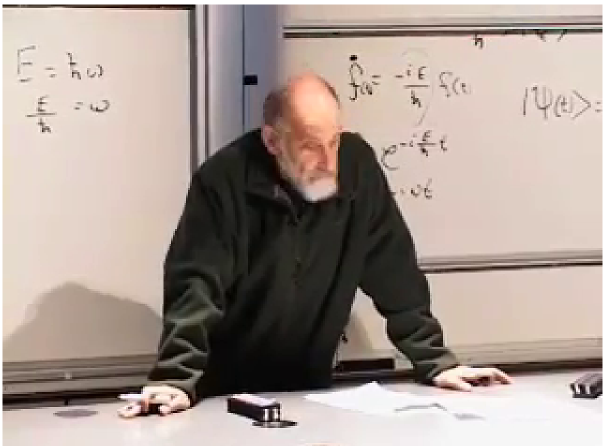

Figure 9.3: The phase solution and the energy–frequency relation are developed on adjacent boards.

Lecture frame: 00:36:19.

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[a] **Question from the audience.** Why were the factors i and $\hbar$ inserted in the infinitesimal operator before the calculation required them?

Response. They are conventions chosen with the answer in view. Without the i, unitarity would make the generator anti-Hermitian. Multiplication by i turns it into a Hermitian observable with real eigenvalues. The factor $\hbar$ gives those eigenvalues the units and normalization convention of energy. The physical statement is unitarity; these factors put its generator into standard form.

---

[a]Lecture timestamp: 00:34:43.

[a] **Question from the audience.** Does $E = \hbar\omega$ belong only to electromagnetic waves and photons?

Response. No. Photons are one important case, but the relation is more general. Any Hamiltonian eigenstate acquires a phase with angular frequency $E/\hbar$. Connecting the frequency of an electromagnetic field mode to the frequency of its photon requires more physics, but the energy–phase relation itself already follows from quantum time evolution.

[a] Lecture timestamp: 00:36:18.

9.5 Superposition in the Energy Basis

A single energy eigenstate changes only by an overall phase, and an overall phase cannot affect a probability. Interesting dynamics begins when a state contains several energy eigenvectors. Since a Hermitian Hamiltonian has a complete orthonormal eigenbasis,

$$H\,|\psi_n\rangle = E_n\,|\psi_n\rangle\,, \qquad |\psi(0)\rangle = \sum_n \alpha_n\,|\psi_n\rangle\,. \tag{9.28}$$

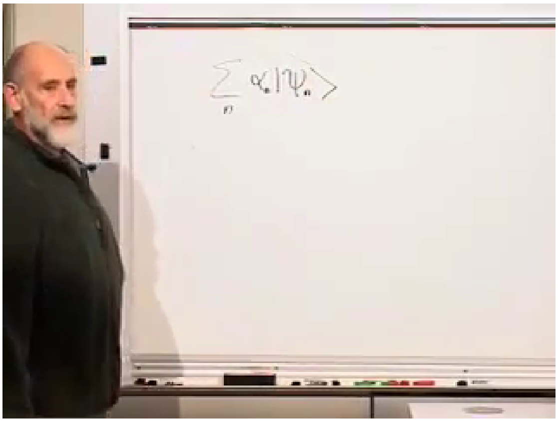

Figure 9.4: Expansion of an arbitrary state in Hamiltonian eigenvectors.

Lecture frame: 00:38:35.

Linearity permits each component to evolve separately:

$$\boxed{|\psi(t)\rangle = \sum_n \alpha_n e^{-iE_n t/\hbar} |\psi_n\rangle}. \tag{9.29}$$

High-energy components wind rapidly around the complex plane; low-energy components wind more slowly. What changes measurable probabilities is not a phase shared by the whole state, but the changing relative phases between components with different E_n.

In the energy basis,

$$\begin{aligned} H &= \begin{pmatrix} E_1 & 0 & \cdots \\ 0 & E_2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}, \\ U(t) &= \begin{pmatrix} e^{-iE_1 t/\hbar} & 0 & \cdots \\ 0 & e^{-iE_2 t/\hbar} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}. \end{aligned} \tag{9.30}$$

The same statement is summarized without choosing a basis by

$$\boxed{U(t) = e^{-iHt/\hbar}}, \tag{9.31}$$

for a time-independent Hamiltonian.

---

[a] **Question from the audience.** Is this decomposition essentially a Fourier series, and is it analogous to decompositions used in heat conduction?

Response. It is indeed a Fourier-like decomposition in time: each energy eigenvector carries a definite frequency. Fourier and Laplace methods also separate modes in many classical equations, including diffusion. The mathematical resemblance is real, but claims about which physical quantity corresponds to frequency must be checked case by case. In quantum mechanics the specific relation is fixed by the Schrödinger equation.

---

[a] Lecture timestamp: 00:40:17.

9.5.1 Combining Systems

For two noninteracting systems A and B, the combined Hamiltonian is the sum of the separate generators on the product space:

$$H_{AB} = H_A \otimes I_B + I_A \otimes H_B. \tag{9.32}$$

If

$$H_A |a\rangle = E_a |a\rangle, \qquad H_B |b\rangle = E_b |b\rangle, \tag{9.33}$$

then

$$H_{AB} |a, b\rangle = (E_a + E_b) |a, b\rangle. \tag{9.34}$$

Energies, and therefore phase frequencies, add for separated noninteracting systems. An interaction contributes another term H_{int}, so this simple sum no longer gives the whole energy.

---

[a] **Question from the audience.** If two separate systems contain components with the same frequency, what happens when the systems are brought together?

Response. For noninteracting systems, the combined energy eigenvalues are sums of the separate eigenvalues. Equality of two frequencies may produce degeneracy, but it does not by itself mix the corresponding eigenvectors. Mixing requires an interaction term or a choice of basis inside a degenerate subspace. Once the systems interact, the enlarged Hamiltonian must be diagonalized as a new problem.

---

[a] Lecture timestamp: 00:56:20.

9.6 Expectation Values and Commutators

We have followed the state while keeping observables fixed. This is the Schrödinger picture. Let A be an observable with no explicit time dependence. Its expectation value is

$$\langle A\rangle_t = \langle\psi(t)| A |\psi(t)\rangle. \tag{9.35}$$

Differentiate both the bra and the ket:

$$\frac{d}{dt}\langle A\rangle_t = \left\langle\dot{\psi}\right| A |\psi\rangle + \langle\psi| A \left|\dot{\psi}\right\rangle. \tag{9.36}$$

The Schrödinger equation and its Hermitian conjugate give

$$\left|\dot{\psi}\right\rangle = -\frac{i}{\hbar} H \left|\psi\right\rangle, \qquad \left\langle\dot{\psi}\right| = \frac{i}{\hbar} \left\langle\psi\right| H. \tag{9.37}$$

Hence

$$\frac{d}{dt}\langle A\rangle_t = \frac{i}{\hbar} \left\langle\psi\right| HA \left|\psi\right\rangle - \frac{i}{\hbar} \left\langle\psi\right| AH \left|\psi\right\rangle \tag{9.38}$$

$$= \frac{i}{\hbar} \left\langle\psi\right| (HA - AH) \left|\psi\right\rangle. \tag{9.39}$$

Define the commutator

$$[H, A] = HA - AH. \tag{9.40}$$

Then the equation of motion for an expectation value is

$$\boxed{\frac{d}{dt}\langle A\rangle_t = \frac{i}{\hbar}\langle [H, A]\rangle_t}. \tag{9.41}$$

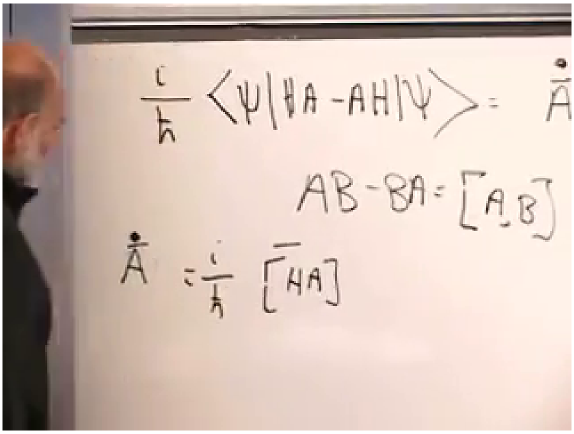

Figure 9.5: The expectation-value equation and the commutator $AB - BA = [A, B]$.

Lecture frame: 00:51:05.

[a] **Question from the audience.** Why is the commutator in this equation itself inside an expectation value?

Response. The commutator is an operator. Any operator placed between $\langle\psi|$ and $|\psi\rangle$ has an expectation value, so $\langle\psi|\,[H, A]\,|\psi\rangle$ is simply the average of that commutator in the current state.

[a] Lecture timestamp: 00:49:13.

9.6.1 Conservation Laws

Set $A = H$. An operator always commutes with itself:

$$[H, H] = HH - HH = 0. \tag{9.42}$$

Therefore

$$\boxed{\frac{d}{dt}\langle H\rangle_t = 0}. \tag{9.43}$$

The expectation value of the Hamiltonian is conserved. This is the promised structural reason for identifying H as energy. Every isolated system has a Hamiltonian, and every time-independent Hamiltonian supplies this conservation law.

More generally,

$$[H, A] = 0 \quad \Longrightarrow \quad \frac{d}{dt}\langle A\rangle_t = 0. \tag{9.44}$$

[a] **Question from the audience.** What is the physical interpretation of the collection of all operators that commute with H, and does it include quantities beyond energy?

Response. They are the conserved observables of the system. Which ones occur depends on the Hamiltonian. A free particle conserves momentum, while a force can destroy that conservation. Energy is always present because H commutes with itself; additional commuting operators express additional symmetries and conservation laws.

[a] Lecture timestamp: 00:51:41.

9.7 One Spin in a Magnetic Field

Consider a spin with Pauli operators

$$\boldsymbol{\sigma} = (\sigma_1, \sigma_2, \sigma_3) \tag{9.45}$$

in a fixed external magnetic field $\mathbf{B}$. The magnetic moment is proportional to the spin. In the convention used here,

$$\boxed{H = \frac{\mu}{2}\boldsymbol{\sigma} \cdot \mathbf{B}}. \tag{9.46}$$

The factor 1/2, the sign, and the charge-dependent scale may be absorbed into the definition of μ. The field is treated classically: a large stable magnet supplies $\mathbf{B}$, and we do not model the quantum state of that magnet.

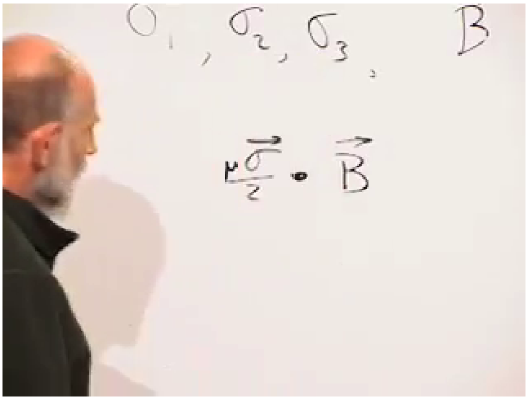

Figure 9.6: The magnetic-moment operator couples to the external magnetic field.

Lecture frame: 01:01:45.

Choose the field along the third axis,

$$\mathbf{B} = B\hat{\mathbf{z}}, \qquad H = \frac{\mu B}{2}\sigma_3. \tag{9.47}$$

We shall track the expectation values of the three spin components. To shorten the equations, write

$$s_i(t) = \langle \sigma_i \rangle_t. \tag{9.48}$$

The longitudinal component obeys

$$\dot{s}_3 = \frac{i}{\hbar}\langle [H, \sigma_3] \rangle = 0, \tag{9.49}$$

because H is proportional to σ_3. The projection of the spin along the field is fixed. Geometrically, the tip of the average spin must remain on a cone of fixed opening angle about the field axis.

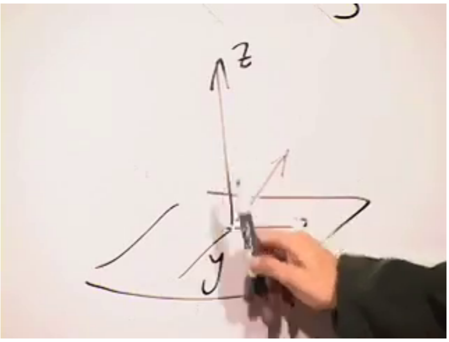

Figure 9.7: Fixed longitudinal spin leaves motion on a cone about the magnetic-field axis.

Lecture frame: 01:04:45.

For the first transverse component,

$$\dot{s}_1 = \frac{i\mu B}{2\hbar}\langle [\sigma_3, \sigma_1] \rangle. \tag{9.50}$$

The commutator can be checked directly from

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{9.51}$$

Indeed,

$$\sigma_3\sigma_1 = i\sigma_2, \qquad \sigma_1\sigma_3 = -i\sigma_2, \tag{9.52}$$

so

$$[\sigma_3, \sigma_1] = 2i\sigma_2. \tag{9.53}$$

Consequently,

$$\dot{s}_1 = -\frac{\mu B}{\hbar} s_2. \tag{9.54}$$

Likewise, using $[\sigma_3, \sigma_2] = -2i\sigma_1$,

$$\dot{s}_2 = \frac{\mu B}{\hbar} s_1. \tag{9.55}$$

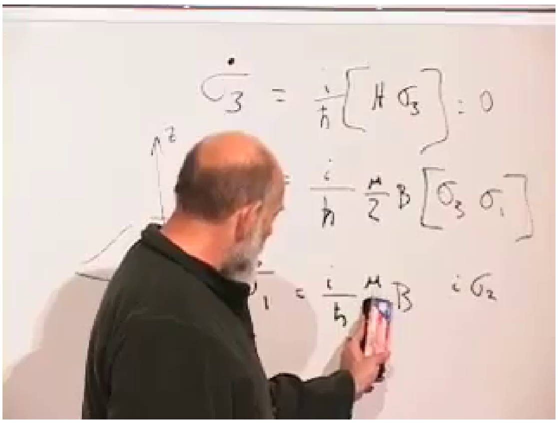

Figure 9.8: The Pauli commutator calculation leading to the transverse equation of motion.

Lecture frame: 01:09:35.

Let

$$\Omega = \frac{\mu B}{\hbar}. \tag{9.56}$$

The three equations are

$$\dot{s}_1 = -\Omega s_2, \qquad \dot{s}_2 = \Omega s_1, \qquad \dot{s}_3 = 0. \tag{9.57}$$

A solution with transverse amplitude $s_\perp$ is

$$s_1(t) = s_\perp \cos(\Omega t + \varphi), \qquad s_2(t) = s_\perp \sin(\Omega t + \varphi), \tag{9.58}$$

with s_3 constant.

The average spin therefore precesses around the magnetic field like a gyroscope. Choosing the field along z was only

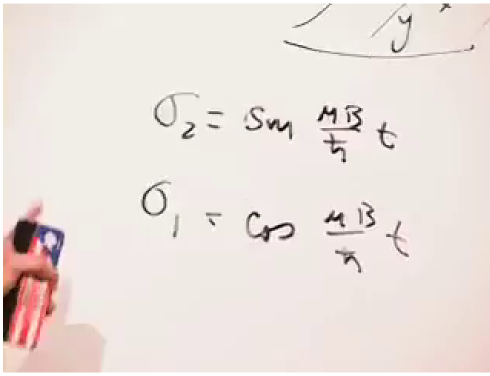

Figure 9.9: The transverse spin components form a sine–cosine pair with angular frequency $\mu B/\hbar$.

Lecture frame: 01:11:45.

a convenience. For a field in any direction, the component parallel to **B** is conserved and the perpendicular component rotates around it.

---

[a] **Question from the audience.** Should the sine and cosine solution contain an additional phase?

Response. Yes. The phase φ specifies the initial direction in the transverse plane, or equivalently the choice of the instant called $t = 0$. Setting it to zero merely chooses the spin to begin along one particular transverse axis.

---

[a]Lecture timestamp: 01:13:02.

9.7.1 Preparation and Measurement

Suppose an ensemble is prepared in the $+1$ eigenstate of σ_1. At $t = 0$, a measurement of σ_1 is certain to return $+1$, while measurements of σ_2 or σ_3 give equal probabilities for ± 1. After a quarter precession period, the state has rotated to the $+1$ eigenstate of σ_2: σ_2 is then certain and σ_1 is maximally uncertain. After another quarter period it reaches the -1 eigenstate of σ_1. The rotating expectation values therefore have direct operational meaning.

[a] **Question from the audience.** If expectation values require many electrons, will the members of the ensemble not carry different phases and wash out the precession?

Response. Not when the same experiment is performed on every member. Identical preparation gives every electron the same initial state and hence the same phase evolution. If their initial phases are uncontrolled, the ensemble is not a pure preparation; it is a mixed state, and dephasing can reduce the transverse average.

[a]Lecture timestamp: 01:14:04.

9.8 Two Coupled Spins

Now consider two spins. Merely saying that they are entangled is not enough to predict what they will do. Dynamics requires a Hamiltonian.

[a] **Question from the audience.** If two electrons are entangled so that opposite results occur along any common axis, does that mean their motions are always 180° out of phase?

Response. No dynamical conclusion follows from entanglement alone. The stated anticorrelation describes the singlet state, but its subsequent evolution depends on the energy operator. We must first specify whether there is an external field, an interaction between the spins, or both.

[a]Lecture timestamp: 01:16:49.

Take the simple isotropic interaction in which each spin supplies the effective field felt by the other:

$$\boxed{H = J\,\boldsymbol{\sigma}\cdot\boldsymbol{\tau} = J\left(\sigma_1\tau_1 + \sigma_2\tau_2 + \sigma_3\tau_3\right)}. \tag{9.59}$$

The operators $\boldsymbol{\sigma}$ act on the first spin and $\boldsymbol{\tau}$ on the second. The coupling J sets the energy scale; its value and sign are left unspecified.

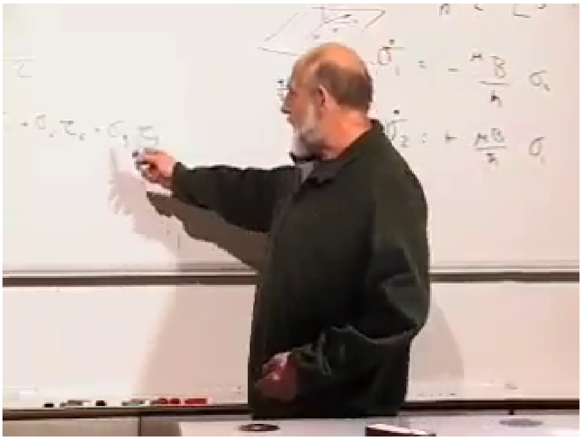

Figure 9.10: The isotropic two-spin interaction $H \propto \boldsymbol{\sigma} \cdot \boldsymbol{\tau}$.

Lecture frame: 01:19:20.

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[a] **Question from the audience.** Would placing two entangled electrons in a uniform magnetic field necessarily disentangle them?

Response. Not necessarily. A field changes the Hamiltonian and may tend to align the spins, especially if it is strong, but unitary evolution in a uniform field does not by itself guarantee loss of entanglement. The answer depends on the complete Hamiltonian and the initial state.

---

[a]Lecture timestamp: 01:17:49.

9.8.1 Triplet and Singlet Energies

The two-spin Hilbert space has dimension four. The interaction is diagonal in the triplet–singlet basis:

$$|T_+\rangle = |{\uparrow\uparrow}\rangle\,, \tag{9.60}$$

$$|T_0\rangle = \frac{|{\uparrow\downarrow}\rangle + |{\downarrow\uparrow}\rangle}{\sqrt{2}}, \tag{9.61}$$

$$|T_-\rangle = |{\downarrow\downarrow}\rangle\,, \tag{9.62}$$

$$|S\rangle = \frac{|{\uparrow\downarrow}\rangle - |{\downarrow\uparrow}\rangle}{\sqrt{2}}. \tag{9.63}$$

The three triplet states share one energy E_T; the singlet has another energy E_S. Thus there are four independent eigenvectors but only two distinct energy levels. The triplet level is threefold degenerate.

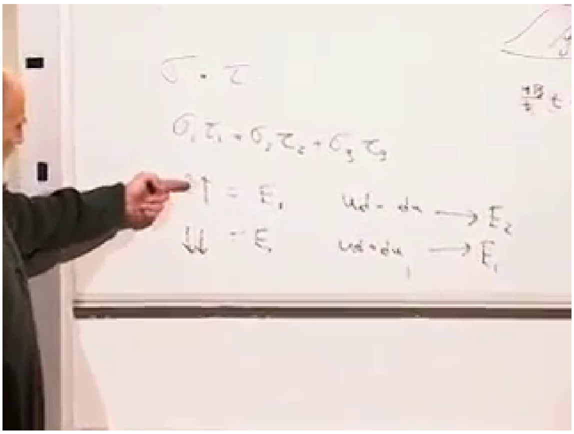

Figure 9.11: The coupled-spin Hamiltonian has a degenerate triplet level and a separate singlet level.

Lecture frame: 01:20:30.

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[a] **Question from the audience.** What happens if the initial state is already the singlet energy eigenstate?

Response. It remains the singlet and acquires only the phase $e^{-iE_S t/\hbar}$. As with any single energy eigenstate, its measurable properties do not change merely because the whole ket is multiplied by that common phase.

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[a] Lecture timestamp: 01:21:16.

[a] **Question from the audience.** Does an external magnetic field supply energy to the two-spin system during this calculation?

Response. There is no external field in the example being solved. The only field under discussion is the mutual interaction represented by $J\boldsymbol{\sigma}\cdot\boldsymbol{\tau}$. Adding an external field would add another term to H and define a different problem.

[a]Lecture timestamp: 01:21:54.

9.8.2 Begin with One Spin Up and One Down

An energy eigenstate would only gather a phase, so choose a product state that is not an energy eigenstate:

$$|\psi(0)\rangle = |\uparrow\downarrow\rangle\,. \tag{9.64}$$

Decompose it into the two relevant eigenvectors:

$$\boxed{|\uparrow\downarrow\rangle = \frac{1}{\sqrt{2}}\,|T_0\rangle + \frac{1}{\sqrt{2}}\,|S\rangle}. \tag{9.65}$$

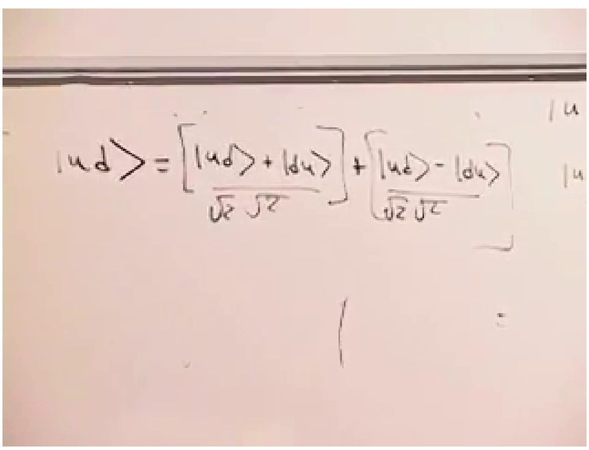

Figure 9.12: The product state $|\uparrow\downarrow\rangle$ decomposed into triplet and singlet components.

Lecture frame: 01:24:30.

Each energy component now receives its own phase:

$$|\psi(t)\rangle = \frac{e^{-iE_T t/\hbar}}{\sqrt{2}} |T_0\rangle + \frac{e^{-iE_S t/\hbar}}{\sqrt{2}} |S\rangle . \tag{9.66}$$

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[a] **Question from the audience.** Which is larger, the triplet energy E_T or the singlet energy E_S?

Response. That depends on the sign of the coupling. For $H = J\boldsymbol{\sigma}\cdot\boldsymbol{\tau}$ with $J > 0$, the singlet is lower and the triplet is higher; reversing the sign reverses the ordering. Adding a constant to both energies changes only an overall phase, so one may set either level to zero without changing the transition probabilities.

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[a] Lecture timestamp: 01:25:59.

Return to the product basis:

$$|\psi(t)\rangle = \frac{1}{2}\left(e^{-iE_T t/\hbar} + e^{-iE_S t/\hbar}\right)|{\uparrow\downarrow}\rangle \tag{9.67}$$

$$+ \frac{1}{2}\left(e^{-iE_T t/\hbar} - e^{-iE_S t/\hbar}\right)|{\downarrow\uparrow}\rangle . \tag{9.68}$$

At $t = 0$, the coefficient of $|{\downarrow\uparrow}\rangle$ vanishes, as it must. At later times the relative phase between the singlet and triplet components makes this coefficient grow. Define the mean energy and splitting,

$$\overline{E} = \frac{E_T + E_S}{2}, \qquad \Delta E = E_T - E_S. \tag{9.69}$$

Factoring out the mean-energy phase gives

$$\boxed{\begin{aligned}|\psi(t)\rangle = e^{-i\overline{E}t/\hbar}\Bigg[&\cos\left(\frac{\Delta E\, t}{2\hbar}\right)|{\uparrow\downarrow}\rangle \\ &- i\sin\left(\frac{\Delta E\, t}{2\hbar}\right)|{\downarrow\uparrow}\rangle\Bigg]\end{aligned}}. \tag{9.70}$$

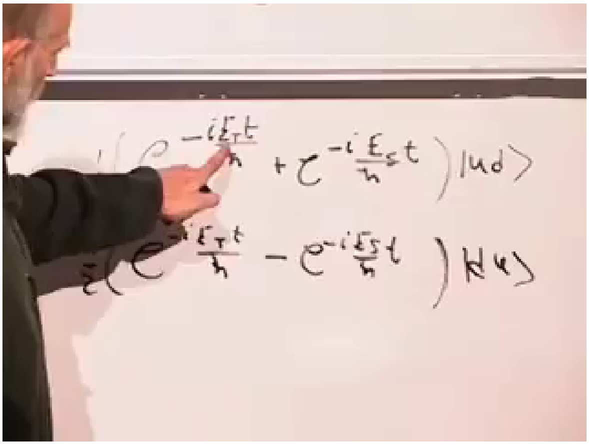

Figure 9.13: The evolved state reassembled in the $|\uparrow\downarrow\rangle$, $|\downarrow\uparrow\rangle$ basis.

Lecture frame: 01:30:40.

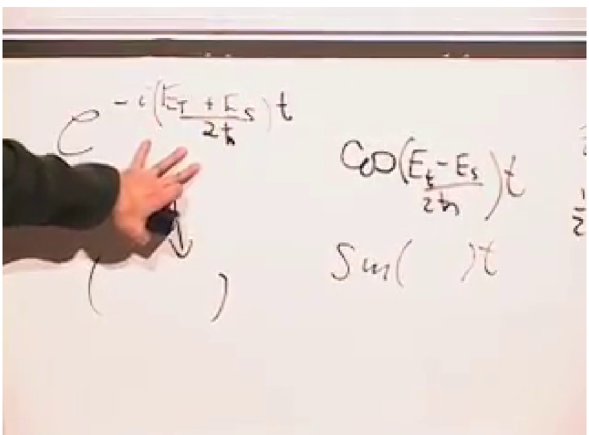

Figure 9.14: The mean energy becomes a common phase, while the singlet–triplet splitting appears in the sine and cosine amplitudes.

Lecture frame: 01:35:15.

The factor $e^{-i\overline{E}t/\hbar}$ is common to the entire ket and drops out of every probability. The measurable probabilities are

$$P_{\uparrow\downarrow}(t) = \cos^2\!\left(\frac{\Delta E\, t}{2\hbar}\right), \qquad P_{\downarrow\uparrow}(t) = \sin^2\!\left(\frac{\Delta E\, t}{2\hbar}\right). \quad (9.71)$$

The state oscillates from $|\uparrow\downarrow\rangle$ to $|\downarrow\uparrow\rangle$ and back, at a rate determined only by the energy difference. At intermediate times both amplitudes are nonzero, and except at special instants the two spins are entangled.

The calculation illustrates a general method. Choose the basis in which the physical question is posed, expand the initial state in eigenvectors of the Hamiltonian, let each eigenvector acquire its own phase, and then recombine the pieces in the measurement basis. Quantum dynamics becomes systematic once the Hamiltonian has been diagonalized; all observable motion is carried by relative phase.